Neural Networks

Some of these slides from
makeyourownneuralnetwork.blogspot.co.uk
See also
http://iamtrask.github.io/2015/07/12/basic-python-network/

Humans and (old fashioned) computers do not excel at the same tasks

locate people in this photo

add these numbers

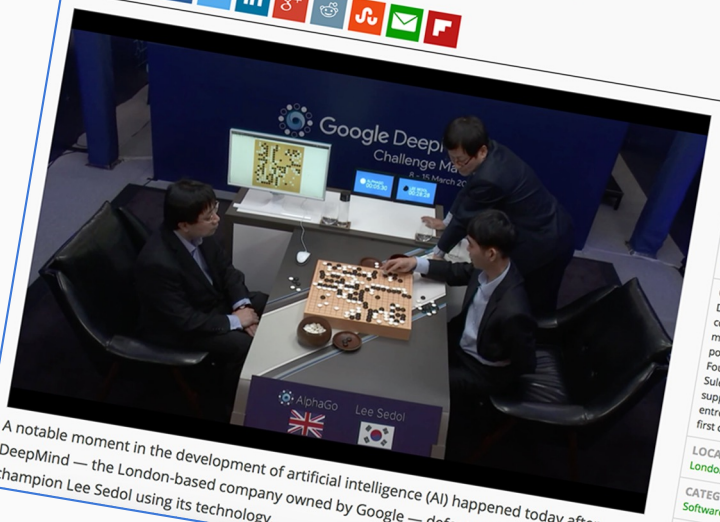
```
   2403343781289312
+  2843033712837981
+  2362142787897881
+  3256541312323213
+  9864479802118978
+  8976677987987897
+  8981257890087988

                  = ?
```

Humans and computers do not excel at same tasks

Imitate brains: NNs at the forefront of recent AI revolution (“Deep Learning”)

Google’s DeepMind chalks up AI landmark after Go world champion Lee Sedol

Posted Mar 9, 2016 by Jon Russell (@jonrussell)



A notable moment in the development of artificial intelligence (AI) happened today after DeepMind — the London-based company owned by Google — defeated Go world champion Lee Sedol using its technology.

CrunchBase

DeepMind

FOUNDED 2011

OVERVIEW DeepMind is a cutting edge arti company. We combine the best machine learning and systems n powerful general-purpose learni Founded by Demis Hassabis, Sha Suleyman, the company is based supported by some of the most ic entrepreneurs and investors of the first commercial ...

LOCATION London, H9

CATEGORIES Software

FOUNDERS Demis Hassabis, Shane Legg, Mustafa S

WEBSITE http://deepmind.com

nature International weekly journal of science

Home | News & Comment | Research | Careers & Jobs | Current Issue | Archive | Audio & Video | For

Archive > Volume 529 > Issue 7587 > Articles > Article

ARTICLE PREVIEW

view full access options

NATURE | ARTICLE

日本語要約

Mastering the game of Go with deep neural networks and tree search

David Silver, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, Sander Dieleman, Dominik Grewe, John Nham, Nal Kalchbrenner, Ilya Sutskever, Timothy Lillicrap, Madeleine Leach, Koray Kavukcuoglu, Thore Graepel & Demis Hassabis

Affiliations | Contributions | Corresponding authors

Imitate brains: NNs at the forefront of recent AI revolution (“Deep Learning”)

New Scientist

HOME NEWS TECHNOLOGY SPACE PHYSICS HEALTH EARTH HUMANS LIFE TOPICS EVENTS JOBS MAGAZINE

Home | News | Health | Technology

TECHNOLOGY NEWS 29 April 2016

Revealed: Google AI has access to huge haul of NHS patient data

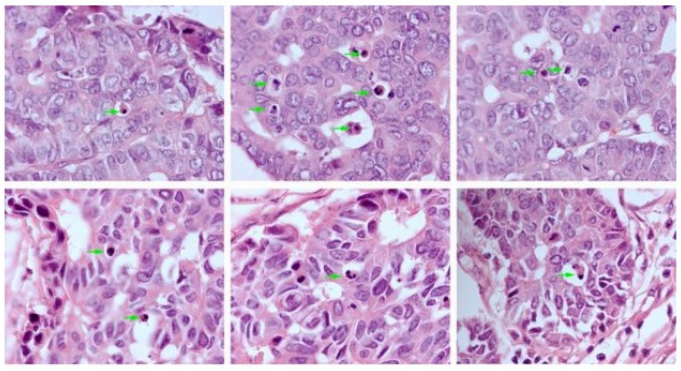
A data-sharing agreement obtained by **New Scientist** shows that Google DeepMind's collaboration with the NHS goes far beyond what it has publicly announced

HOME › EXTREME › AI BEATS DOCTORS AT VISUAL DIAGNOSIS, OBSERVES MANY TIMES MORE LUNG CANCER SIGNALS

AI beats doctors at visual diagnosis, observes many times more lung cancer signals

By Graham Templeton on August 18, 2016 at 1:00 pm | 25 Comments

677 shares



DANIELA HERNANDEZ, KAISER HEALTH NEWS BUSINESS 06.02.14 6:30 AM

ARTIFICIAL INTELLIGENCE IS NOW TELLING DOCTORS HOW TO TREAT YOU

IBM's Watson AI saved a woman from leukemia

It discovered a rare illness that doctors had missed.

Jon Fingas , @jonfingas
08.07.16 in Medicine

8 Comments

16560 Shares

TECHNOLOGY

TECHNOLOGY | RECODE | MOBILE | SOCIAL MEDIA | ENTERPRISE | GAMING

AI will eliminate 6 percent of jobs in five years, says report

Harriet Taylor | @Harri8t
Monday, 12 Sep 2016 | 5:03 PM ET

CNBC

Imitate brains: NNs at the forefront of recent AI revolution ("Deep Learning")

New Scientist

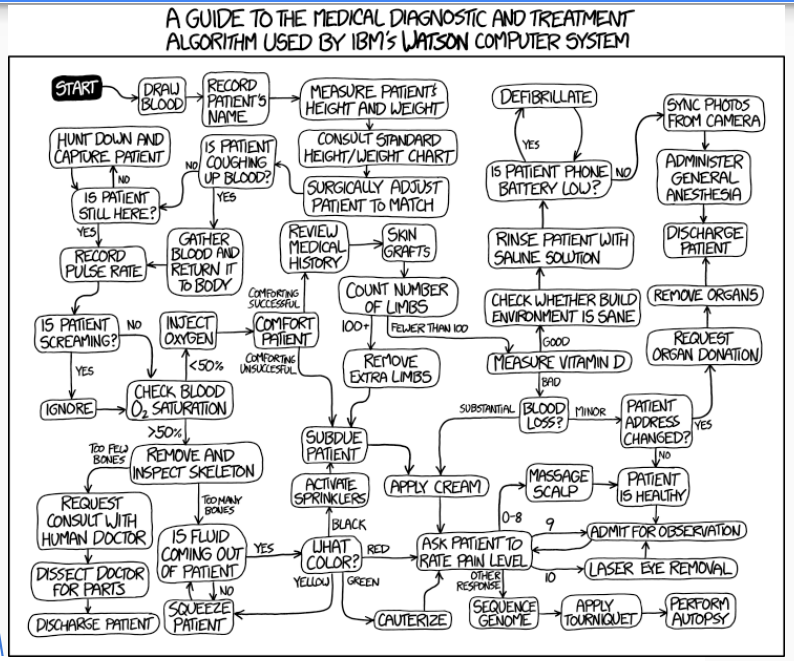
HOME NEWS TECHNOLOGY SPACE PHYSICS HEALTH EARTH HUMANS LIFE TOPICS EVENTS JOBS MAGAZINE

Home | News | Health | Technology

TECHNOLOGY NEWS 29 April 2016

Revealed: Google AI has access to huge haul of NHS patient data

A data-sharing agreement obtained by **New Scientist** shows that Google DeepMind's collaboration with the NHS goes far beyond what it has publicly announced

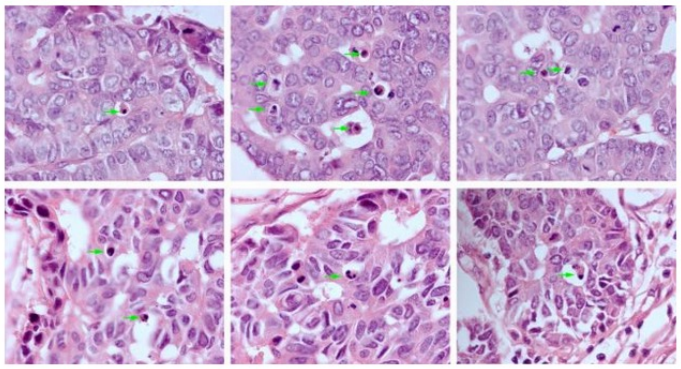


HOME › EXTREME › AI BEATS DOCTORS AT VISUAL DIAGNOSIS, OBSERVES MANY TIMES MORE LUNG CANCER SIGNALS

AI beats doctors at visual diagnosis, observes many times more lung cancer signals

By Graham Templeton on August 18, 2016 at 1:00 pm | 25 Comments

677 shares



IBM's Watson AI saved a woman from leukemia

It discovered a rare illness that doctors had missed.

Jon Fingas , @jonfingas
08.07.16 in Medicine

8 Comments

16560 Shares

TECHNOLOGY

TECHNOLOGY | RECODE | MOBILE | SOCIAL MEDIA | ENTERPRISE | GAMING

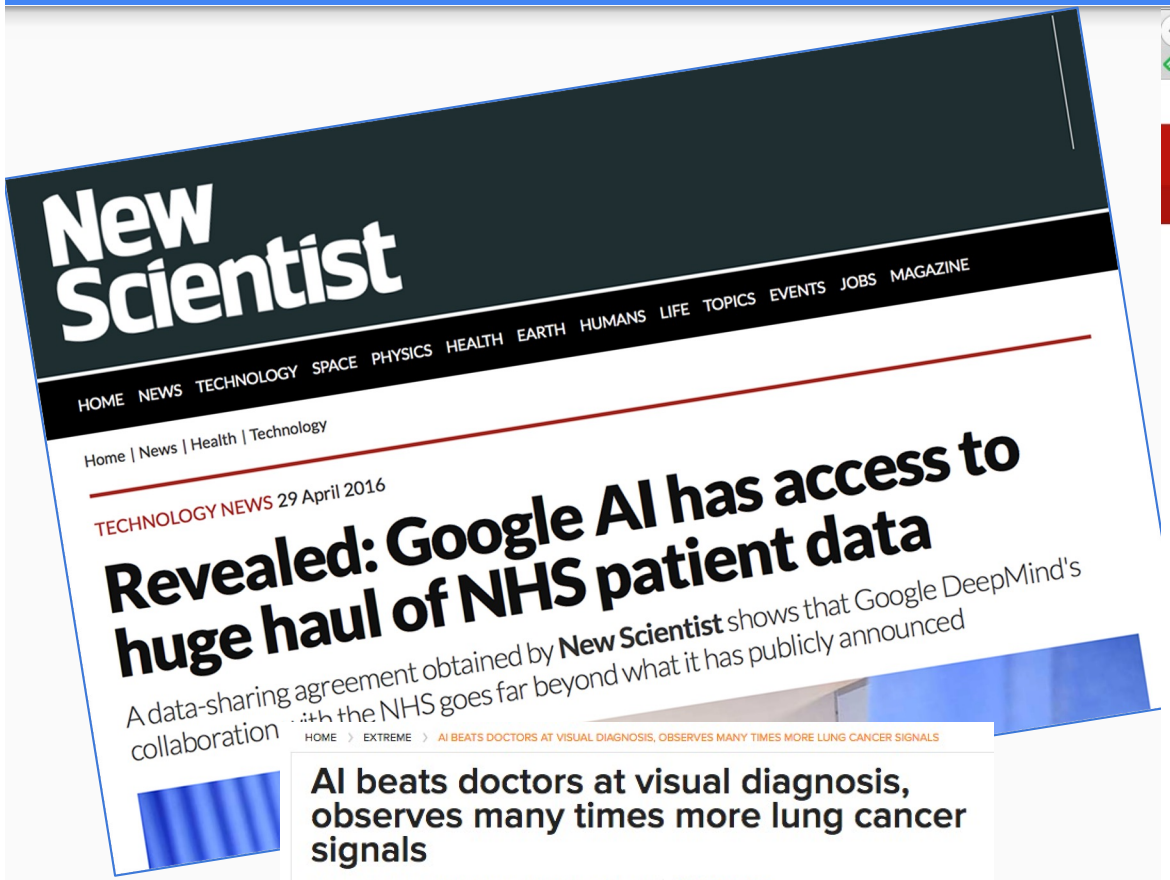
AI will eliminate 6 percent of jobs in five years, says report

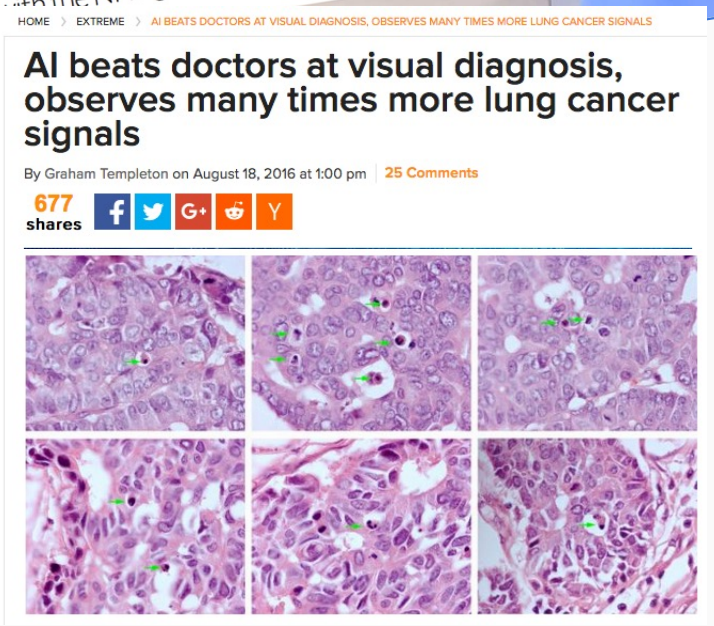
Harriet Taylor | @Harri8t

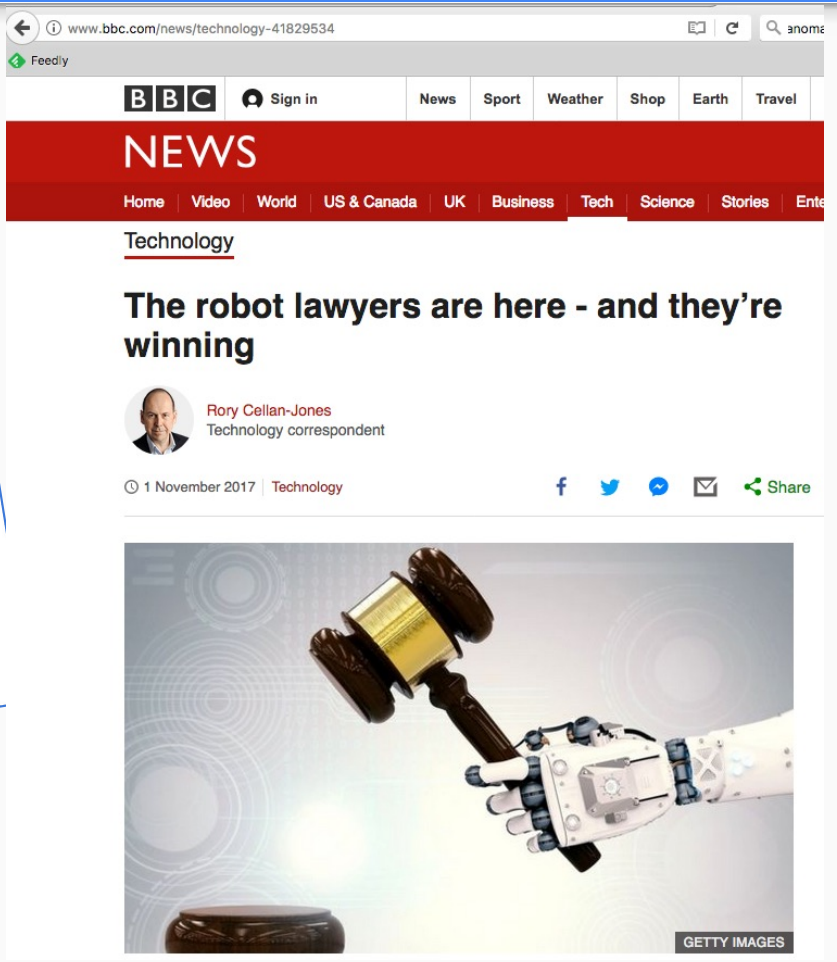
Monday, 12 Sep 2016 | 5:03 PM ET

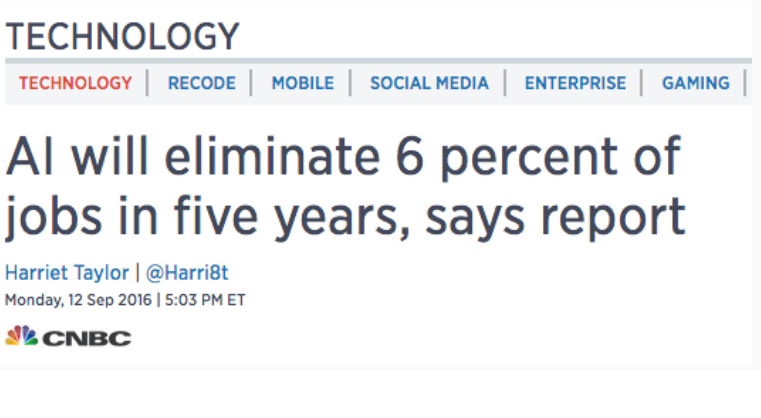
CNBC

Imitate brains: NNs at the forefront of recent AI revolution (“Deep Learning”)





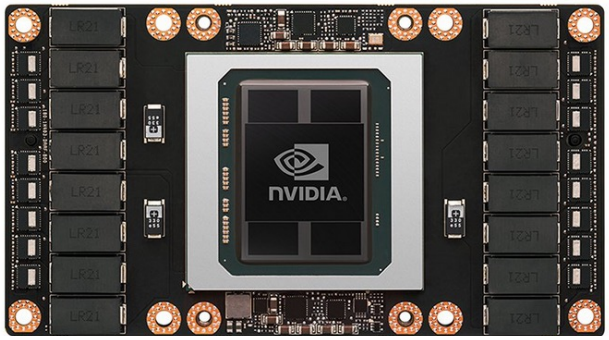




Altering the ecosystem: from NN chip design to drastic societal changes

TECHNOLOGY

Nvidia Unveils Artificial-Intelligence Tech For Baidu Self-Driving Cars



Nvidia releases Pascal GPUs for neural networks

Nvidia has launched a pair of GPUs that the chip giant claims are four times faster than last year's offering.

CRUNCH NETWORK

The driverless truck is coming, and it's going to automate millions of jobs

Posted Apr 25, 2016 by Ryan Petersen (@typesfast)

Pittsburgh residents get ready for driverless Uber cars

Business

Self-driving cars set to disrupt UK's £14bn motor insurance industry

share

Self-driving cars will result in an 80pc drop in crashes by 2035 CREDIT: ALAMY

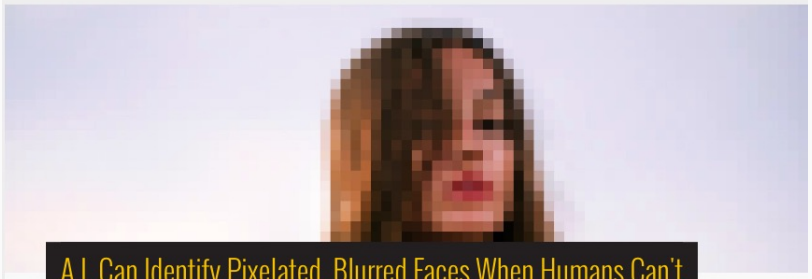
MOTHERBOARD Watch Sections

The Tradeoffs of Imbuing Self-Driving Cars With Human Morality

August 25, 2016 // 07:00 AM EST

NNs catching up on everything we can do, and then some

Explore INVERSE Follow



A.I. Can Identify Pixelated, Blurred Faces When Humans Can't

AT&T Top 1	2.50	95.00	95.00	96.25	95.00	96.25
AT&T Top 5	12.50	100	100	100	98.75	98.75
FaceScrub Top 1	0.19	75.49	71.53	69.91	65.25	57.56
FaceScrub Top 5	0.94	86.06	83.74	82.08	79.13	72.23

On the top, visual examples of two industry-standard obfuscation methods. On the bottom, the neural networks' accuracy. Each number corresponds with the obfuscation level above, left to right (or non-obfuscated to most-obfuscated).

Feedback | Like 5.8M | Follow @MailOnline | DailyMail | Wednesday, Sep 14th

Daily Mail .com

Home | U.K. | News | Sports | U.S. Showbiz | Australia | Femail | Health | Science | Money

Latest Headlines | Science | Pictures

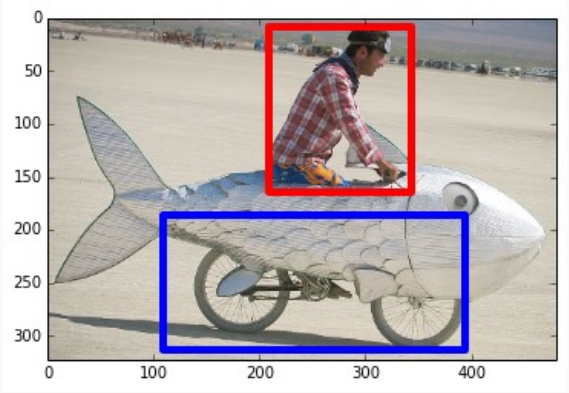
The phone keyboard that 'thinks' like a human: SwiftKey reveals predictive system that uses a neural network to know exactly what you want to type

- The Neural Alpha keyboard can complete whole sentences for you
- It works by looking at clusters of words to find meaning of a sentence
- The keyboard has gone live today and can be downloaded on Android

By ELLIE ZOLFAGHARIFARD FOR DAILYMAIL.COM

PUBLISHED: 16:24 EST, 8 October 2015 | **UPDATED:** 15:03 EST, 15 October 2015

Share 65 shares 29 View comments



NNs catching up on everything we can do, and then some

ars TECHNICA BIZ & IT TECH SCIENCE POLICY CARS GAMING & CULTURE FORUMS SIGN IN

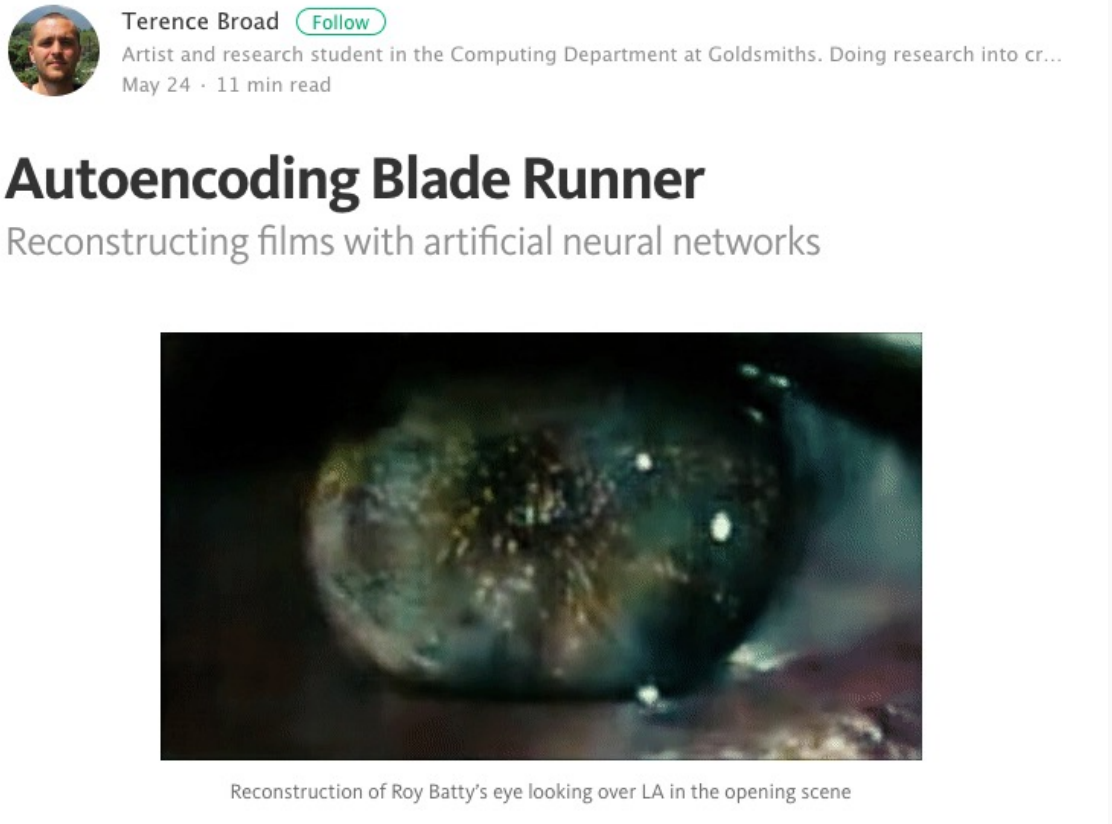
THE MULTIVERSE —

Movie written by algorithm turns out to be hilarious and intense

For *Sunspring*'s exclusive debut on Ars, we talked to the filmmakers about collaborating with an AI.

ANNALEE NEWITZ - 6/9/2016, 5:30 AM

Sunspring, a short science fiction movie written entirely by AI, debuts exclusively on Ars today.



Terence Broad Follow

Artist and research student in the Computing Department at Goldsmiths. Doing research into cr...

May 24 · 11 min read

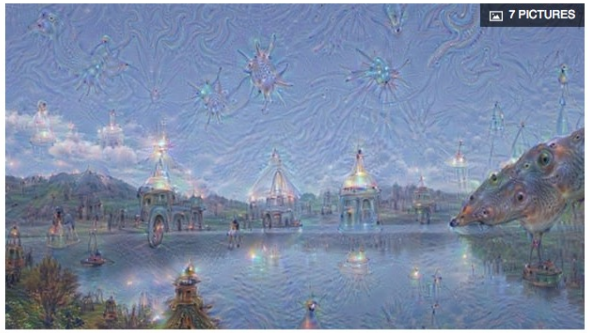
Autoencoding Blade Runner

Reconstructing films with artificial neural networks

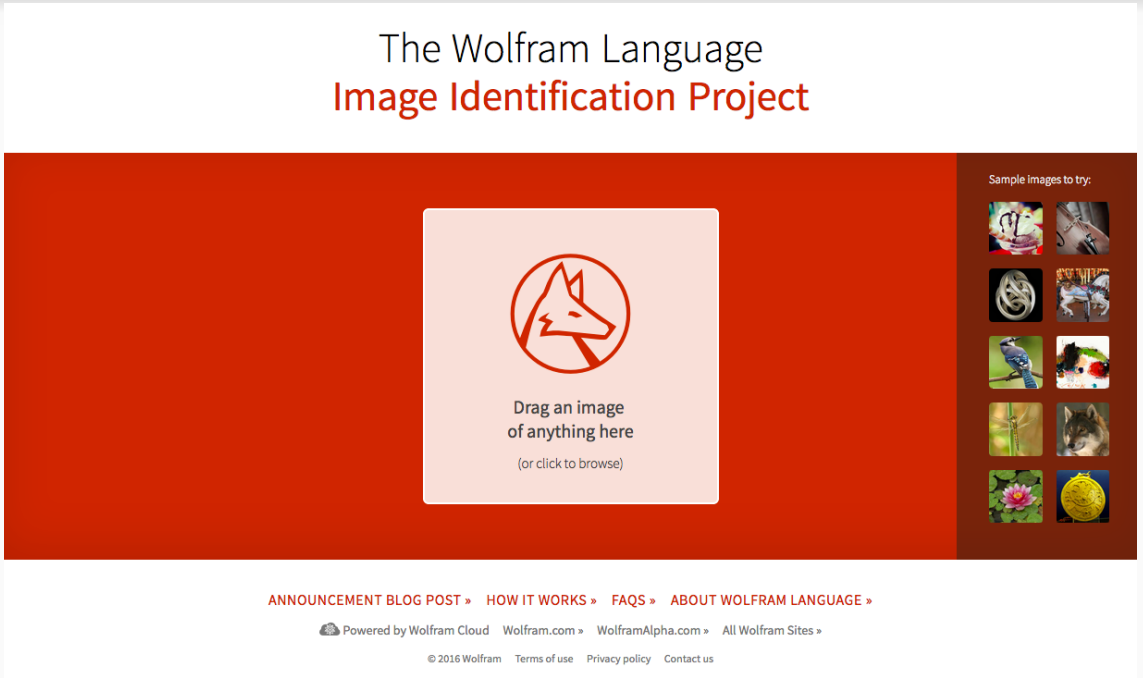
Reconstruction of Roy Batty's eye looking over LA in the opening scene

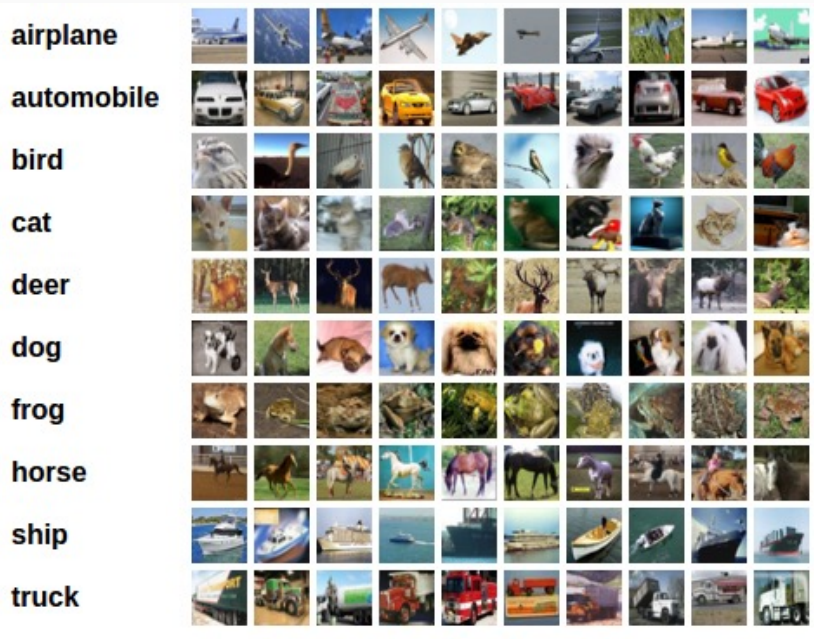
Google's neural networks create bizarre "Inceptionism" art

David Szondy | June 28, 2015

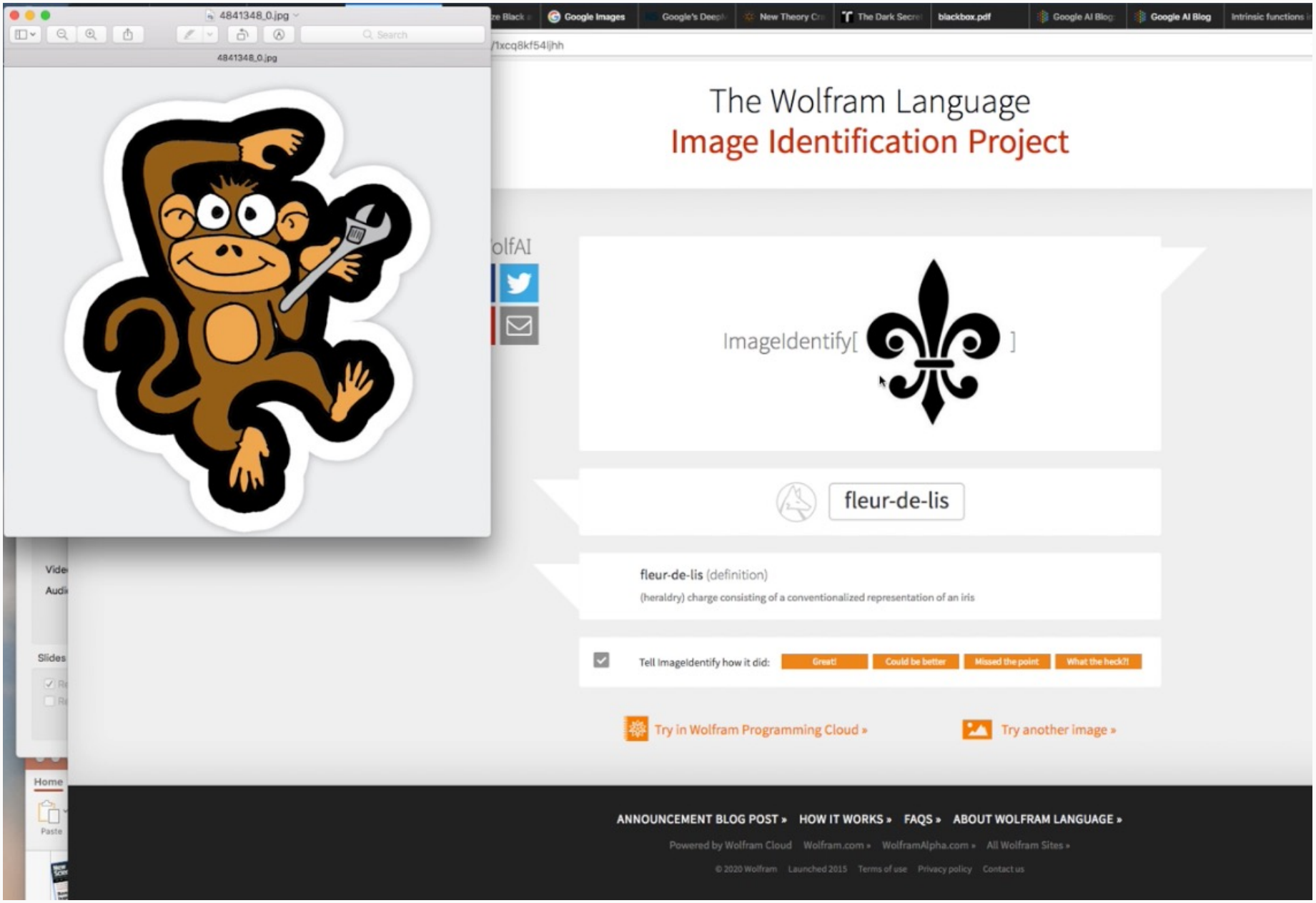


NNs need training! (not always: supervised vs. unsupervised learning)

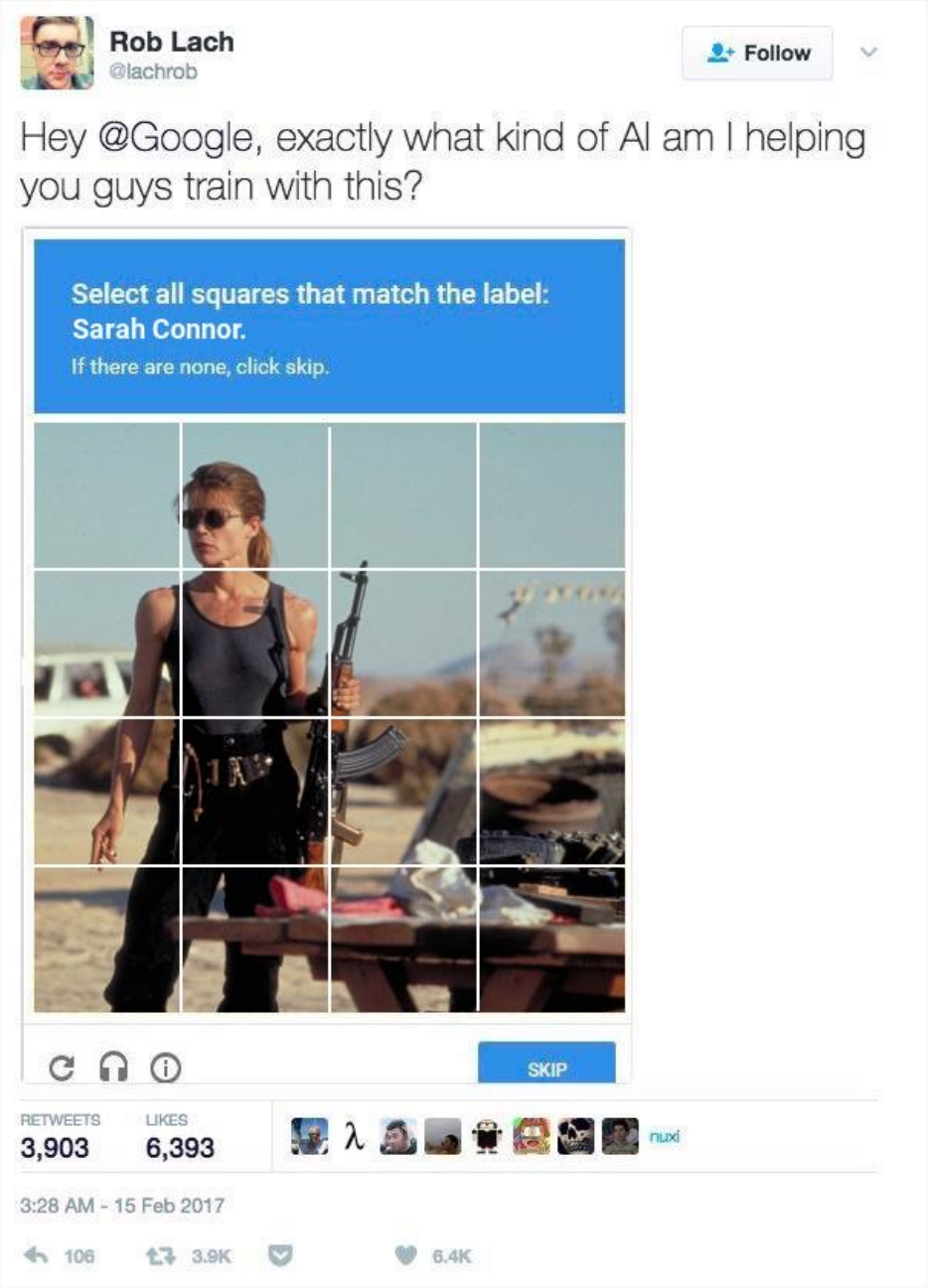




NNs need training! (not always: supervised vs. unsupervised learning)



NNs need training! (not always: supervised vs. unsupervised learning)



You may be *uploaded to* a neural network one day…

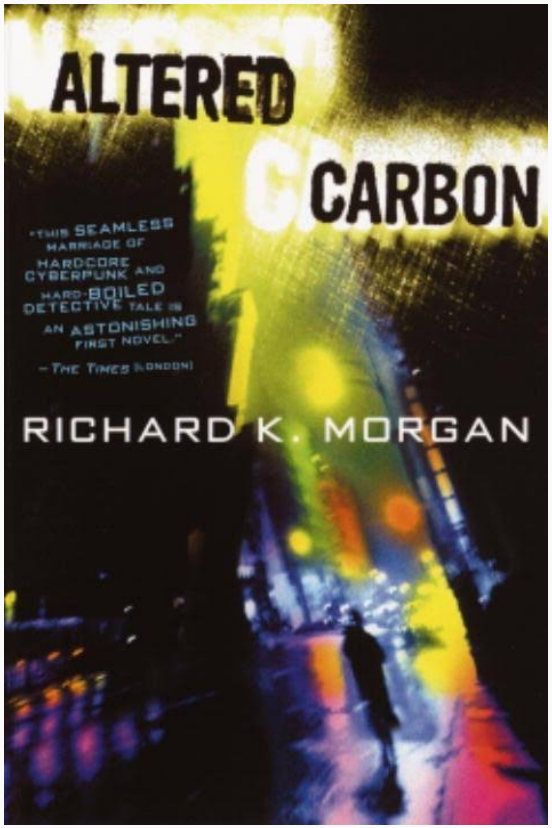
THE CHRONICLE OF HIGHER EDUCATION NEWS OPINION DAT

SECTIONS FEATURED: Do Better on Diversity The New Cheating Economy Where Your Freshmen Come From The Trenc

THE CHRONICLE REVIEW

The Strange Neuroscience of Immortality

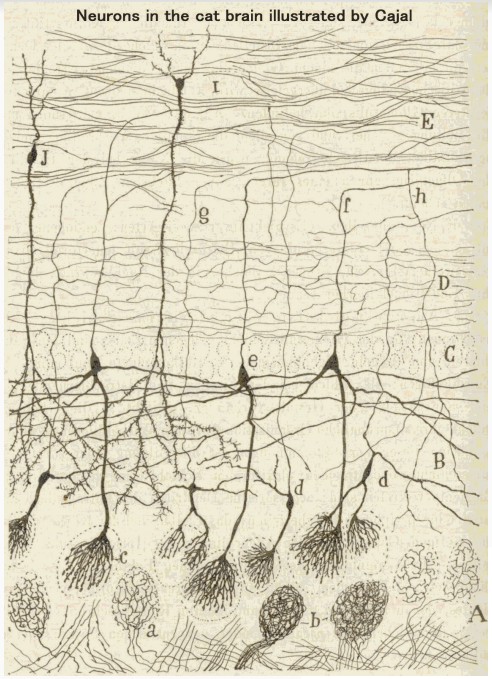
Harry Campbell for The Chronicle Review

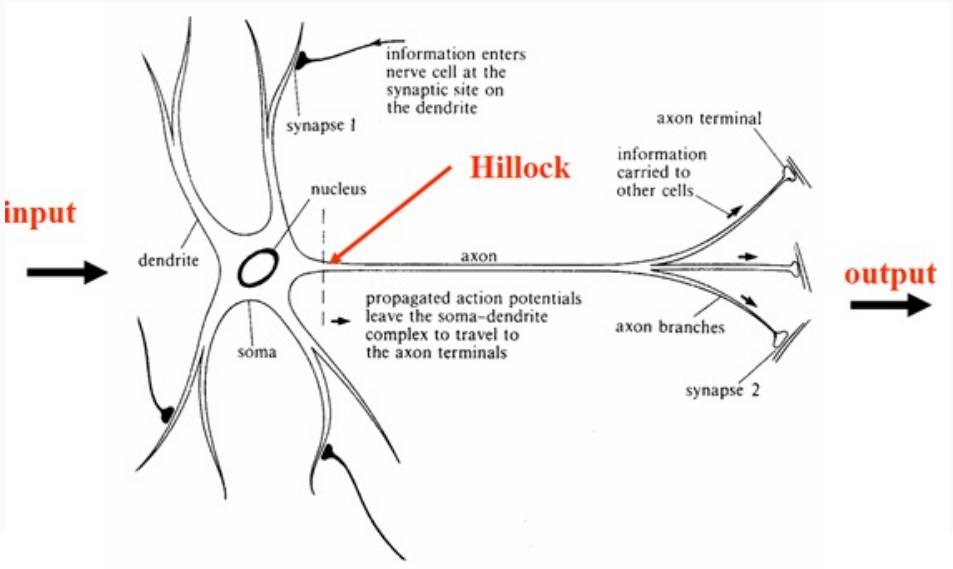


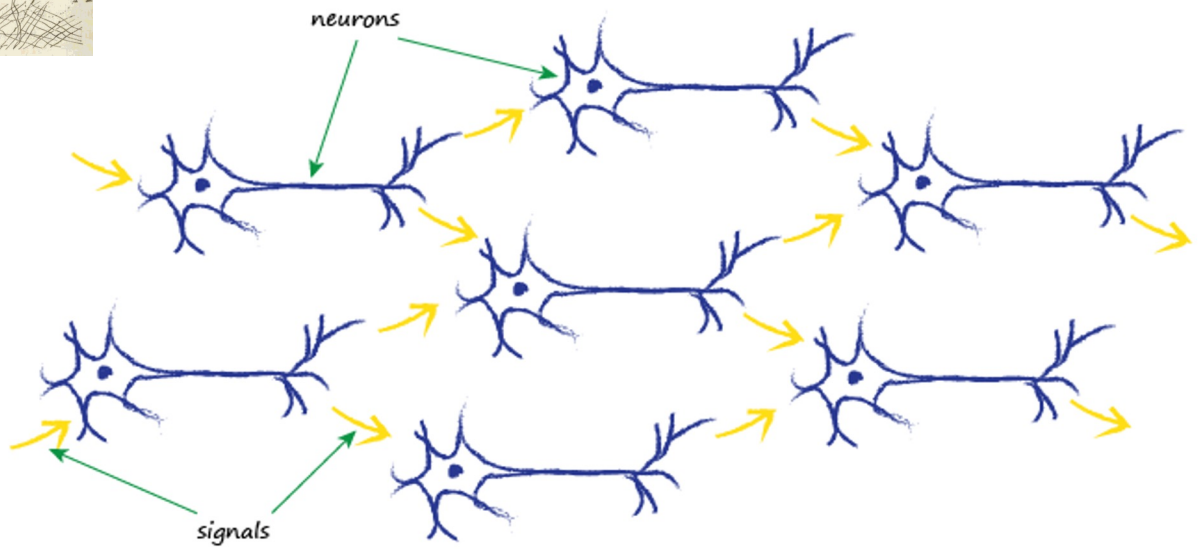
(pretty much covers all permutations)

(try uploading this image to Google Image Search)…. https://research.googleblog.com/

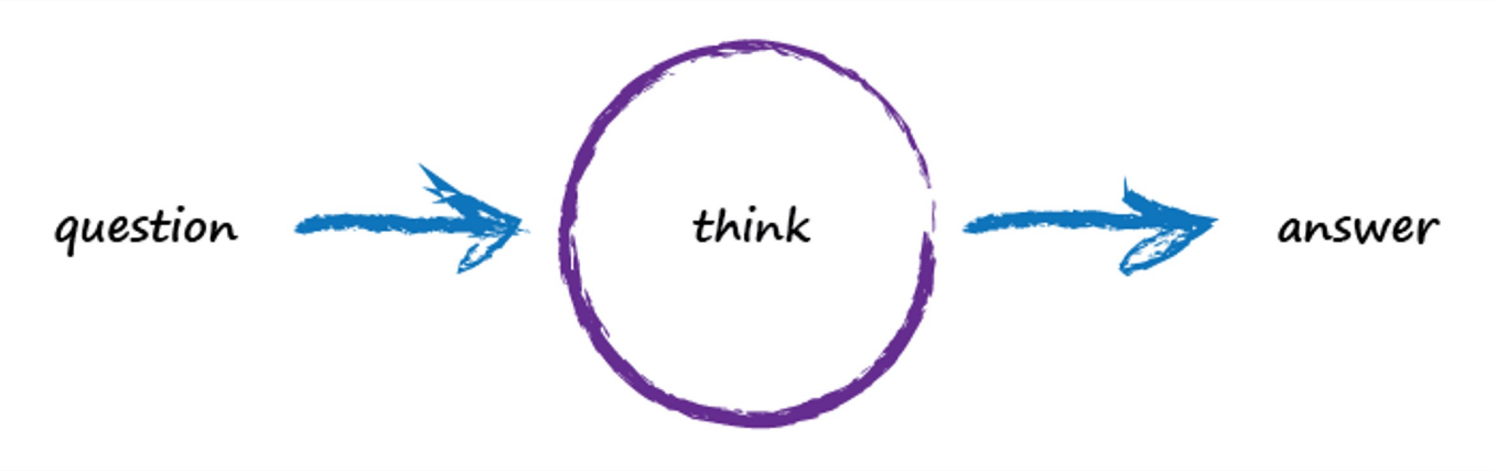
Brains in Nature

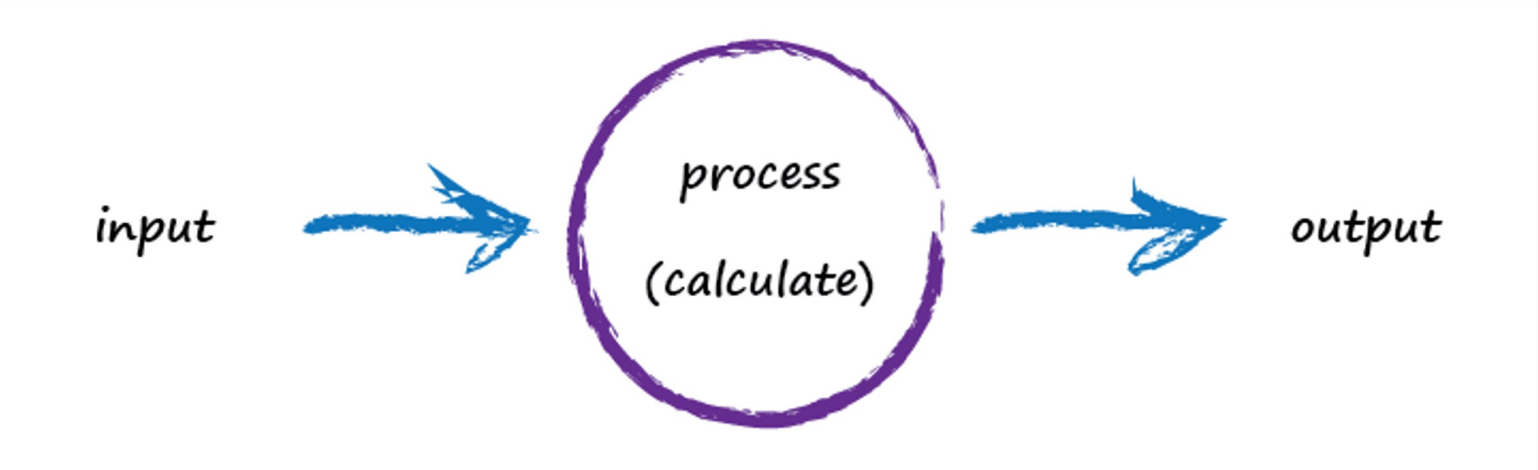




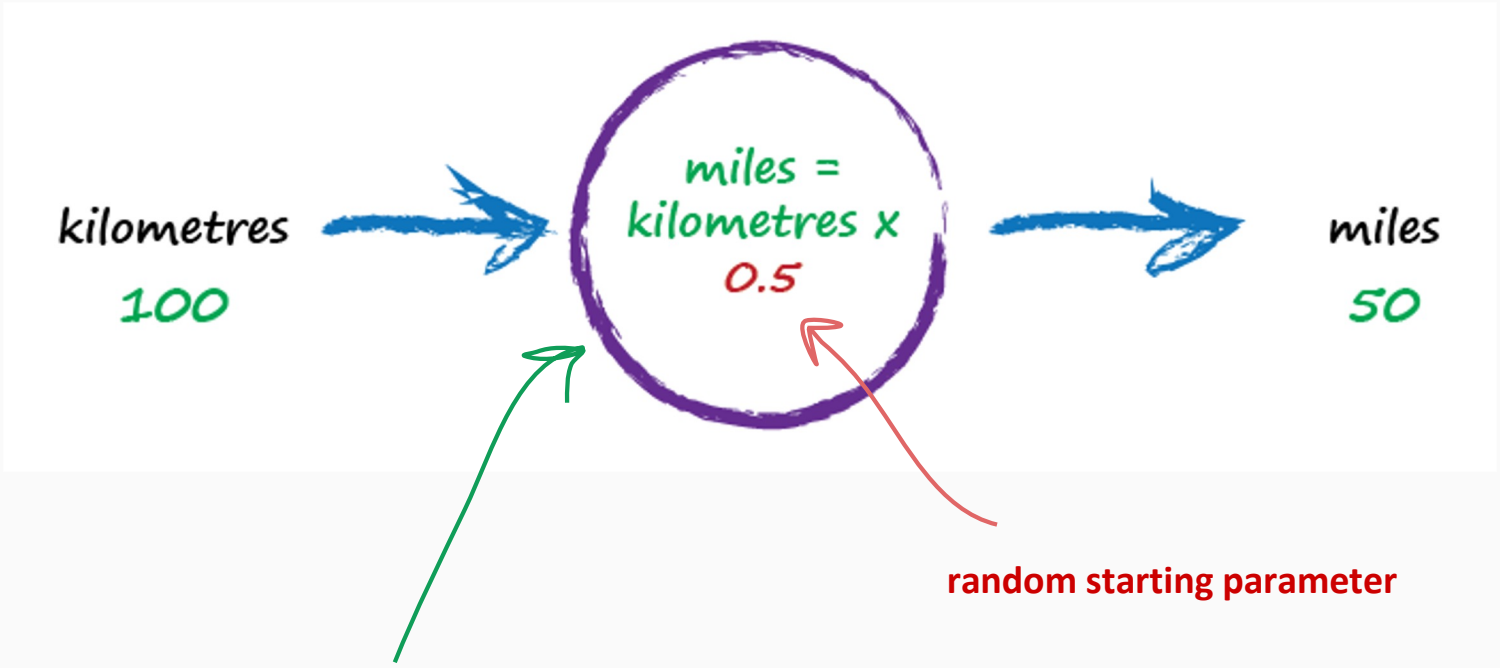


Simple Predicting Machine

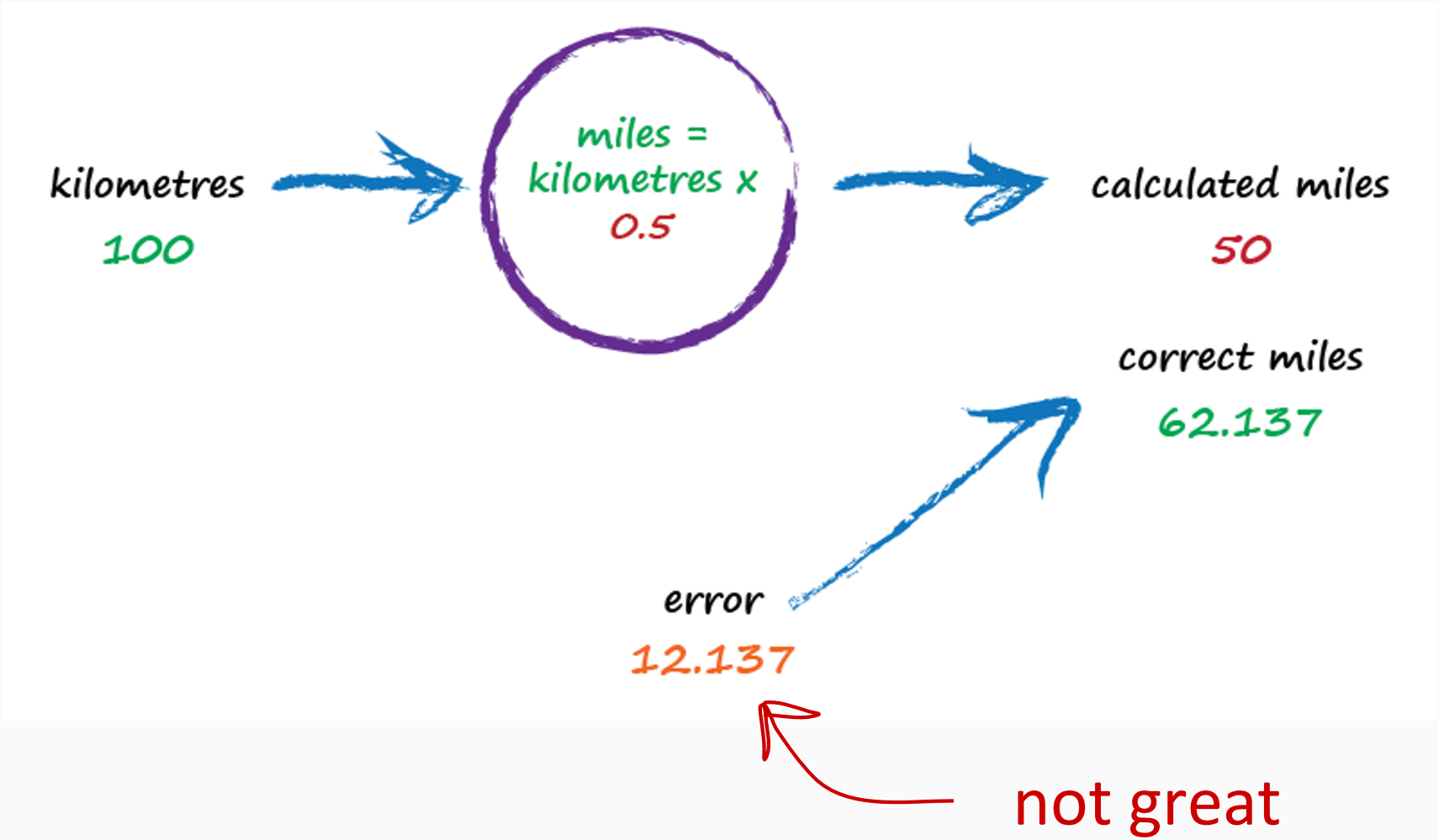




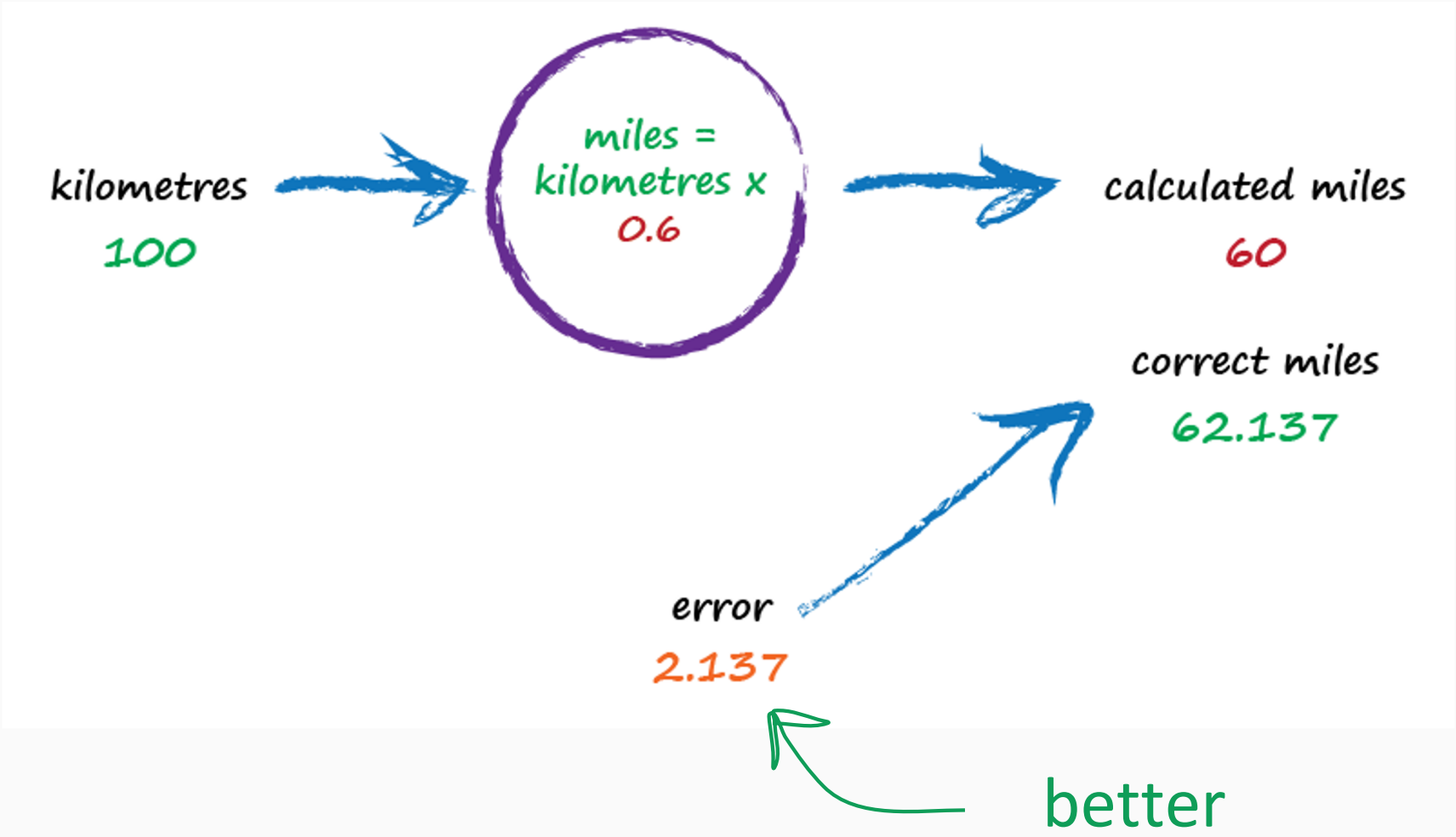
Kilometres to Miles



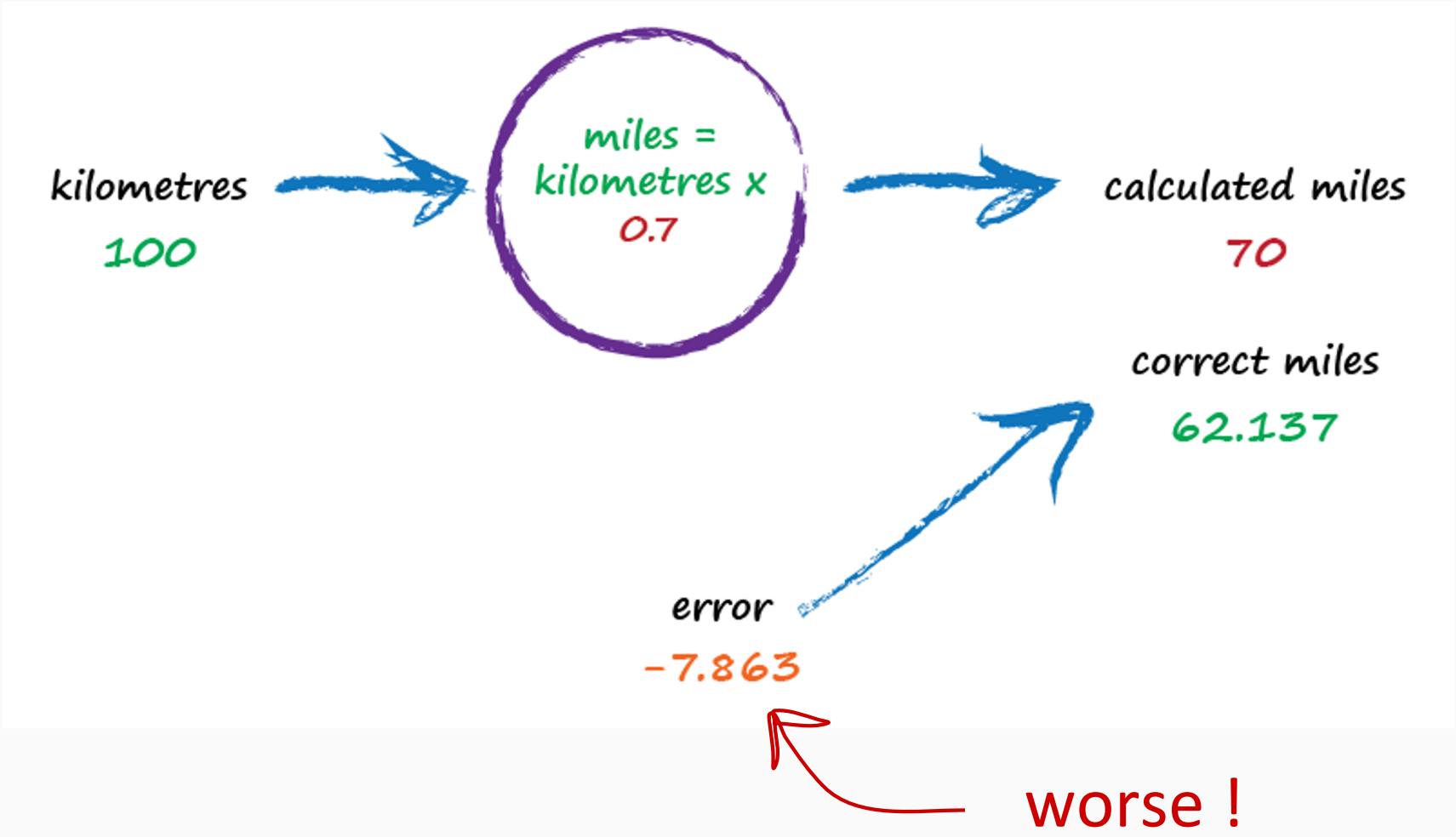
Kilometres to Miles



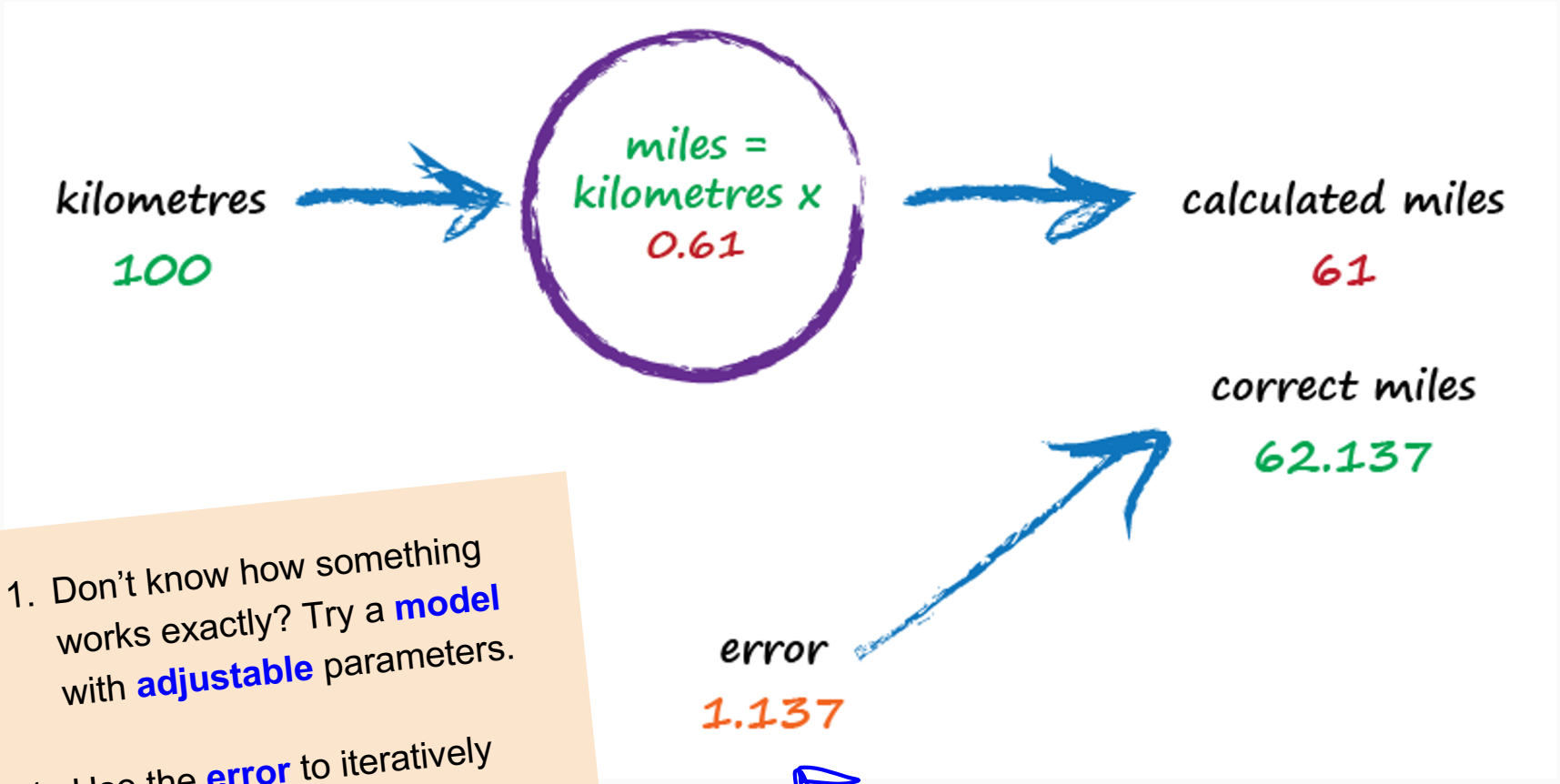
Kilometres to Miles



Kilometres to Miles



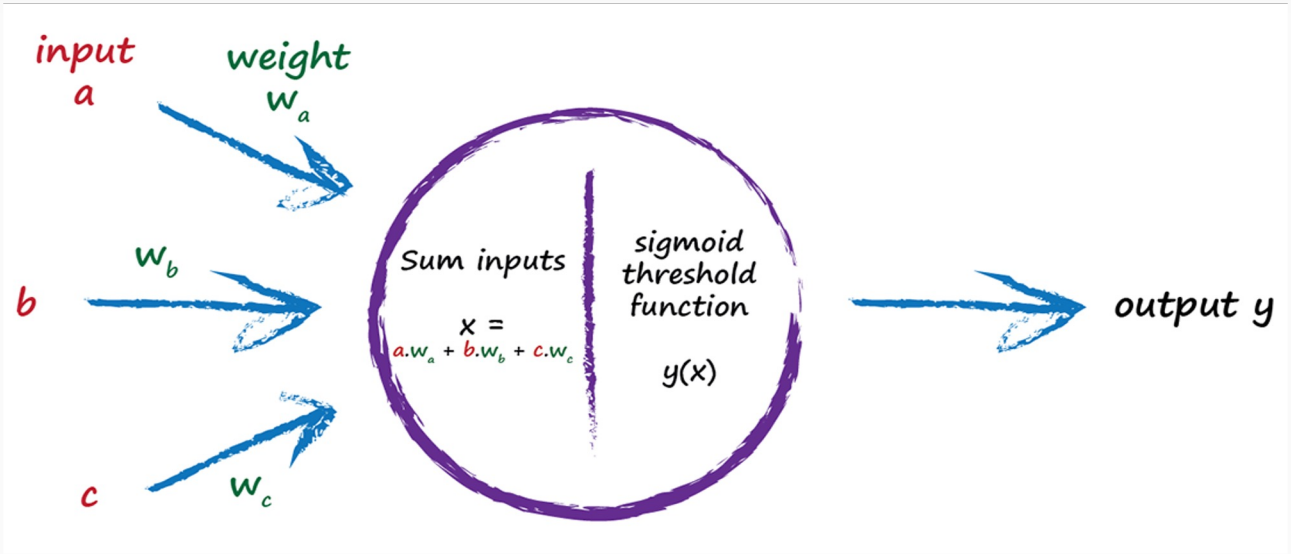
Kilometres to Miles

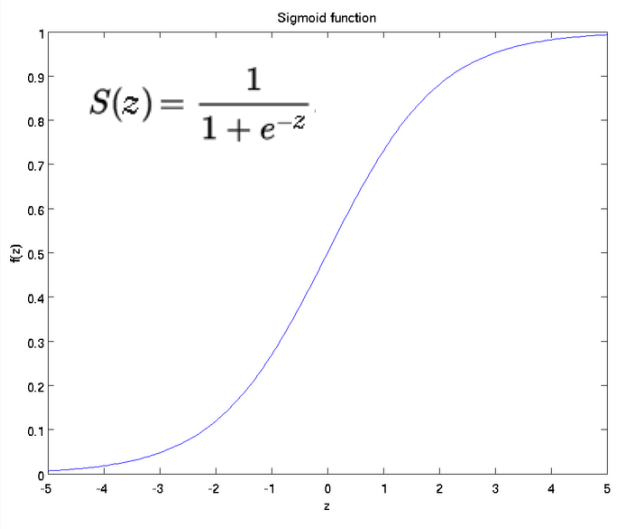


1. Don't know how something works exactly? Try a **model** with **adjustable** parameters.

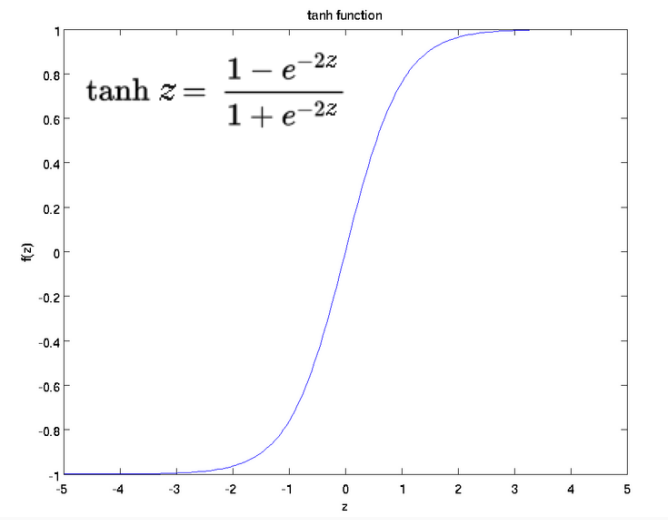
1. Use the **error** to iteratively refine the parameters.

Artificial Neuron: transfer or activation functions





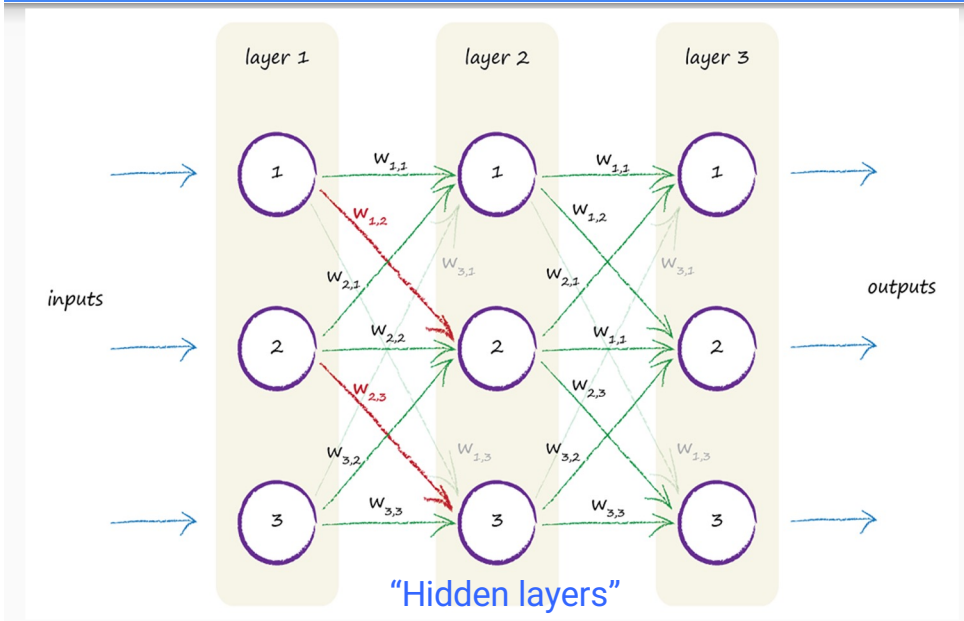
$$f'(z) = f(z)(1 - f(z)).$$



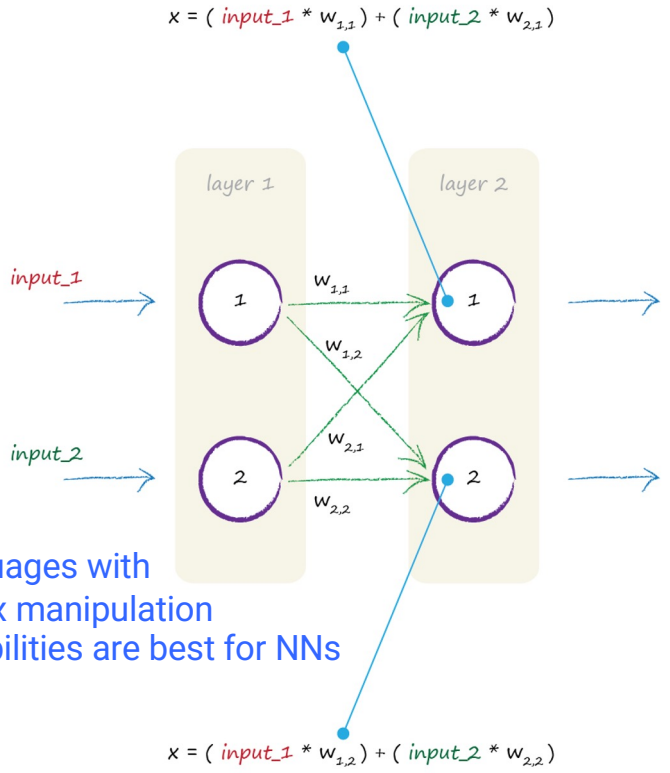
$$f'(z) = 1 - (f(z))^2$$

Non-linear and bounded (to mimic biological neurons) and differentiable (for backpropagation)

Feedforward networks (as opposed to recurrent NNs)

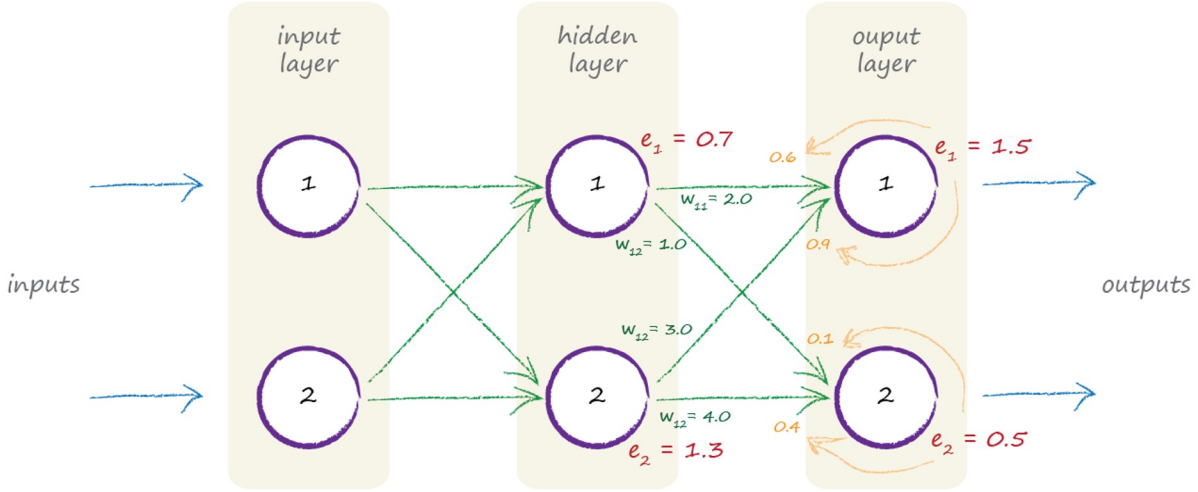


"Hidden layers"
(NNs have varied architecture, depending on application)



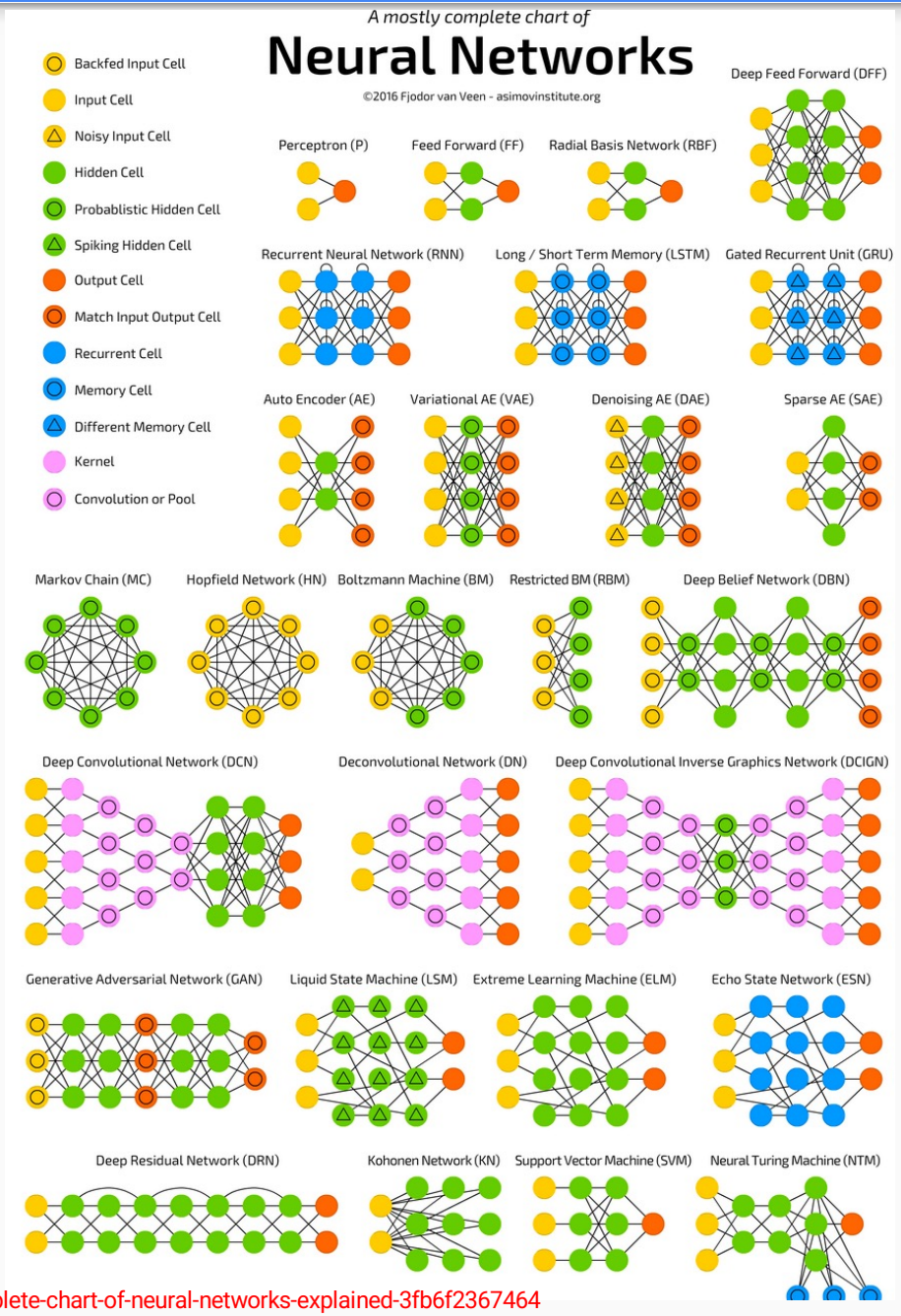
Languages with matrix manipulation capabilities are best for NNs

"Backpropagation"
(the secret sauce: **brains are plastic**)



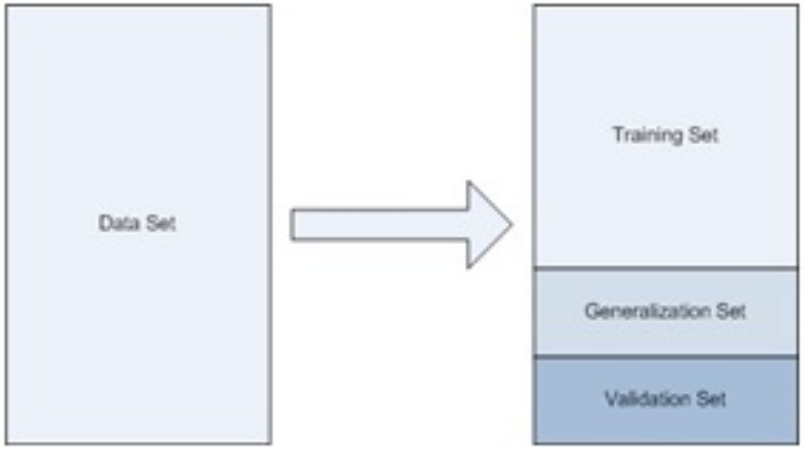
1. Remember we use the **error** to guide how we refine a model's parameter – the weights.
2. The error at the output nodes is easy - the difference between the **desired** and **actual (training)** outputs.
3. The error at internal nodes isn't obvious. A **heuristic** approach is to split it in **proportion** to the link weights.
4. … and back propagating the error can be expressed as a **matrix** multiplication too!

Many possible NN architectures.
(the human brain seems dramatically under-exploited, in this respect)



https://towardsdatascience.com/the-mostly-complete-chart-of-neural-networks-explained-3fb6f2367464

Practically speaking, when do you stop training? (no fixed rule, when good enough)



Stopping Conditions

There are several measures used to decide when to stop training. I'm going to list the various measures, what they mean and how to calculate them.

- **Maximum Epochs Reached** – The first measure is really easy, all it means is that the NN will stop once a set number of epochs have elapsed.
- **Training Set Accuracy** – This is the number of correctly classified patterns over the total number of patterns in the training set for an epoch. Obviously the higher the accuracy the better. But remember that this is the accuracy on previously seen patterns so you can use this alone to stop your training.
- **Generalization Set Accuracy** – This is the number of correctly classified patterns over the total number of patterns in the generalization set for an epoch. This gets calculated at the end of an epoch once all the weight changes have been completed. This represents the accuracy of the network in dealing with unseen patterns. Again you can't use this measure alone since this could have a much higher accuracy that the training set error.
- **Training Set Mean Squared Error (MSE)** – this is the average of the sum of the squared errors (desired – actual) for each pattern in the training set. This gives a more detailed measure of the current networks accuracy, the smaller the MSE the better the network performs.
- **Generalization Set Mean Squared Error (MSE)** – this is the average of the sum of the squared errors (desired – actual) for each pattern in the generalization set.

$$MSE = \frac{\sum_{i=0}^{n}(desired\ value\ i - actual\ value\ i)^2}{n}$$

$$where\ n = number\ of\ patterns\ in\ set$$

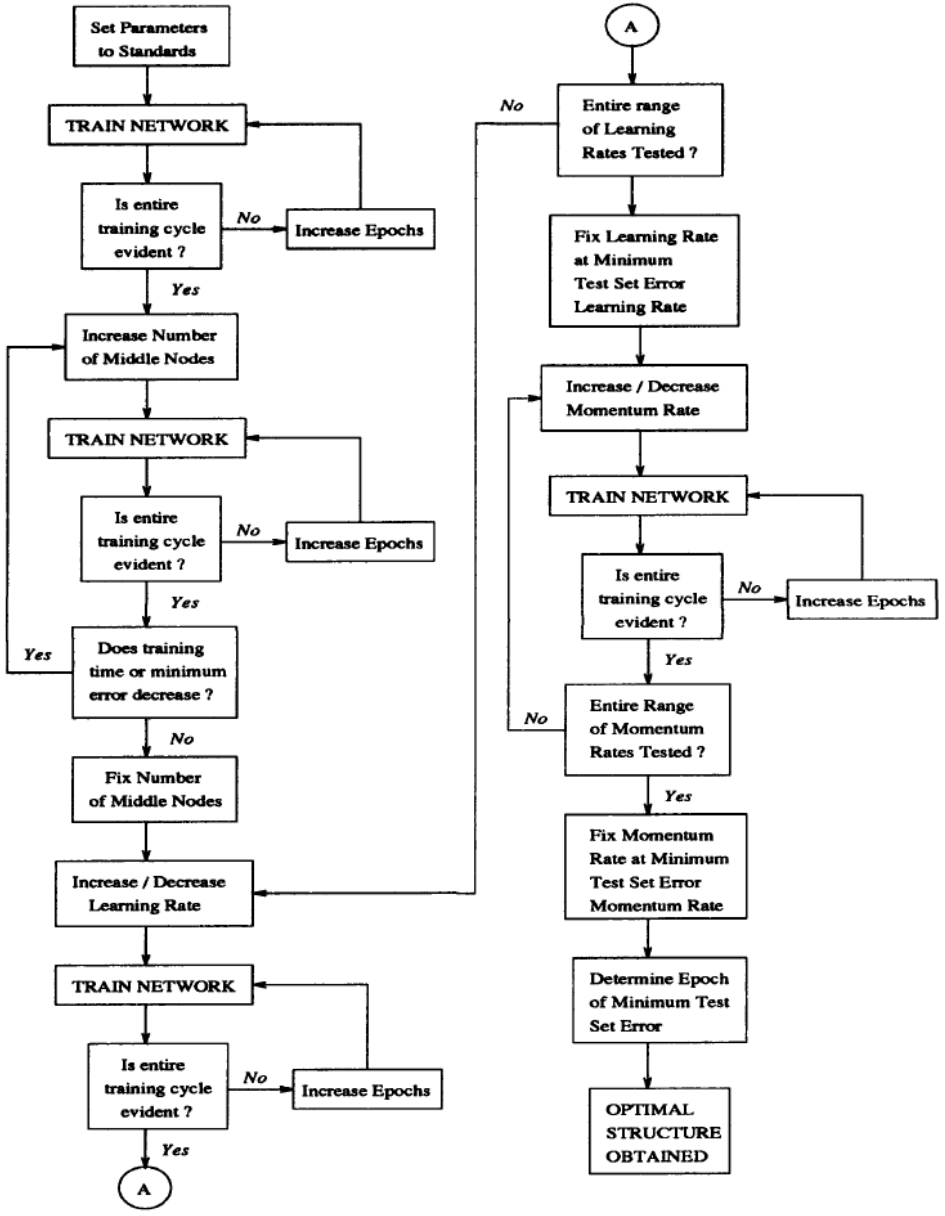
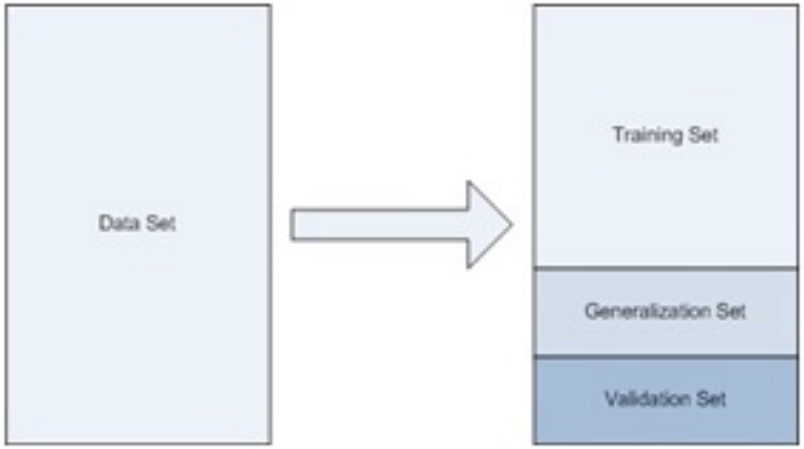
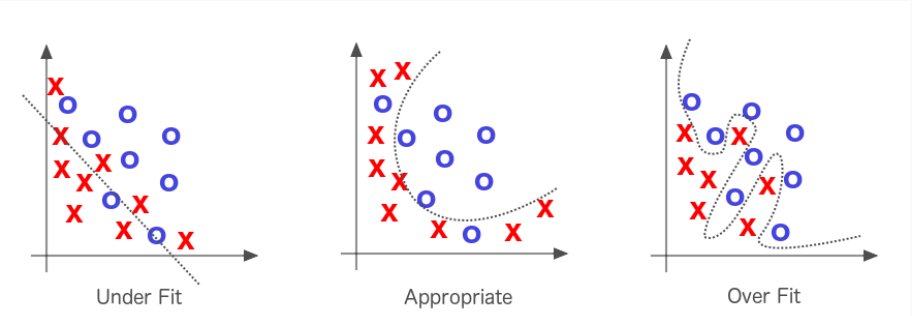
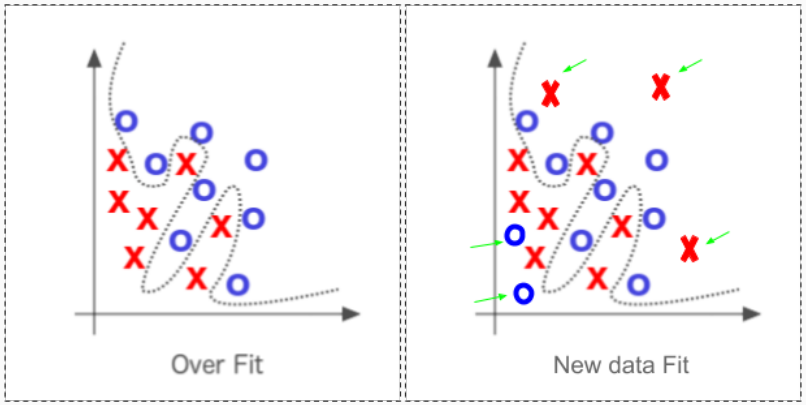


Figure 2. Building An Optimal Network

Practically speaking, when do you stop training? (no fixed rule, when good enough)







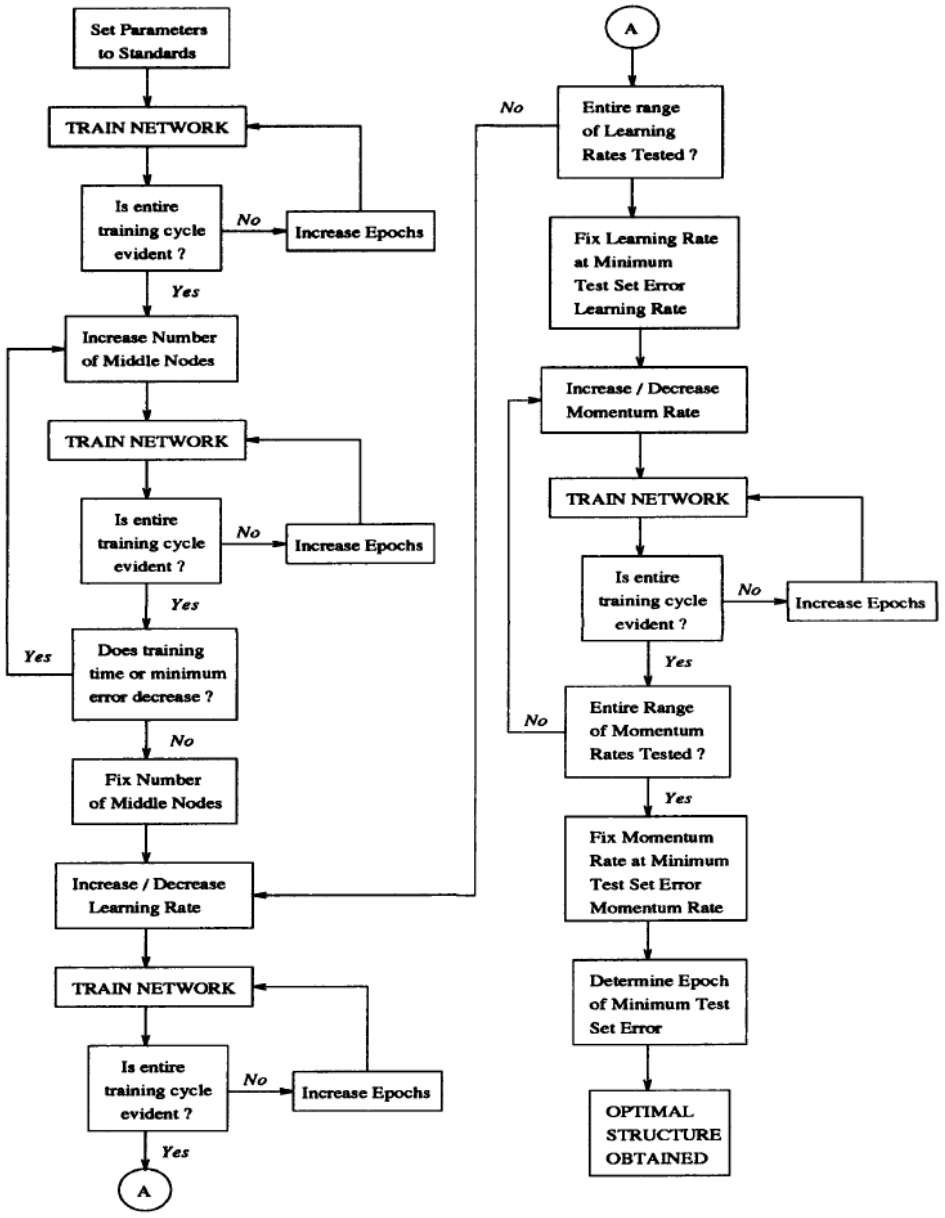


Figure 2. Building An Optimal Network

FORTRAN 90: We'll use matrix algebra capabilities, MATMUL, TRANSPOSE, SUM…

Fortran 90: Array Operations

Let's start with a simple example. Suppose we wish to add two arrays `A` and `B` and put the result in `C`. In Fortran 77 , we would have something like

```
do i=1,n
do j=1,n
C(j,i) = A(j,i) + B(j,i)
enddo
enddo
```

```
      GAME OF LIFE:

C *** REFRESH THE BOARD
      DO 37 I=1,N
      DO 34 J=1,N
      STATUS(I,J)=NEXT(I,J)
   34 CONTINUE
   37 CONTINUE

      INSTEAD SIMPLY WRITE STATUS=NEXT
```

In Fortran 90, it is as simple as `C = A + B` .

Most of the intrinsic functions operate component-wise on arrays. `C = sin(A)` is equivalent to (in this case, A is a one dimensional array)

```
do i=1,n
C(i) = sin(A(i))
enddo
```

Element-wise product (not dot product)

Note: `C = A*B` multplies corresponding elements in A and B. It does NOT do matrix multiplication. There are some intrinic functions for matrix multiplication (`matmul`) and dot products (`dot_product`). The matrix multiply would look like `C = matmul(A,B)` . There are a number of other intrinic subroutines and functions for finding the size and rank of an array, reshaping an array, converting an array to vector and back, tranposes, and many more.

Remember, must be consistent: (m,k) x (k,n) = (m,n)
(the compiler won't let you make this boo-boo)

Array sections

Along with the ability to operate on whole arrays comes the ability to operate on just part of an array. The following examples show operation on the rows, columns, or a sub-matrix of a matrix.

```
real*8 :: A(3,3)
real*8 :: B(2,2)
real*8 :: v(3)

 v(:) = A(:,1)   ! set v equal to the first column of A
 v(:) = A(1,:)   ! set v equal to the first row of A
 B(:,:) = A(1:2,1:2) ! B is the upper 2x2 part of A
```

In general, a section of an array is specified by `v`(*start:end:stride*) A bare colon (:) specifies the entire dimension, as shown in the examples above.

Obtaining the diagonal of a matrix requires converting the matrix to an array, and then using a stride that is one greater than the dimension.

https://courses.physics.illinois.edu/phys466/sp2013/comp_info/array.html

Our wee single-layer NN (a.k.a. “perceptron”):

Our training set. Can you find the rule joining input & output?
(we are going to teach a CPU to do what your brain just did, only so much faster)

Inputs			Output
0	0	1	0
1	1	1	1
1	0	1	1
0	1	1	0

after training, we will
test our NN with the 4 remaining cases

Our perceptron (exactly this architecture):

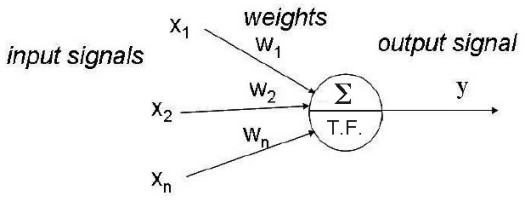
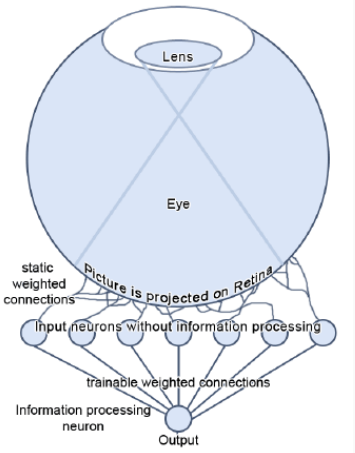


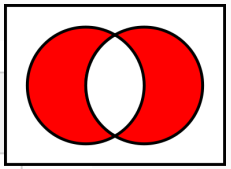
Fig. 1: An artificial neuron

NN revolution in mid-80’s
(power of multi-layer NNs)



Your brain did this 1st,
followed by character recognition
(another NN -TBD-)

XOR



Inputs			Output
0	0	1	0
0	1	1	1
1	0	1	1
1	1	1	0

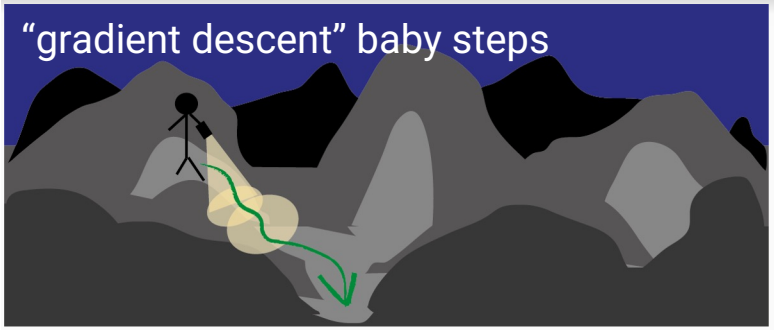
Input		Output
A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

Slightly more complicated.
Would require two-layer NN
(already enough for image recognition!)

Our wee single-layer NN (a.k.a. “perceptron”):

Bit of perspective:

Yes, But How Do We Actually Update The Weights?



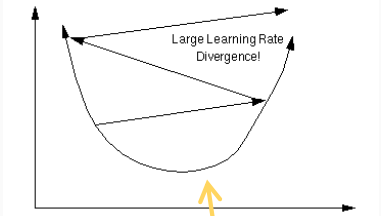
move w_{jk} in the opposite direction to the slope

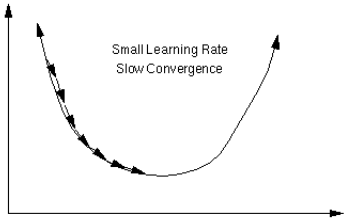
Error (difference between NN output and training output, squared)

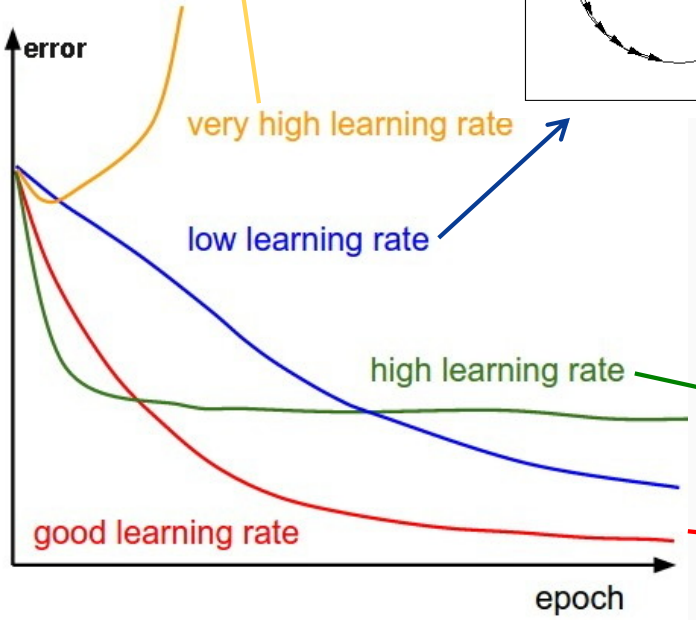
$$\text{new } w_{jk} = \text{old } w_{jk} - \alpha \cdot \frac{\partial E}{\partial w_{jk}}$$

"learning rate" (size of your steps)

Learning rate Goldilocks:







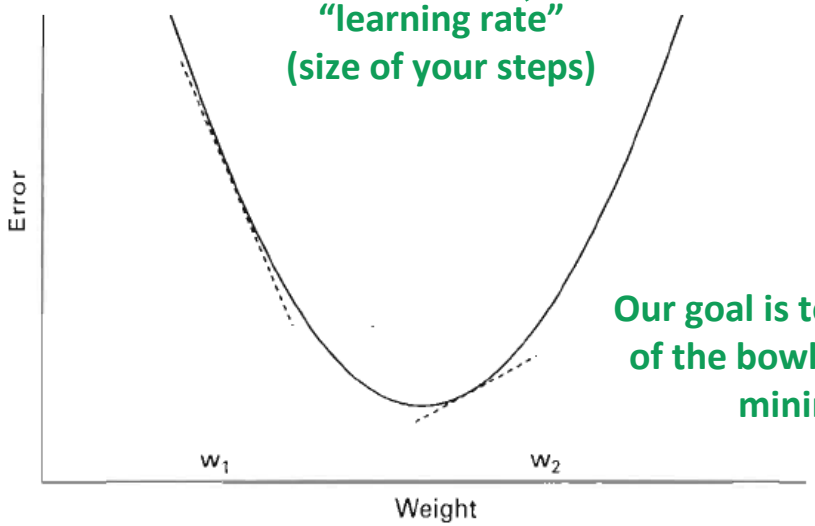
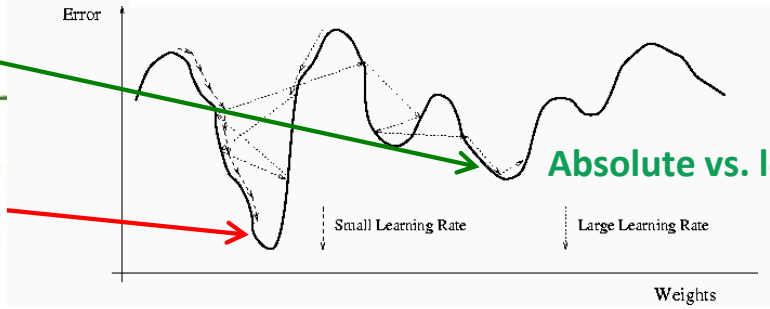


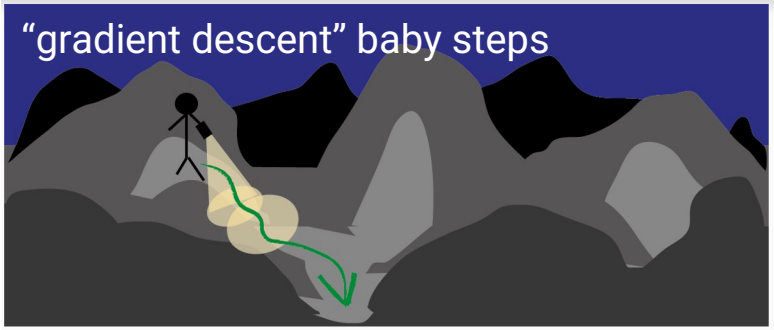
Figure 2.7
A hypothetical function relating the value of a single network weight to error. Slopes of the error function at two weight values (w_1 and w_2) are shown with dashed lines.

Our goal is to get to the bottom of the bowl (find weights that minimize errors)



Absolute vs. local minima…

Yes, But How Do We Actually Update The Weights?



move w_{jk} in the opposite direction to the slope

Error (difference between NN output and training output, squared)

$$\text{new } w_{jk} = \text{old } w_{jk} - \alpha \cdot \frac{\partial E}{\partial w_{jk}}$$

"learning rate" (size of your steps)

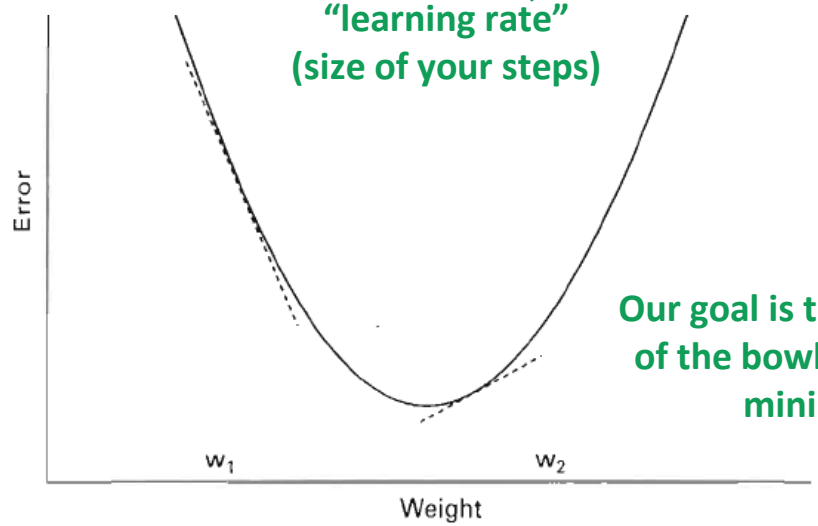


Figure 2.7
A hypothetical function relating the value of a single network weight to error. Slopes of the error function at two weight values (w_1 and w_2) are shown with dashed lines.

Our goal is to get to the bottom of the bowl (find weights that minimize errors)

The curse of overfitting:

- Training should stop as soon an increase in generalization error is observed (***Early Stopping***).

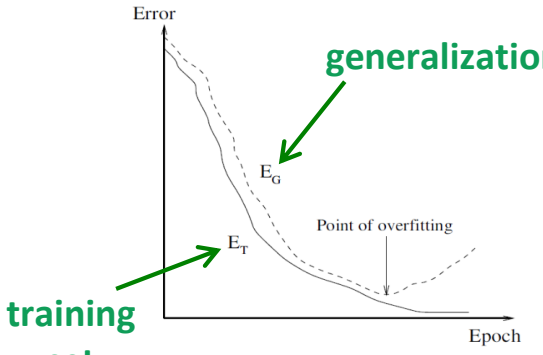
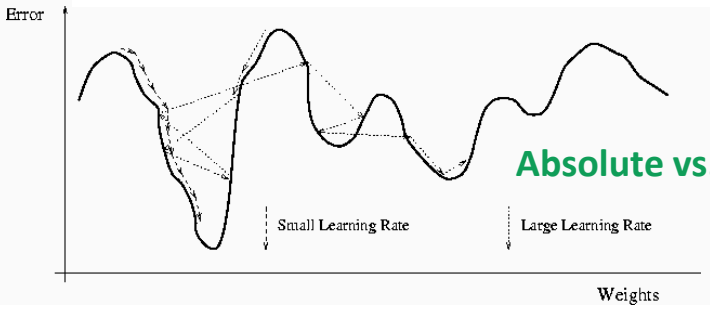


Figure 7.1 Illustration of Overfitting



Absolute vs. local minima…

More formally (the “delta rule”)

$$o_k = \frac{1}{1 + e^{-\sum_{j=1}^{3}(w_{j,k}\cdot\frac{1}{1+e^{-\sum_{i=1}^{3}(w_{i,j}\cdot x_i)}})}}$$

Derivation [edit]

Since backpropagation uses the gradient descent method, one needs to calculate the derivative of the squared error function with respect to the weights of the network. Assuming one output neuron,[note 2] the squared error function is:

$$E = \tfrac{1}{2}(t - y)^2,$$

Wikipedia on “Backpropagation”
is as clear on the subject as it gets

where

E is the squared error,
t is the target output for a training sample, and
y is the actual output of the output neuron.

The factor of $\frac{1}{2}$ is included to cancel the exponent when differentiating. Later, the expression will be multiplied with an arbitrary learning rate, so that it doesn't matter if a constant coefficient is introduced now.

For each neuron j, its output o_j is defined as

$$o_j = \varphi(\text{net}_j) = \varphi\left(\sum_{k=1}^{n} w_{kj} o_k\right).$$

The input net_j to a neuron is the weighted sum of outputs o_k of previous neurons. If the neuron is in the first layer after the input layer, the o_k of the input layer are simply the inputs x_k to the network. The number of input units to the neuron is n. The variable w_{ij} denotes the weight between neurons i and j.

The activation function φ is in general non-linear and differentiable. A commonly used activation function is the logistic function:

logistic = sigmoid

$$\varphi(z) = \frac{1}{1 + e^{-z}}$$

which has a nice derivative of:

$$\frac{d\varphi}{dz}(z) = \varphi(z)(1 - \varphi(z))$$

Finding the derivative of the error [edit]

Calculating the partial derivative of the error with respect to a weight w_{ij} is done using the chain rule twice:

$$\frac{\partial E}{\partial w_{ij}} = \frac{\partial E}{\partial o_j}\frac{\partial o_j}{\partial \text{net}_j}\frac{\partial \text{net}_j}{\partial w_{ij}}$$

In the last term of the right-hand side of the following, only one term in the sum net_j depends on w_{ij}, so that

$$\frac{\partial \text{net}_j}{\partial w_{ij}} = \frac{\partial}{\partial w_{ij}}\left(\sum_{k=1}^{n} w_{kj} o_k\right) = o_i. \quad (1)$$

If the neuron is in the first layer after the input layer, o_i is just x_i. (x_i are the inputs to NN)

The derivative of the output of neuron j with respect to its input is simply the partial derivative of the activation function (assuming here that the logistic function is used):

$$\frac{\partial o_j}{\partial \text{net}_j} = \frac{\partial}{\partial \text{net}_j}\varphi(\text{net}_j) = \varphi(\text{net}_j)(1 - \varphi(\text{net}_j)) \quad (2)$$

This is the reason why backpropagation requires the activation function to be differentiable.

The first term is straightforward to evaluate if the neuron is in the output layer, because then $o_j = y$ and

$$\frac{\partial E}{\partial o_j} = \frac{\partial E}{\partial y} = \frac{\partial}{\partial y}\frac{1}{2}(t - y)^2 = y - t \quad (3)$$

However, if j is in an arbitrary inner layer of the network, finding the derivative E with respect to o_j is less obvious.

Considering E as a function of the inputs of all neurons $L = u, v, \ldots, w$ receiving input from neuron j,

$$\frac{\partial E(o_j)}{\partial o_j} = \frac{\partial E(\text{net}_u, \text{net}_v, \ldots, \text{net}_w)}{\partial o_j}$$

(fortunately not our concern)

and taking the total derivative with respect to o_j, a recursive expression for the derivative is obtained:

$$\frac{\partial E}{\partial o_j} = \sum_{l \in L}\left(\frac{\partial E}{\partial \text{net}_l}\frac{\partial \text{net}_l}{\partial o_j}\right) = \sum_{l \in L}\left(\frac{\partial E}{\partial o_l}\frac{\partial o_l}{\partial \text{net}_l} w_{jl}\right)$$

Therefore, the derivative with respect to o_j can be calculated if all the derivatives with respect to the outputs o_l of the next layer – the one closer to the output neuron – are known.

Putting it all together:

$$\frac{\partial E}{\partial w_{ij}} = \delta_j o_i$$

product of terms (3) and (2) is “delta”

with

$$\delta_j = \frac{\partial E}{\partial o_j}\frac{\partial o_j}{\partial \text{net}_j} = \begin{cases} (o_j - t_j) o_j (1 - o_j) & \text{if } j \text{ is an output neuron,} \\ (\sum_{l \in L} \delta_l w_{jl}) o_j (1 - o_j) & \text{if } j \text{ is an inner neuron.} \end{cases}$$

To update the weight w_{ij} using gradient descent, one must choose a learning rate, α. The change in weight, which is added to the old weight, is equal to the product of the learning rate and the gradient, multiplied by -1:

$$\Delta w_{ij} = -\alpha\frac{\partial E}{\partial w_{ij}} = \begin{cases} -\alpha o_i (o_j - t_j) o_j (1 - o_j) & \text{if } j \text{ is an output neuron,} \\ -\alpha o_i (\sum_{l \in L} \delta_l w_{jl}) o_j (1 - o_j) & \text{if } j \text{ is an inner neuron.} \end{cases}$$

Our case
(single layer NN)

The -1 is required in order to update in the direction of a minimum, not a maximum, of the error function.

For a single-layer network, this expression becomes the Delta Rule. To better understand how backpropagation works, here is an example to illustrate it: The Back Propagation Algorithm, page 20.

$$\text{new } w_{jk} = \text{old } w_{jk} - \alpha \cdot \frac{\partial E}{\partial w_{jk}}$$

$$\Delta w_{ij} = -\alpha\frac{\partial E}{\partial w_{ij}} = -\alpha o_i (o_j - t_j) o_j (1 - o_j)$$

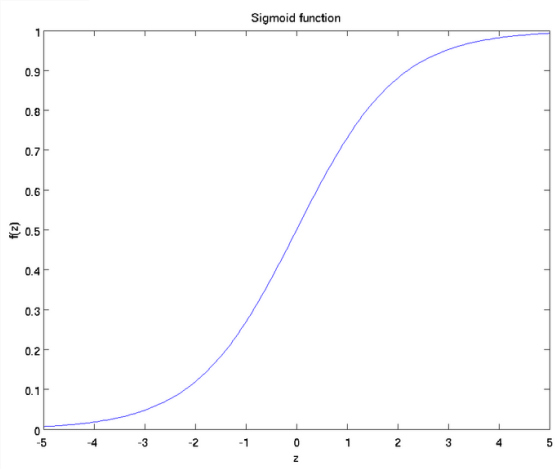
(1) (3) (2)

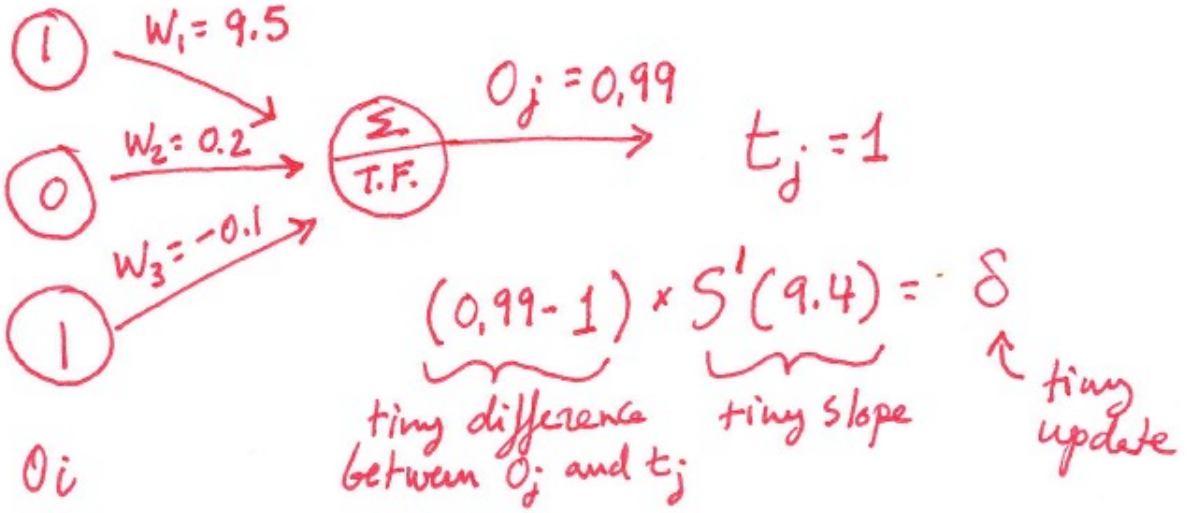
For our single-layer NN:
α is learning rate (=1 for starters)
o_i is (training) inputs to neuron
o_j is its output (result of weighted inputs and transfer function)
t_j is training output
Notice $o_j(1-o_j)$ is derivative of sigmoid, needs update to $1-o_j^2$ if using TANH as the transfer function.
I will stick to same notation in example code.
If you get lost, refresher (in Python, different notation) at
http://iamtrask.github.io/2015/07/12/basic-python-network/

Before we look at code, can we understand what is happening intuitively?

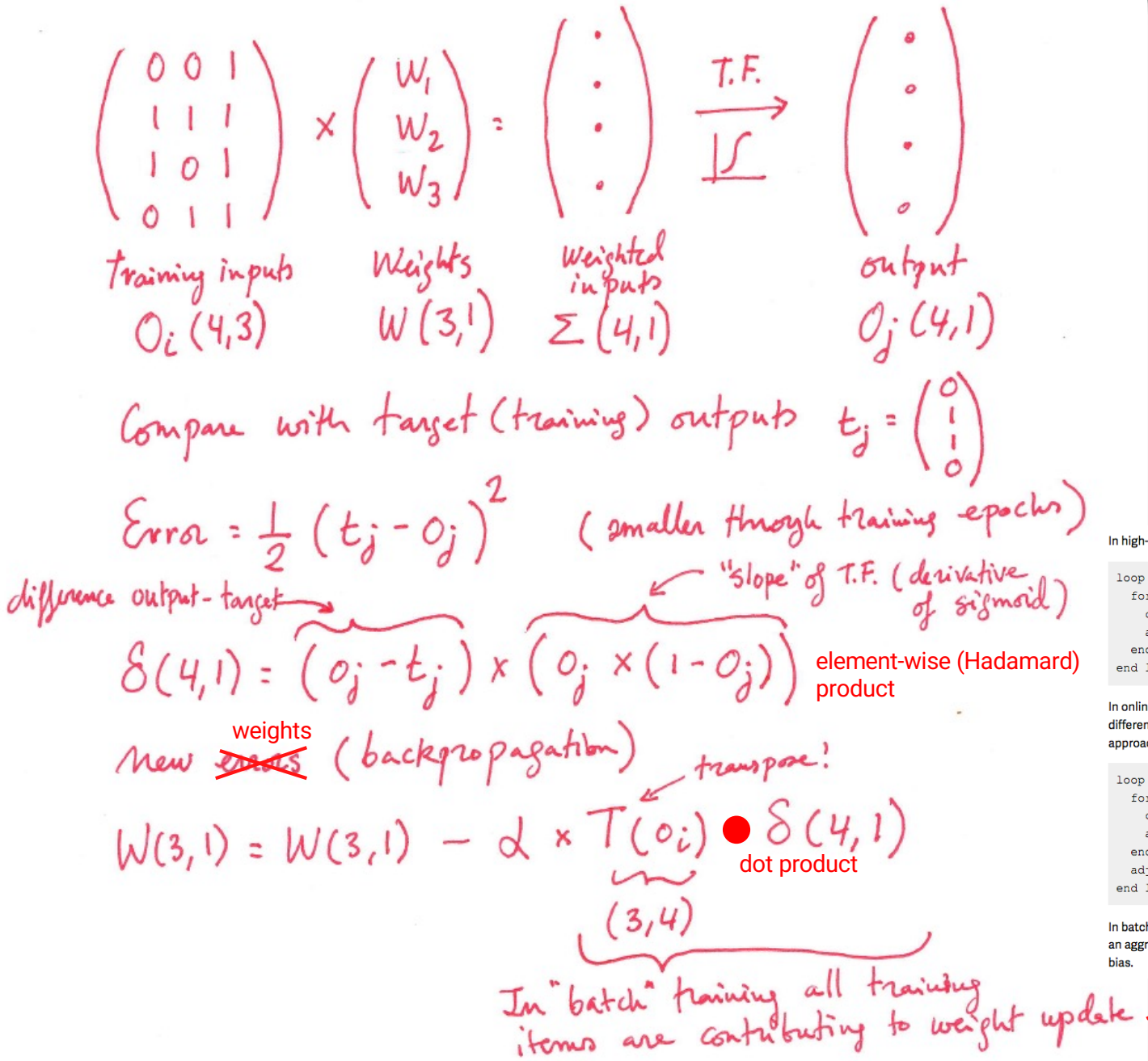
Intuitive meaning of δ:

When we multiply the slope of T.F. by the difference between output and target output, we are reducing the error of high confidence predictions. If the slope is shallow (close to $\emptyset$) then the output was either very high or very low. This means the network was already very confident in the value of the weights. For other "wishy-washy" intermediate values we update the weights most heavily.





A bit of help with your coding:



Inputs			Output
0	0	1	0
1	1	1	1
1	0	1	1
0	1	1	0

In high-level pseudo-code the online training approach is:

```
loop maxEpochs times
  for each training data item
    compute weights and bias deltas for curr item
    adjust weights and bias values using deltas
  end for
end loop
```

In online training, weights and bias values are adjusted for every training item based on the difference between computed outputs and the training data target outputs. The batch approach is:

```
loop maxEpochs times
  for each training item
    compute weights and bias deltas for curr item
    accumulate the deltas
  end for
  adjust weights and bias deltas using accumulated deltas
end loop
```

In batch training the adjustment delta values are accumulated over all training items, to give an aggregate set of deltas, and then the aggregated deltas are applied to each weight and bias.

Take a deep breath, this is the extent of it (in FORTRAN 90):

```
DO J=1,100000                                   ! TRAINING EPOCHS

OJ=1./(1.+EXP(-MATMUL(OI,W)))                   ! FORWARD PROPAGATION USING SIGMOID T.F.

DELTA = (OJ-TJ) * (OJ*(1.-OJ))                  ! FIND DELTA

W = W - ALPHA*MATMUL(TRANSPOSE(OI),DELTA)       ! UPDATE WEIGHTS (BACKPROPAGATION)

END DO                                          ! KEEP AT IT
```

Since we last met, another bastion gone: (old news, these from past PHYS 250)

MIT Technology Review

Topics+ Top Stories

Computing

First Computer to Match Humans in Conversational Speech Recognition

Human-level speech recognition has been a long time coming.

by Emerging Technology from the arXiv October 24, 2016

One by one, the skills that separate us from machines are falling into the machines' column. First there was chess, then *Jeopardy!*, then Go, then object recognition, face recognition, and video gaming in general. You could be forgiven for thinking that humans are becoming obsolete.

arXiv.org > cs > arXiv:1610.05256

Search or Artic

Computer Science > Computation and Language

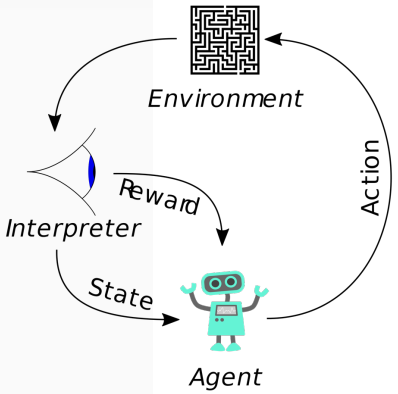
Achieving Human Parity in Conversational Speech Recognition

W. Xiong, J. Droppo, X. Huang, F. Seide, M. Seltzer, A. Stolcke, D. Yu, G. Zweig

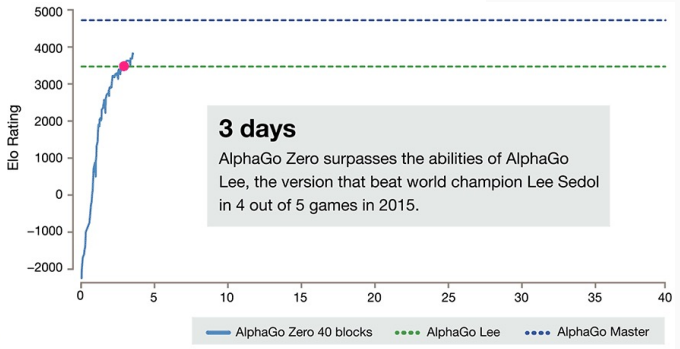
(Submitted on 17 Oct 2016)

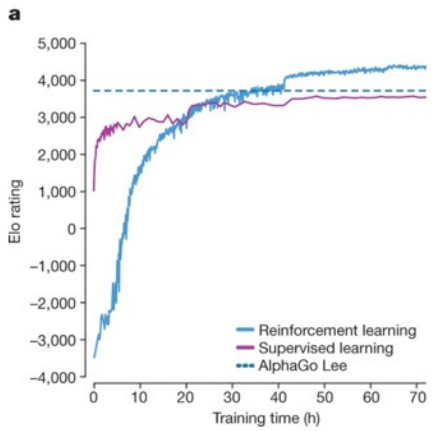
Conversational speech recognition has served as a flagship speech recognition task since the release of the DARPA Switchboard corpus in the 1990s. In this paper, we measure the human error rate on the widely used NIST 2000 test set, and find that our latest automated system has reached human parity. The error rate of professional transcriptionists is 5.9% for the Switchboard portion of the data, in which newly acquainted pairs of people discuss an assigned topic, and 11.3% for the CallHome portion where friends and family members have open-ended conversations. In both cases, our automated system establishes a new state-of-the-art, and edges past the human benchmark. This marks the first time that human parity has been reported for conversational speech. The key to our system's performance is the systematic use of convolutional and LSTM neural networks, combined with a novel spatial smoothing method and lattice-free MMI acoustic training.

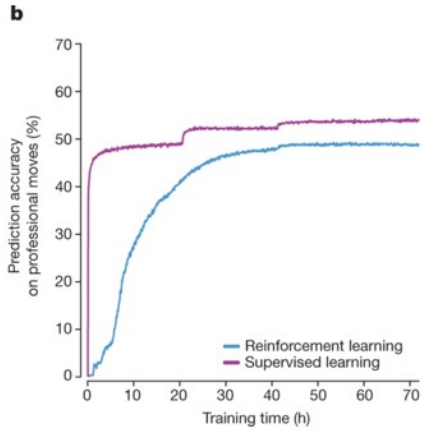
Since we last met, another bastion gone: (old news, these from PHYS 250 in 2017)

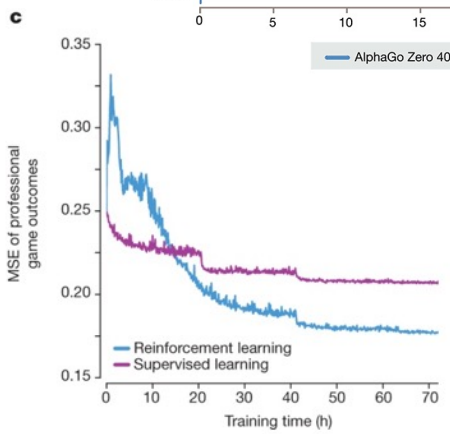


Empirical evaluation of AlphaGo Zero







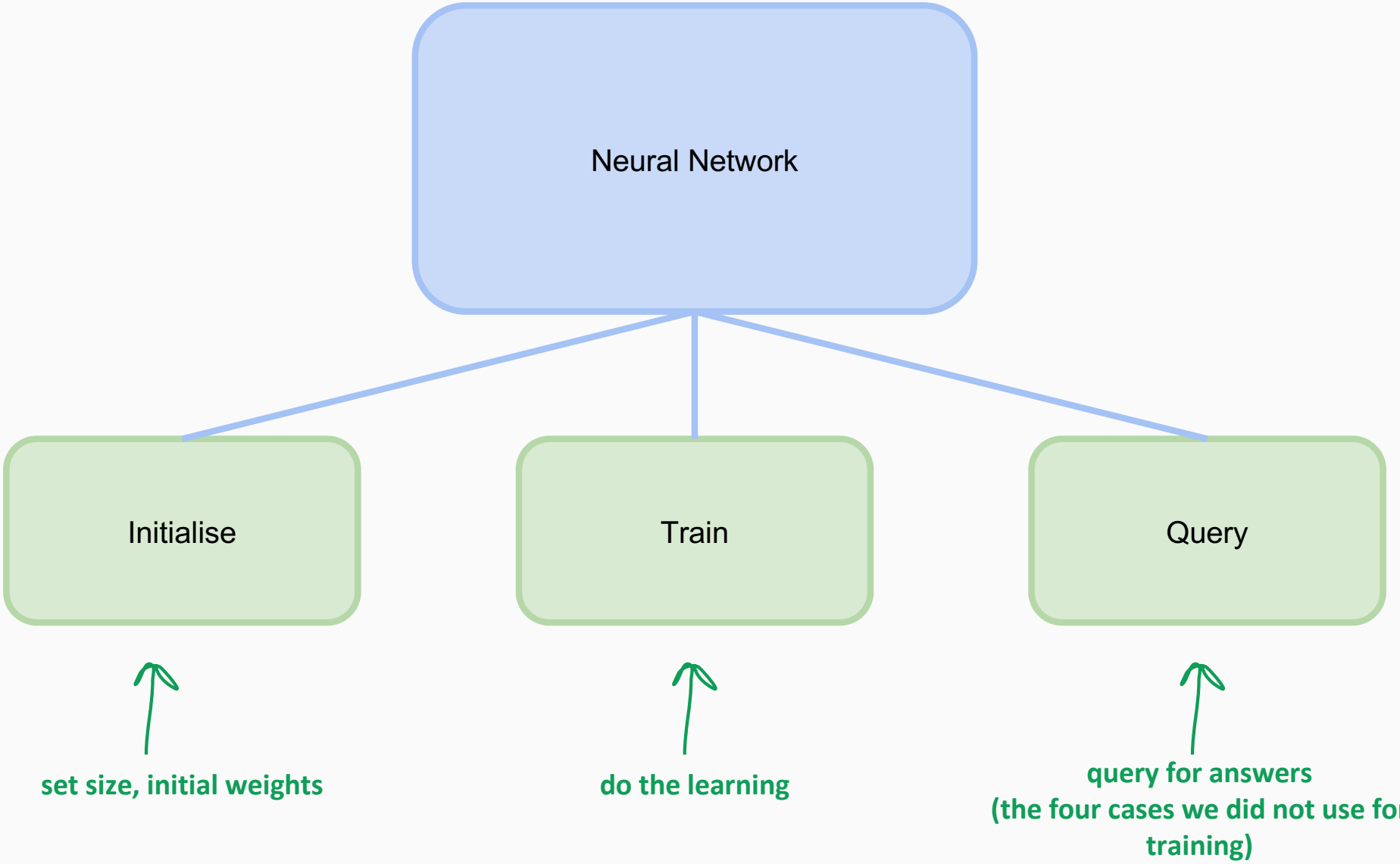


D Silver *et al. Nature* **550,** 354–359 (2017) doi:10.1038/nature24270

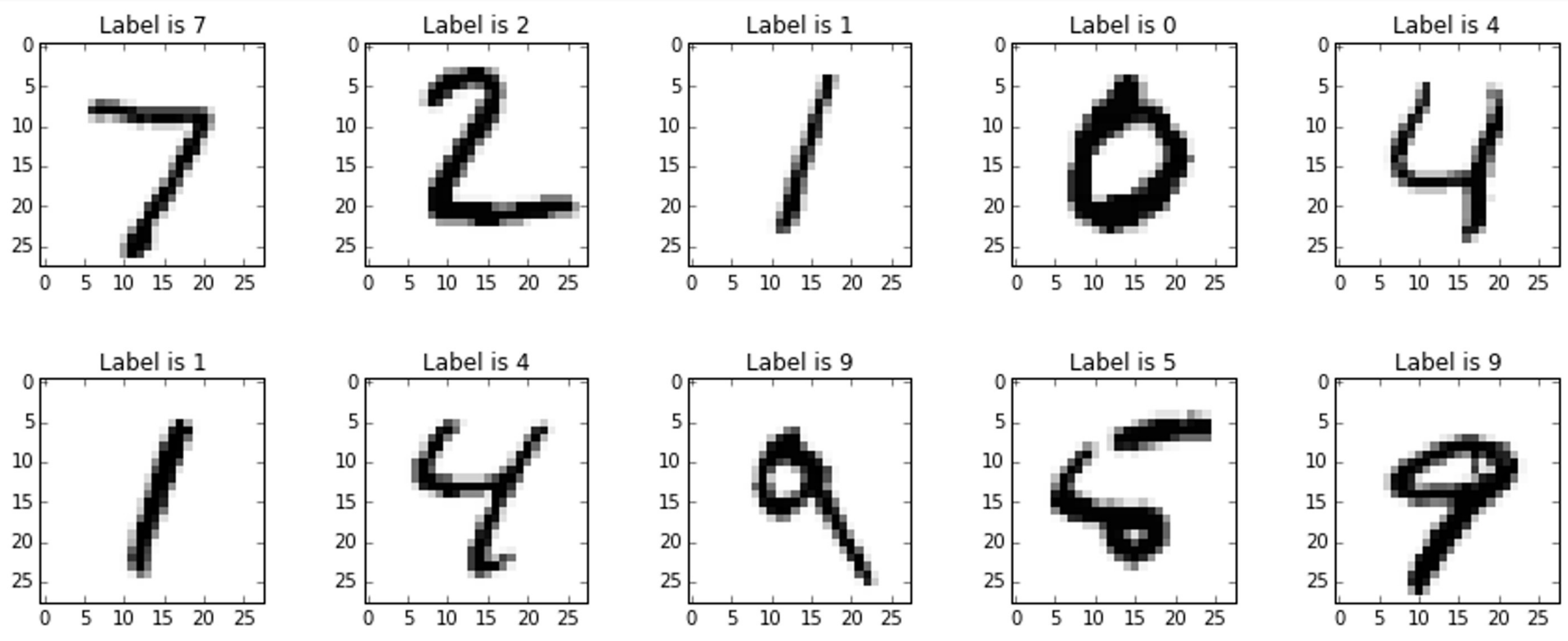
See also:
http://www.kurzweilai.net/alphago-zero-trains-itself-to-be-most-powerful-go-player-in-the-world
https://www.technologyreview.com/s/609141/alphago-zero-shows-machines-can-become-superhuman-without-any-help/

nature

Homework: at a minimum (and extensions using Claude)



Cooler stuff that is really not that much harder. Character recognition (CNNs):



MNIST Datasets

Letter Recognition Data Set

Download: Data Folder, Data Set Description

Abstract: Database of character image features; try to identify the letter

Data Set Characteristics:	Multivariate	**Number of Instances:**	20000	**Area:**	Computer
Attribute Characteristics:	Integer	**Number of Attributes:**	16	**Date Donated**	1991-01-01
Associated Tasks:	Classification	**Missing Values?**	No	**Number of Web Hits:**	174642

Source:

Creator:

David J. Slate
Odesta Corporation; 1890 Maple Ave; Suite 115; Evanston, IL 60201

Donor:

David J. Slate (dave '@' math.nwu.edu) (708) 491-3867

Data Set Information:

The objective is to identify each of a large number of black-and-white rectangular pixel displays as one of the 26 capital letters in the English alphabet. The character images were based on 20 different fonts and each letter within these 20 fonts was randomly distorted to produce a file of 20,000 unique stimuli. Each stimulus was converted into 16 primitive numerical attributes (statistical moments and edge counts) which were then scaled to fit into a range of integer values from 0 through 15. We typically train on the first 16000 items and then use the resulting model to predict the letter category for the remaining 4000. See the article cited above for more details.

MNIST dataset:

60,000 training data examples

10,000 test data examples

```
In [8]: data_file = open("mnist_dataset/mnist_train_100.csv", 'r')
        data_list = data_file.readlines()
        data_file.close()

In [9]: len(data_list)

Out[9]: 100

In [10]: data_list[0]

Out[10]: '5,0,0,0
```

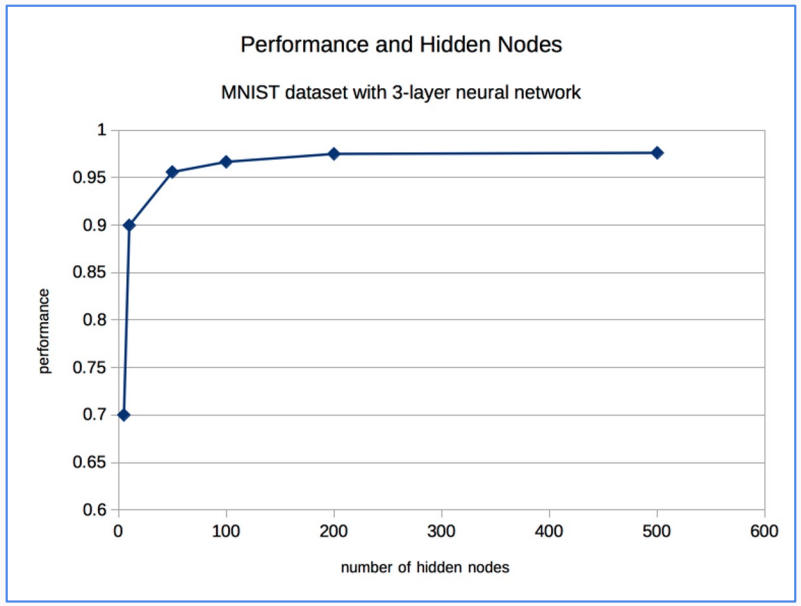
MNIST Datasets

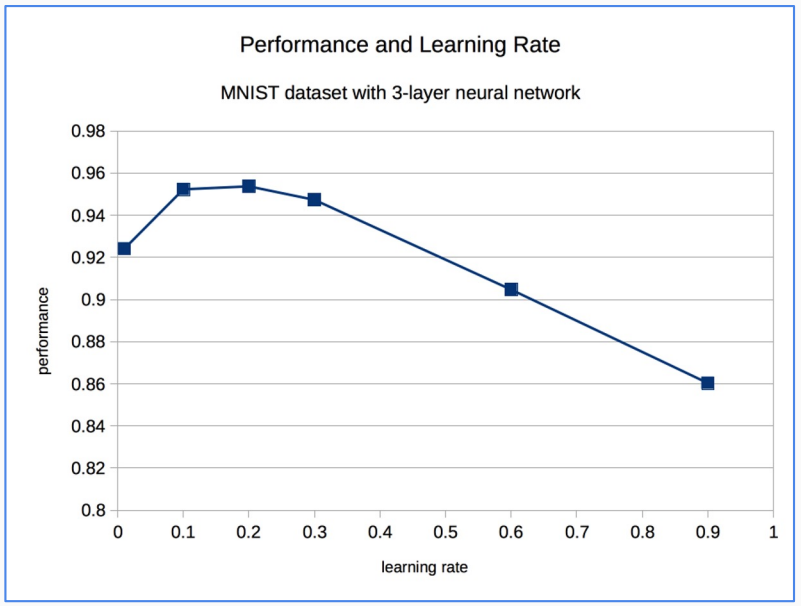
```
In [8]: data_file = open("mnist_dataset/mnist_train_100.csv", 'r')
        data_list = data_file.readlines()
        data_file.close()

In [9]: len(data_list)
Out[9]: 100

In [10]: data_list[0]
Out[10]: '5,0,0,0
```

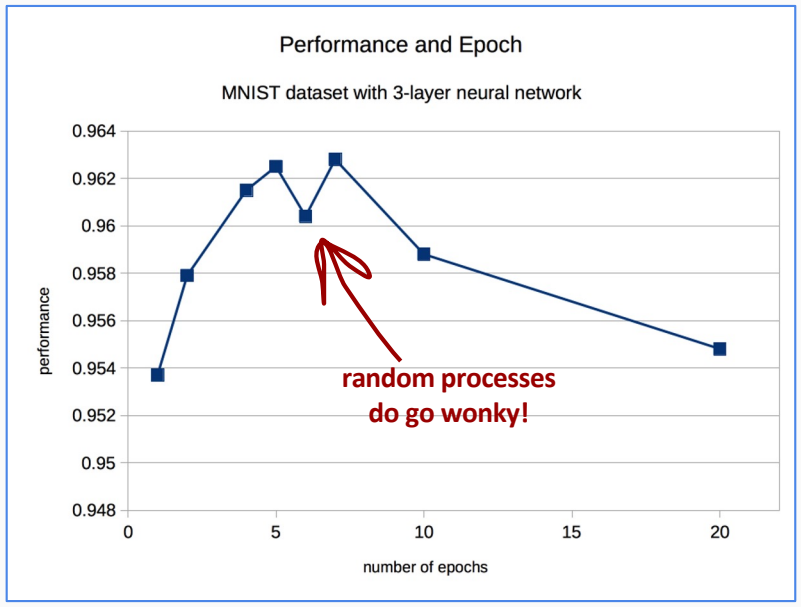
Output Layer Values

output layer	label	example "5"	example "0"	example "9"
0	0	0.00	0.95	0.02
1	1	0.00	0.00	0.00
2	2	0.01	0.01	0.01
3	3	0.00	0.01	0.01
4	4	0.01	0.02	0.40
5	5	0.99	0.00	0.01
6	6	0.00	0.00	0.01
7	7	0.00	0.00	0.00
8	8	0.02	0.00	0.01
9	9	0.01	0.02	0.86

Experiments



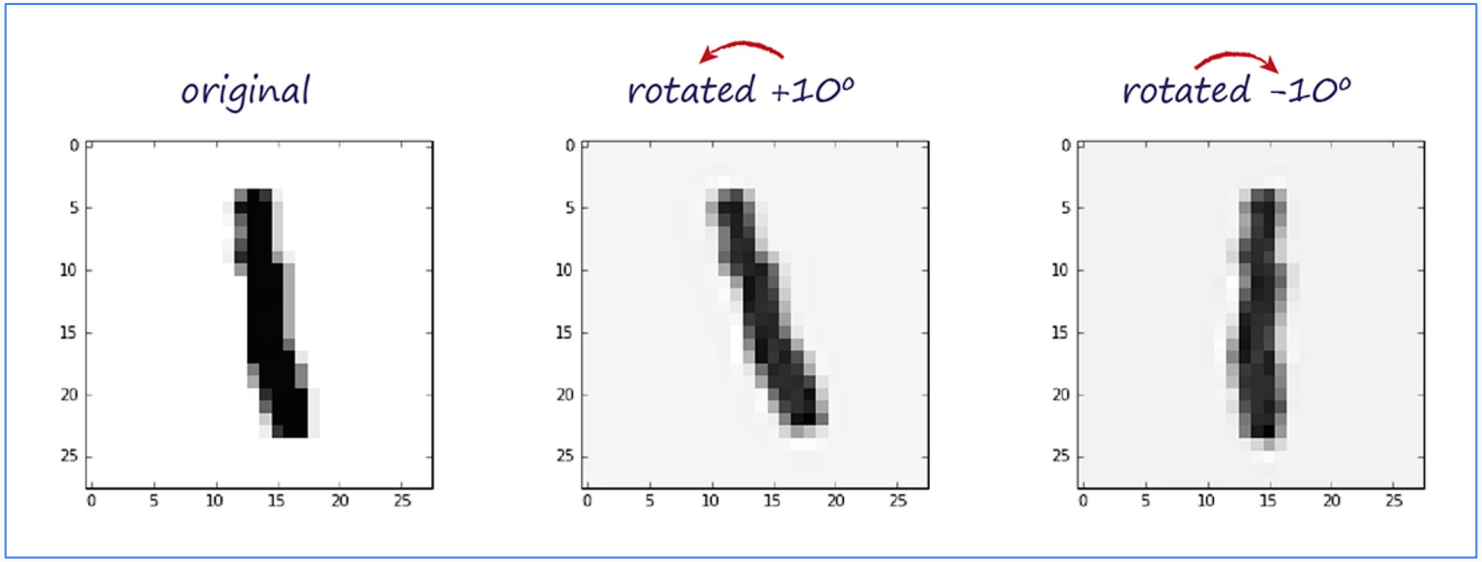


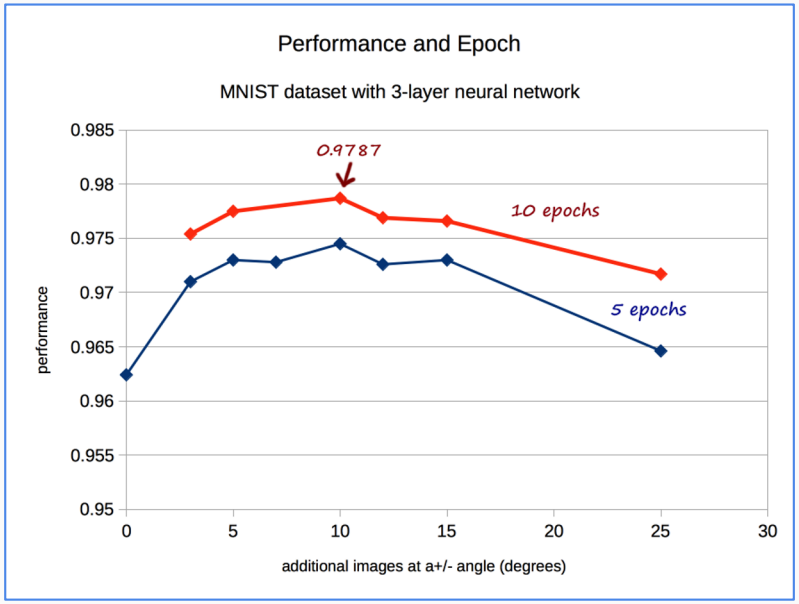


96% is very good!

we've only used simple ideas and code

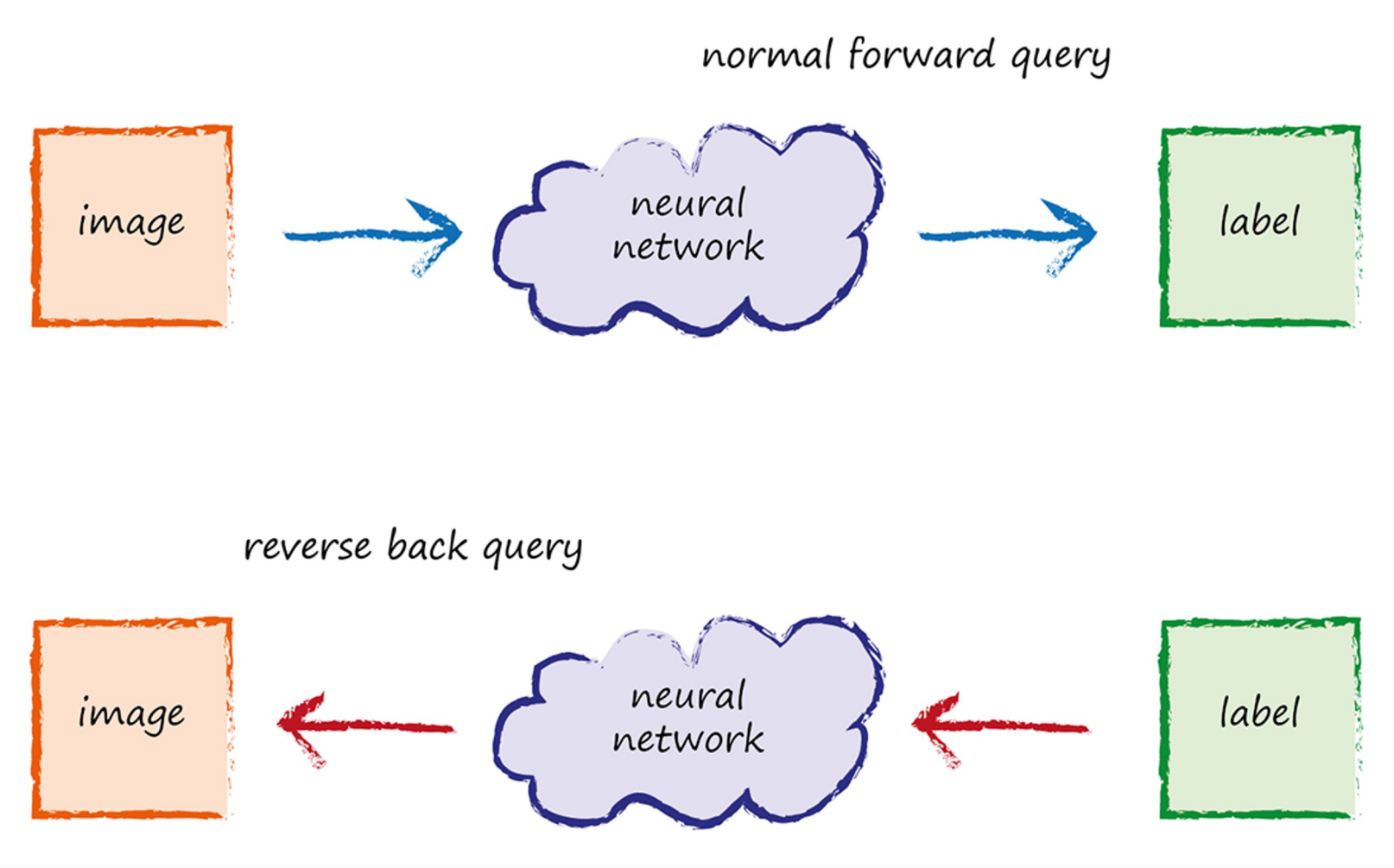
More Experiments



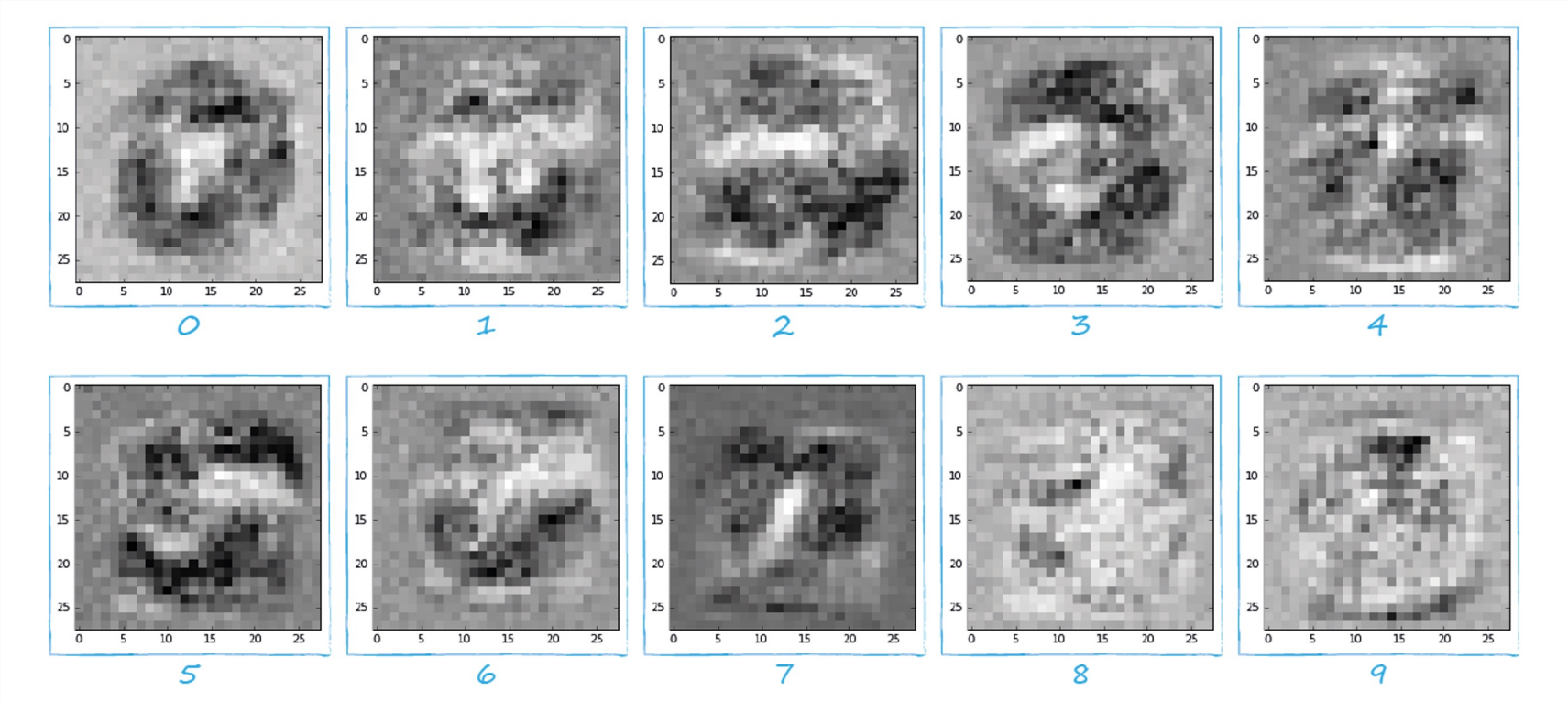


98% is amazing!

Peek Inside The Mind Of a Neural Network?



Peek Inside The Mind Of a Neural Network?

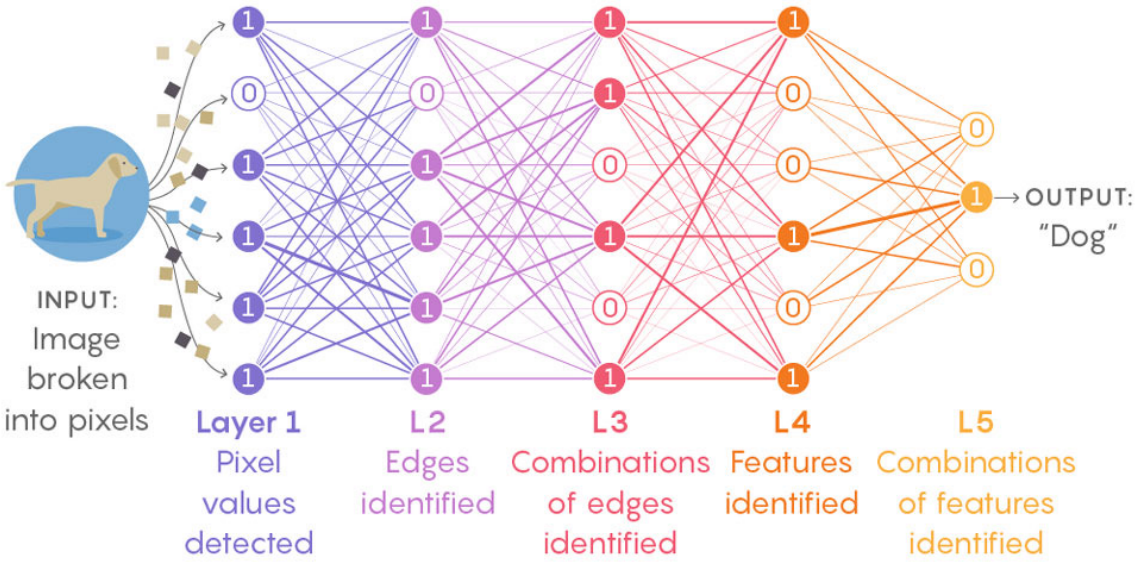


(Remember that autoencoded Bladerunner?)

Deep Learning: more of the same, but turning into a black box, FAST.

Learning From Experience

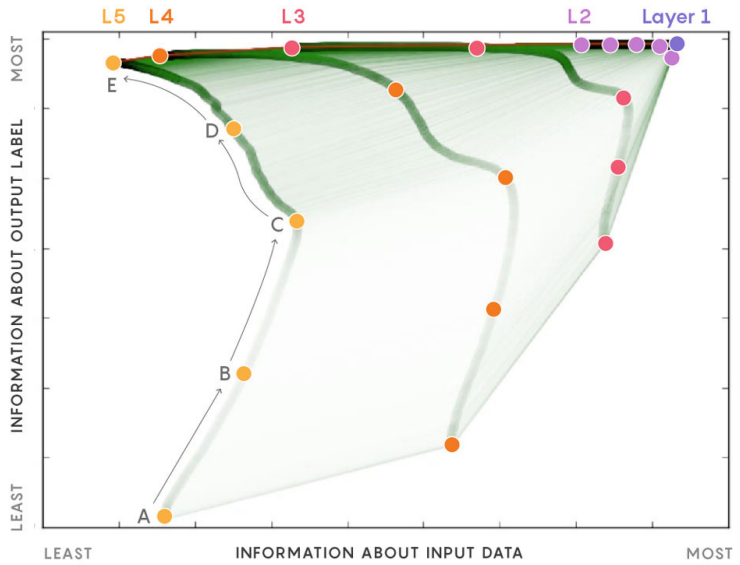
Deep neural networks learn by adjusting the strengths of their connections to better convey input signals through multiple layers to neurons associated with the right general concepts.



When data is fed into a network, each artificial neuron that fires (labeled "1") transmits signals to certain neurons in the next layer, which are likely to fire if multiple signals are received. The process filters out noise and retains only the most relevant features.

Inside Deep Learning

New experiments reveal how deep neural networks evolve as they learn.



A **INITIAL STATE:** Neurons in Layer 1 encode everything about the input data, including all information about its label. Neurons in the highest layers are in a nearly random state bearing little to no relationship to the data or its label.

B **FITTING PHASE:** As deep learning begins, neurons in higher layers gain information about the input and get better at fitting labels to it.

C **PHASE CHANGE:** The layers suddenly shift gears and start to "forget" information about the input.

D **COMPRESSION PHASE:** Higher layers compress their representation of the input data, keeping what is most relevant to the output label. They get better at predicting the label.

E **FINAL STATE:** The last layer achieves an optimal balance of accuracy and compression, retaining only what is needed to predict the label.

See links posted on extricating information from an ANN (example next week)

Have we wandered too far afield from physics?

MIT Technology Review

September 9, 2016

Computing

The Extraordinary Link Between Deep Neural Networks and the Nature of the Universe

Nobody understands why deep neural networks are so good at solving complex problems. Now physicists say the secret is buried in the laws of physics.

The problem: Deep NNs work extraordinarily well, but we don't know why or how, for the same reasons that we do not fully understand how our brain works. See http://kicp.uchicago.edu/~collar/blackbox.pdf

The claim: NNs are exceptionally well-adapted to approximate the few simple mathematical functions that describe the laws of physics governing our universe. These are just a subset of an infinity of possible functions (i.e., we are damn lucky). The reason why NNs work so well is because physics (a.k.a. reality, **nature) is fundamentally simple**. In other words, the **present** NN architecture of our brain (the inspiration for ANNs) is well-adapted to extricate the patterns of information (physical law) governing "reality". (I beg to disagree…)

Have we reached an apex?
(strong element of hubris involved here)

Maybe **we** are simple…

arXiv.org > cond-mat > arXiv:1608.08225

Search or Article-id

Condensed Matter > Disordered Systems and Neural Networks

Why does deep and cheap learning work so well?

Henry W. Lin (Harvard), Max Tegmark (MIT)

(Submitted on 29 Aug 2016)

We show how the success of deep learning depends not only on mathematics but also on physics: although well-known mathematical theorems guarantee that neural networks can approximate arbitrary functions well, the class of functions of practical interest can be approximated through "cheap learning" with exponentially fewer parameters than generic ones, because they have simplifying properties tracing back to the laws of physics. The exceptional simplicity of physics-based functions hinges on properties such as symmetry, locality, compositionality and polynomial log-probability, and we explore how these properties translate into exceptionally simple neural networks approximating both natural phenomena such as images and abstract representations thereof such as drawings. We further argue that when the statistical process generating the data is of a certain hierarchical form prevalent in physics and machine-learning, a deep neural network can be more efficient than a shallow one. We formalize these claims using information theory and discuss the relation to renormalization group procedures. Various "no-flattening theorems" show when these efficient deep networks cannot be accurately approximated by shallow ones without efficiency loss – even for linear networks.

Have we wandered too far afield from physics?

MIT Technology Review

September 9, 2016

Computing

The Extraordinary Link Between Deep Neural Networks and the Nature of the Universe

Nobody understands why deep neural networks are so good at solving complex problems. Now physicists say the secret is buried in the laws of physics.

The problem: Deep NNs work extraordinarily well, but we don't know why or how, for the same reasons that we do not fully understand how our brain works. See http://kicp.uchicago.edu/~collar/blackbox.pdf

The claim: NNs are exceptionally well-adapted to approximate the few simple mathematical functions that describe the laws of physics governing our universe. These are just a subset of an infinity of possible functions (i.e., we are damn lucky). The reason why NNs work so well is because physics (a.k.a. reality, **nature) is fundamentally simple**. In other words, the **present** NN architecture of our brain (the inspiration for ANNs) is well-adapted to extricate the patterns of information (physical law) governing "reality". (I beg to disagree…)

arXiv.org > cond-mat > arXiv:1608.08225

Search or Article-id

Condensed Matter > Disordered Systems and Neural Networks

Why does deep and cheap learning work so well?

Henry W. Lin (Harvard), Max Tegmark (MIT)

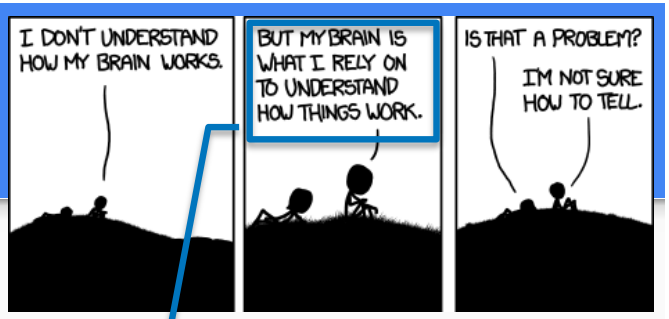
(Submitted on 29 Aug 2016)

We show how the success of deep learning depends not only on mathematics but also on physics: although well-known mathematical theorems guarantee that neural networks can approximate arbitrary functions well, the class of functions of practical interest can be approximated through "cheap learning" with exponentially fewer parameters than generic ones, because they have simplifying properties tracing back to the laws of physics. The exceptional simplicity of physics-based functions hinges on properties such as symmetry, locality, compositionality and polynomial log-probability, and we explore how these properties translate into exceptionally simple neural networks approximating both natural phenomena such as images and abstract representations thereof such as drawings. We further argue that when the statistical process generating the data is of a certain hierarchical form prevalent in physics and machine-learning, a deep neural network can be more efficient than a shallow one. We formalize these claims using information theory and discuss the relation to renormalization group procedures. Various "no-flattening theorems" show when these efficient deep networks cannot be accurately approximated by shallow ones without efficiency loss – even for linear networks.

Have we reached an apex?
(strong element of hubris involved here)

Maybe **we** are simple…

What came first, the chicken or the egg?



THIS

"For reasons that are still not fully understood, our universe can be accurately described by polynomial Hamiltonians of low order."

Lin and Tegmark.

VS.

"Due to the limitations imposed by NN learning, we perceive a reality described by low-order polynomial Hamiltonians only."

Homer Simpson.

Remember: you cannot describe XOR with a single-layer perceptron. Exactly how much mileage do we get from 1E+11 sparsely connected neurons?
https://nautil.us/the-fundamental-limits-of-machine-learning-236116
Putting learning in perspective (we are smarter than a ball of dough, but exactly how much smarter?):
http://www.realclearscience.com/journal_club/2016/10/24/researchers_find_hints_of_learning_in_a_ball_of_dough_109789.html

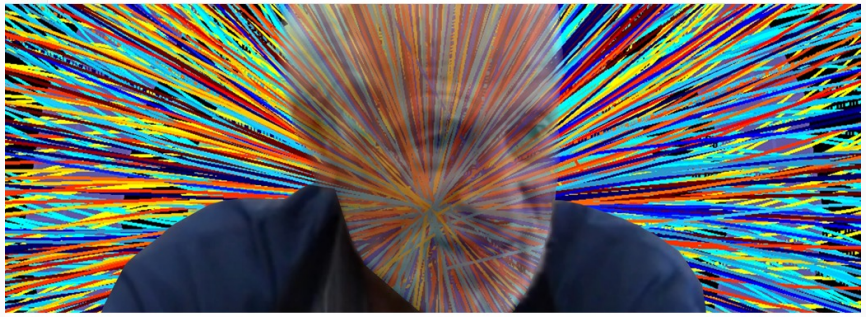
Know your place in the universe: are you sure that NN-based systems represent an evolutionary apex?
Or are they the (temporary?) cookie-cutter that limits our perception/understanding of physics/"reality"?

IMHO: Physics and Neuroscience to make beautiful music together in the years to come

Move along, nothing to see here (old fight between Scientific Realism and Empiricism)

NAUTILUS ISSUES TOPICS BLOG NEWSLETTER STORE PRIME

How I Rewired My Brain to Become Fluent in Math | How Much More Can We Learn About the Universe? | How to Tell If You're a Jerk | Is Artificial Ir Permanently



IDEAS | PHILOSOPHY

Why Science Should Stay Clear of Metaphysics

Meet the philosopher who revived anti-realism.

QUANTA MAGAZINE illuminating science

PHYSICS MATHEMATICS BIOLOGY COMPUTER SCIENCE MULTIMEDIA BLOG ALL SUBSCRIBE

PHYSICS

A Fight for the Soul of Science

String theory, the multiverse and other ideas of modern physics are potentially untestable. At a historic meeting in Munich, scientists and philosophers asked: should we trust them anyway?

Must science be testable?

String wars among physicists have highlighted just how much science needs philosophy – and not just the amateur version

nature International weekly journal of science

Home | News & Comment | Research | Careers & Jobs | Current Issue | Archive | Audio & Video | For A

Archive > Volume 516 > Issue 7531 > Comment > Article

NATURE | COMMENT

Scientific method: Defend the integrity of physics

George Ellis & Joe Silk

16 December 2014

Attempts to exempt speculative theories of the Universe from experimental verification undermine science, argue George Ellis and Joe Silk.

PDF | Rights & Permissions

Subject terms: Physics · Philosophy · Astronomy and astrophysics

"They're getting too big for their boots. We practical physicists will have to take them down a peg or two."

Ernest Rutherford.

My own take: the answer to this riddle will come from neuroscience (psychology of perception) (These couple slides from PHYS 123)

within Immanuel Kant's *Critique of Pure Reason* (1781/1787). Kant argued that space as we know it is a preconscious organizing feature of the human mind, a scaffold upon which we're able to understand the physical world of objects, extension and motion. In a sense, space to Kant was a window into the world, rather than a thing to be perceived in it.

Immanuel Kant, circa 1781: enter Kantian "a prioris"

"Although all our cognition begins with experience, it does not follow that is caused by experience…"

"It is quite possible that our empirical knowledge is a compound of that which receive through impressions, and that which the faculty of cognition supplies from itself."

A prioris, forms of intuition or transcendental conditions deeply seated in our cognitive faculties. The filters through which we perceive the world.

In other words, our hardware (the way our brain is "hardwired") may be distorting an external "reality". Spatial separation, time, causality, may be simple constructs of the specific neural networks that we posses at our present time in evolution.

The experimental evidence is clearly there: QM nonlocality, the superposition state, spooky action-at-a-distance, limited access to information -Heisenberg's uncertainty principle-… essentially everything we gather from QM runs against our naïve "classical" understanding of the world, and yet it is an exceptionally successful theory in its predictive power.

Noam Chomsky on the limits of knowledge:
https://www.youtube.com/watch?v=tc1hsQWzUKc&ab_channel=PhilosophyOverdose

Incipient state of understanding:
Space, time, and the human brain NN architecture
(These couple slides from PHYS 123)

https://science.sciencemag.org/content/358/6362/482
https://blogs.scientificamerican.com/mind-guest-blog/decoding-space-and-time-in-the-brain/
https://www.sciencedirect.com/book/9780123859488/space-time-and-number-in-the-brain#book-description

SPECIAL SECTION CHALLENGES IN NEUROSCIENCE

REVIEW

Space and time in the brain

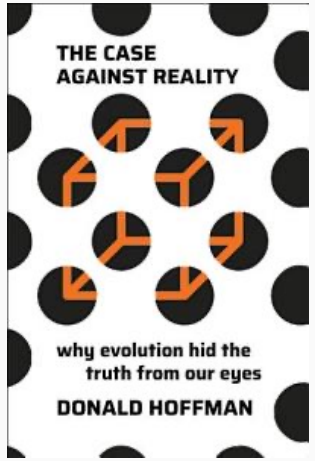
György Buzsáki* and Rodolfo Llinás

Nothing is more intuitive, yet more complex, than the concepts of space and time. In contrast to spacetime in physics, space and time in neuroscience remain separate coordinates to which we attach our observations. Investigators of navigation and memory relate neuronal activity to position, distance, time point, and duration and compare these parameters to units of measuring instruments. Although spatial-temporal sequences of brain activity often correlate with distance and duration measures, these correlations may not correspond to neuronal representations of space or time. Neither instruments nor brains sense space or time. Neuronal activity can be described as a succession of events without resorting to the concepts of space or time. Instead of searching for brain representations of our preconceived ideas, we suggest investigating how brain mechanisms give rise to inferential, model-building explanations.

"...neither clocks nor brains make time per se."

"...we do not directly sense time."

https://www.quantamagazine.org/the-evolutionary-argument-against-reality-20160421/



An example of NN use by a “practical physicist”: pulse-shape discrimination

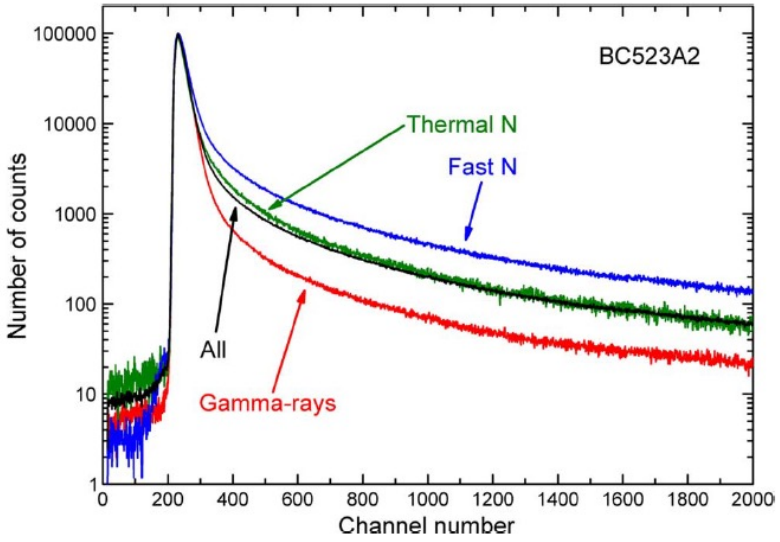
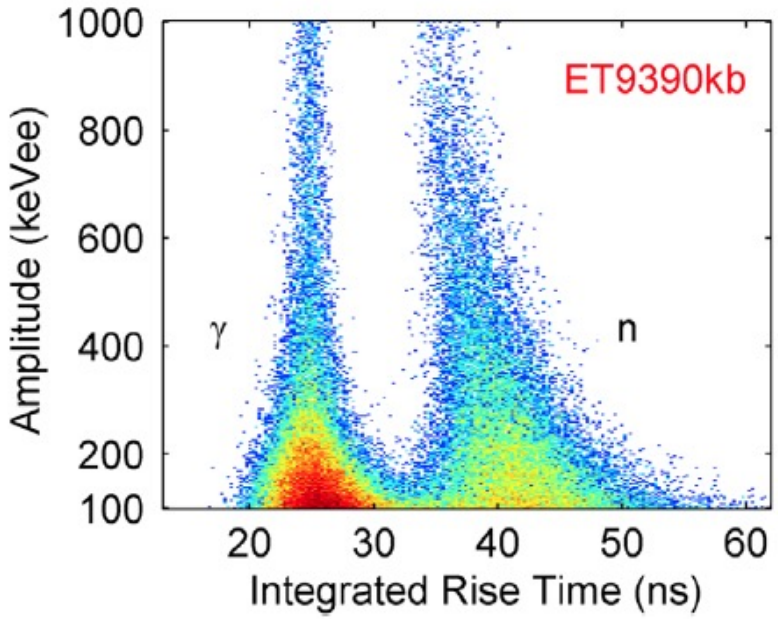


Fig. 4. Normalized light pulse shapes obtained for different gates set at the data recorded with BC523A2 scintillator and at 500 ns TAC range.



IEEE TRANSACTIONS ON NUCLEAR SCIENCE, VOL. 57, NO. 6, DECEMBER 2010

X.L. Luo et al. / Nuclear Instruments and Methods in Physics Research A 767 (2014) 83–91

E. Ronchi et al. / Nuclear Instruments and Methods in Physics Research A 610 (2009) 534–539

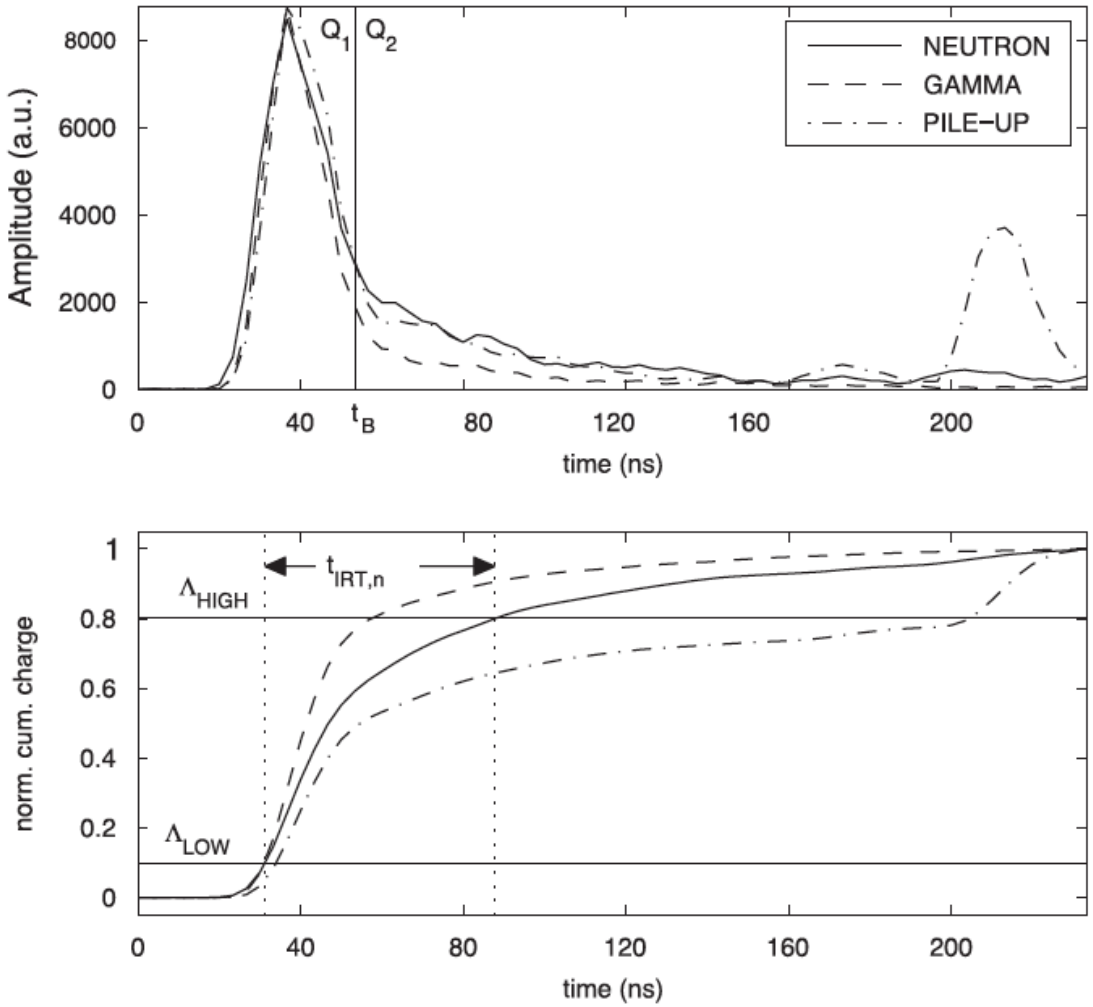


Fig. 1. (above) Example of neutron (solid line), γ (dashed) and pile-up (dash-dotted) waveforms from Ref. [3]. (below) Normalized cumulative charge distribution of the three waveforms.

An example of NN use by a “practical physicist”: pulse-shape discrimination

IEEE TRANSACTIONS ON NUCLEAR SCIENCE, VOL. 57, NO. 6, DECEMBER 2010

X.L. Luo et al. / Nuclear Instruments and Methods in Physics Research A 767 (2014) 83–91

E. Ronchi et al. / Nuclear Instruments and Methods in Physics Research A 610 (2009) 534–539

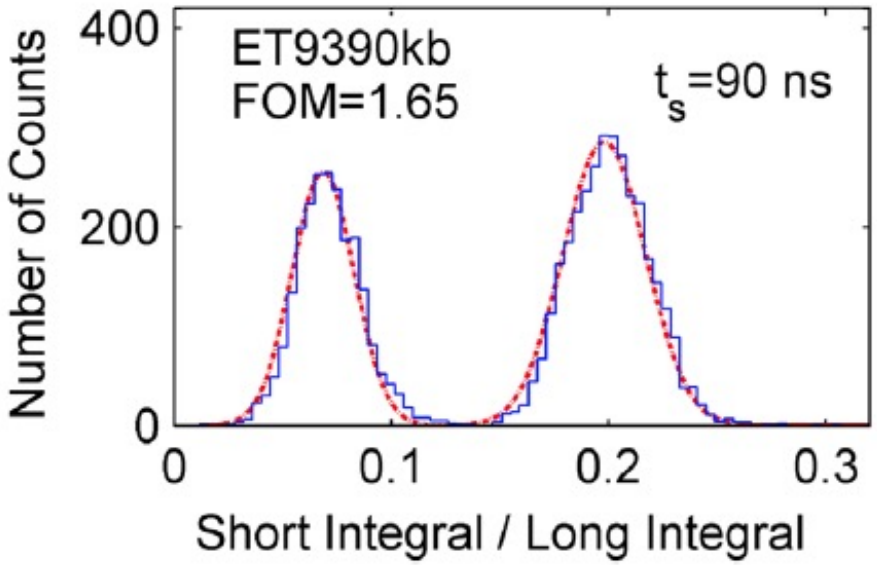


Fig. 6. Neutron–γ discrimination spectra with fitted Gaussian distributions at 320 ± 20 keVee using the CC method for PMT ET9390kb

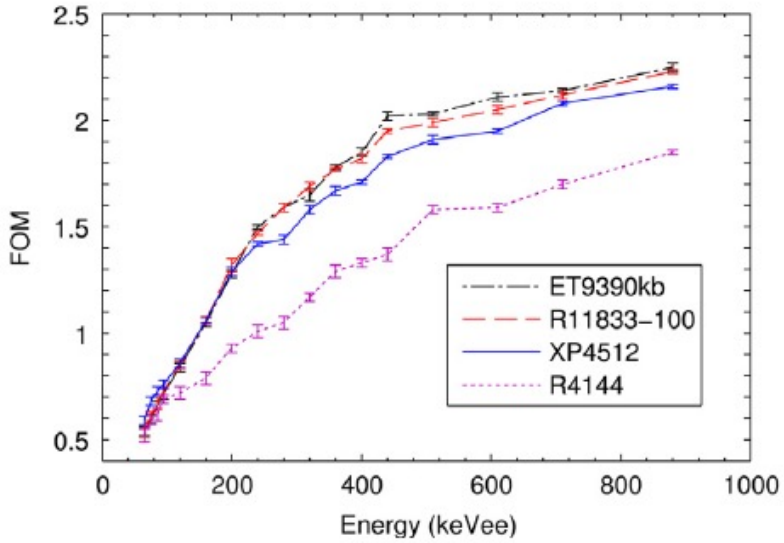
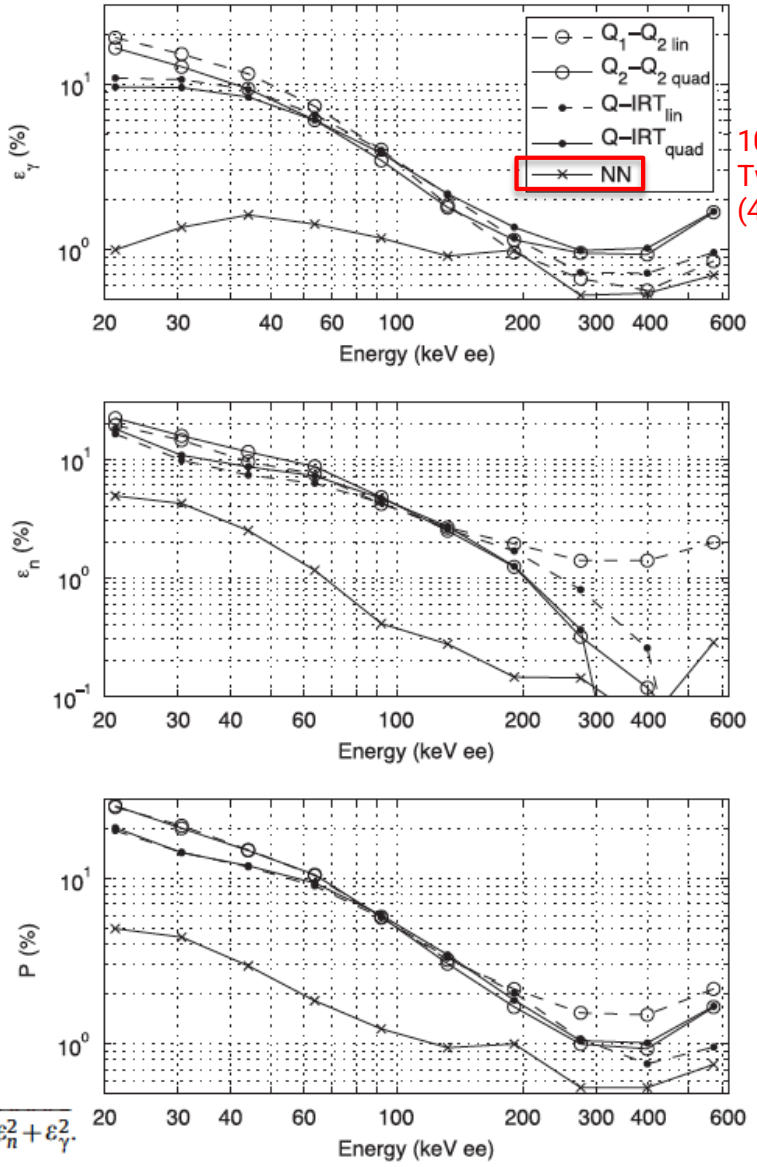


Fig. 7. FOM values of the CC method for PMT ET9390kb, R11833-100, XP4512, and R4144 as a function of energy window (the widths of the windows are 10, 40, and 100 keVee in energy regions of 50–100, 100–500 and 500–1000 keVee, respectively).

$$\mathrm{FOM} = \frac{S}{\mathrm{FWHM_n} + \mathrm{FWHM_\gamma}}$$

An example of NN use by a “practical physicist”: pulse-shape discrimination



106 inputs
Two layers
(40 and 10 neurons)

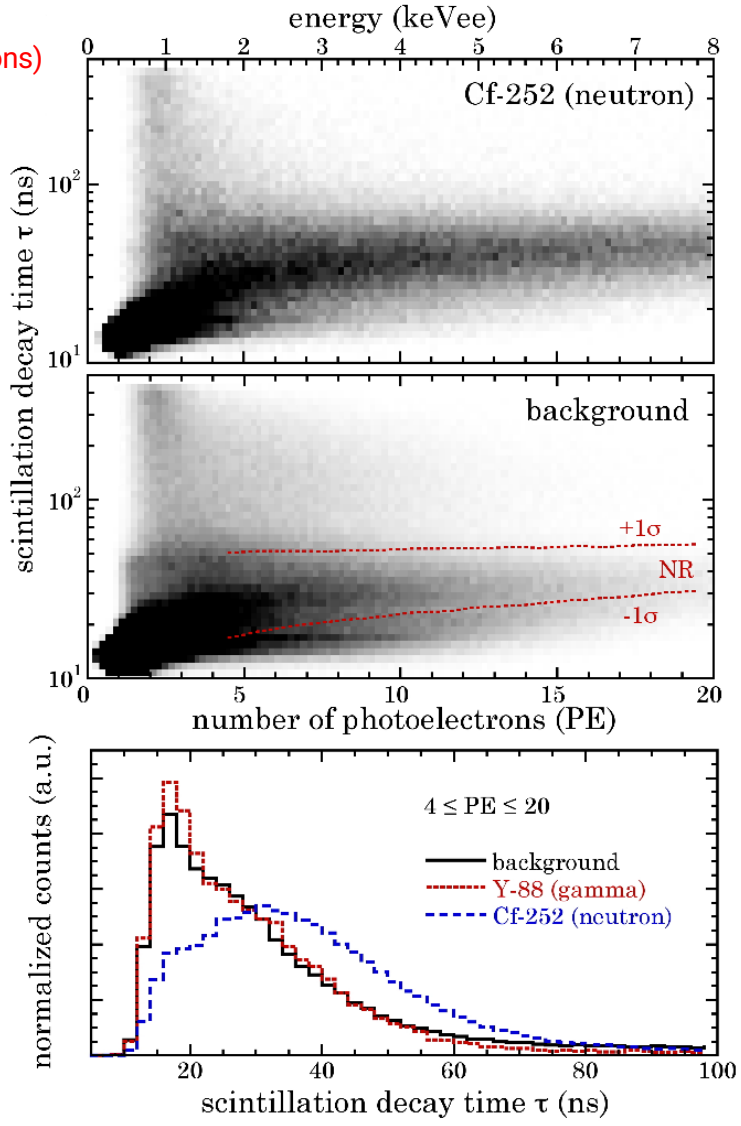
$$P = \sqrt{\varepsilon_n^2 + \varepsilon_\gamma^2}.$$

Fig. 5. Fraction of incorrectly identified γ (top) and neutron (middle) events as a function of the deposited energy for all methods. (bottom) *P* as a function of the deposited energy for all methods.

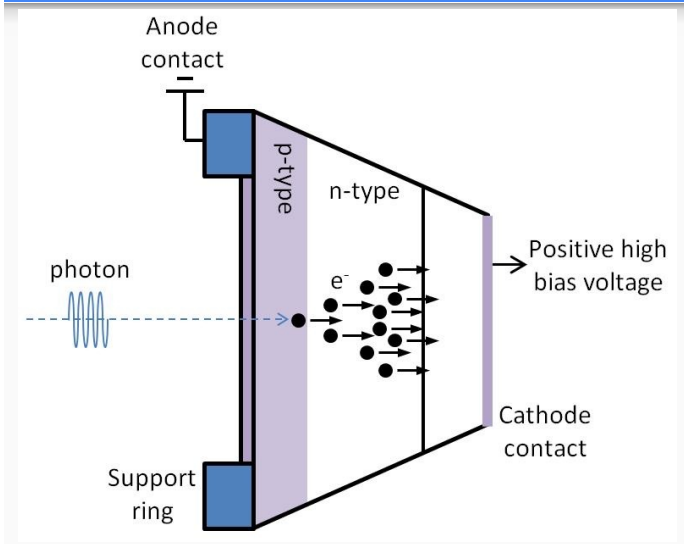
IEEE TRANSACTIONS ON NUCLEAR SCIENCE, VOL. 57, NO. 6, DECEMBER 2010

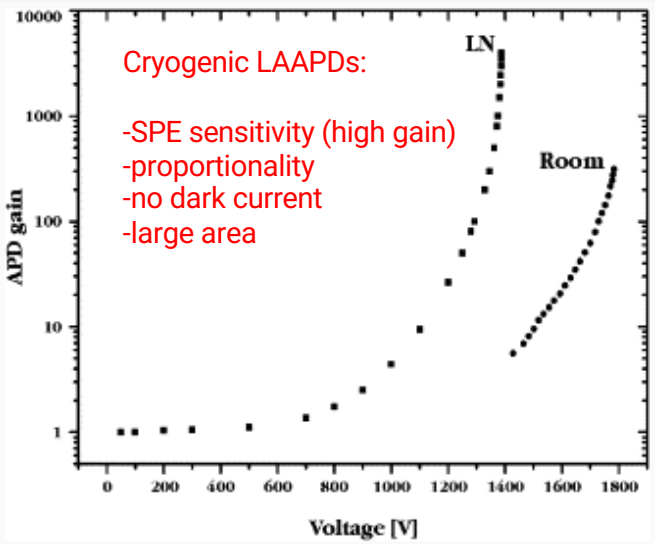
X.L. Luo et al. / Nuclear Instruments and Methods in Physics Research A 767 (2014) 83–91

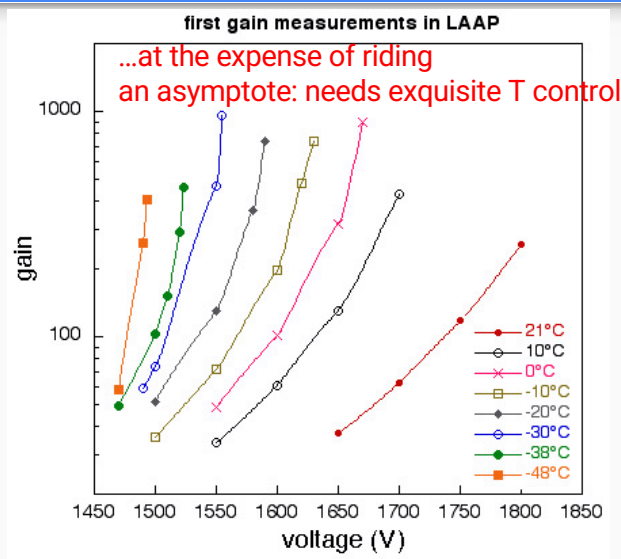
E. Ronchi et al. / Nuclear Instruments and Methods in Physics Research A 610 (2009) 534–539

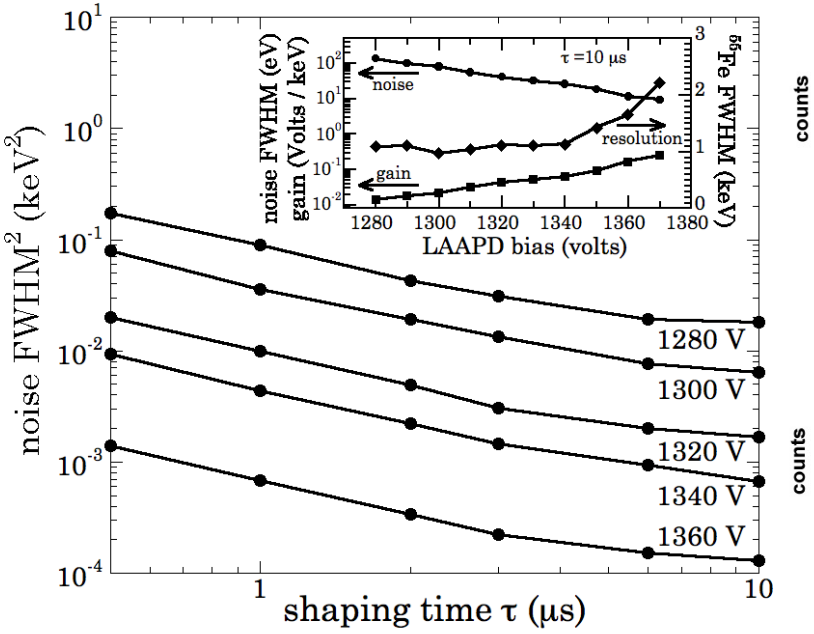


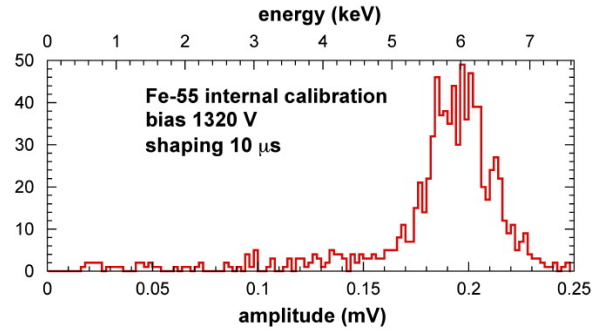
Another example: PID control

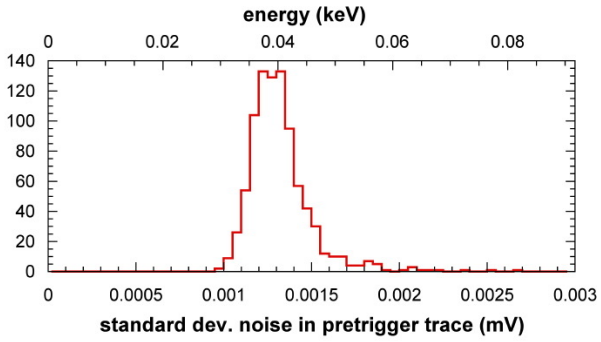




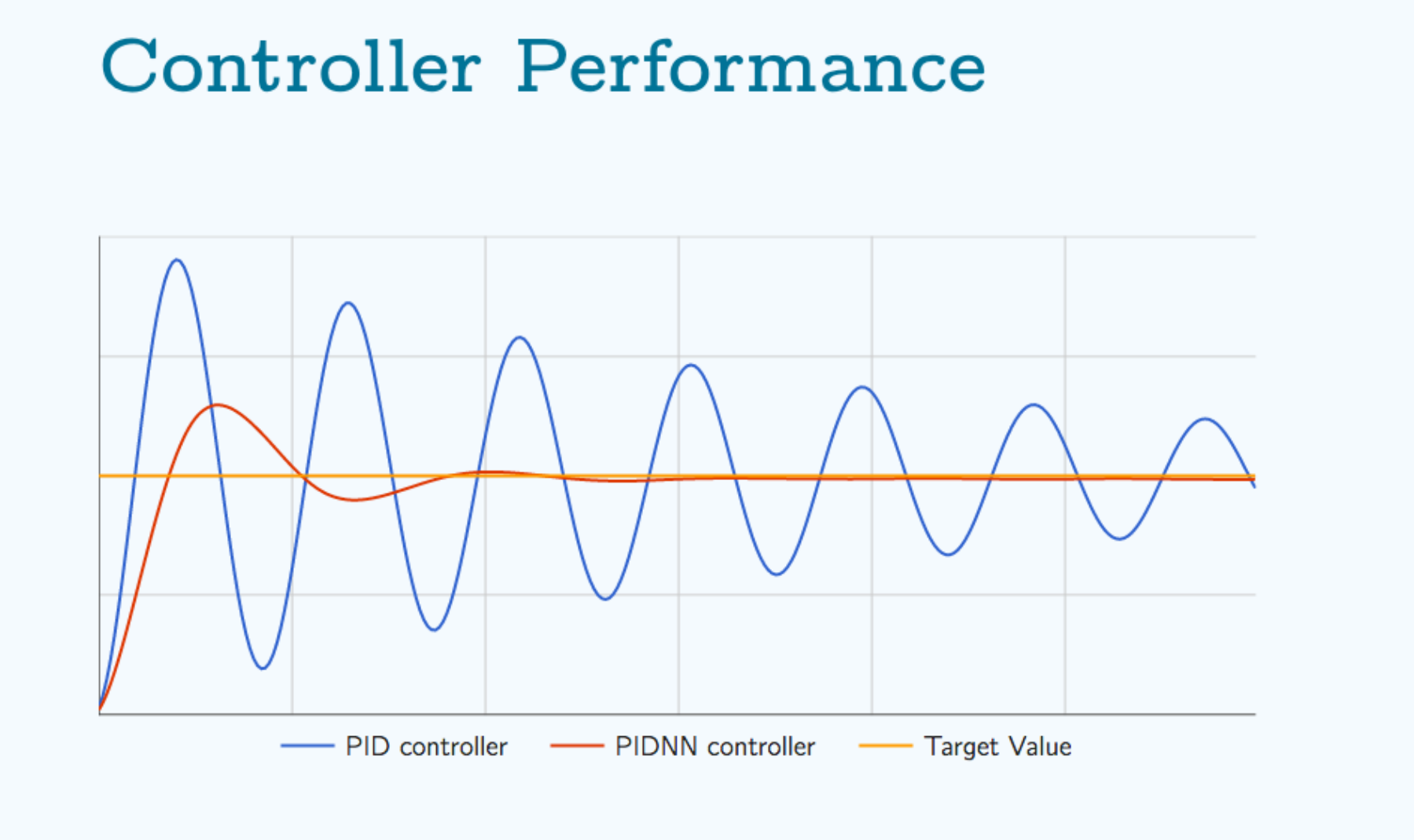




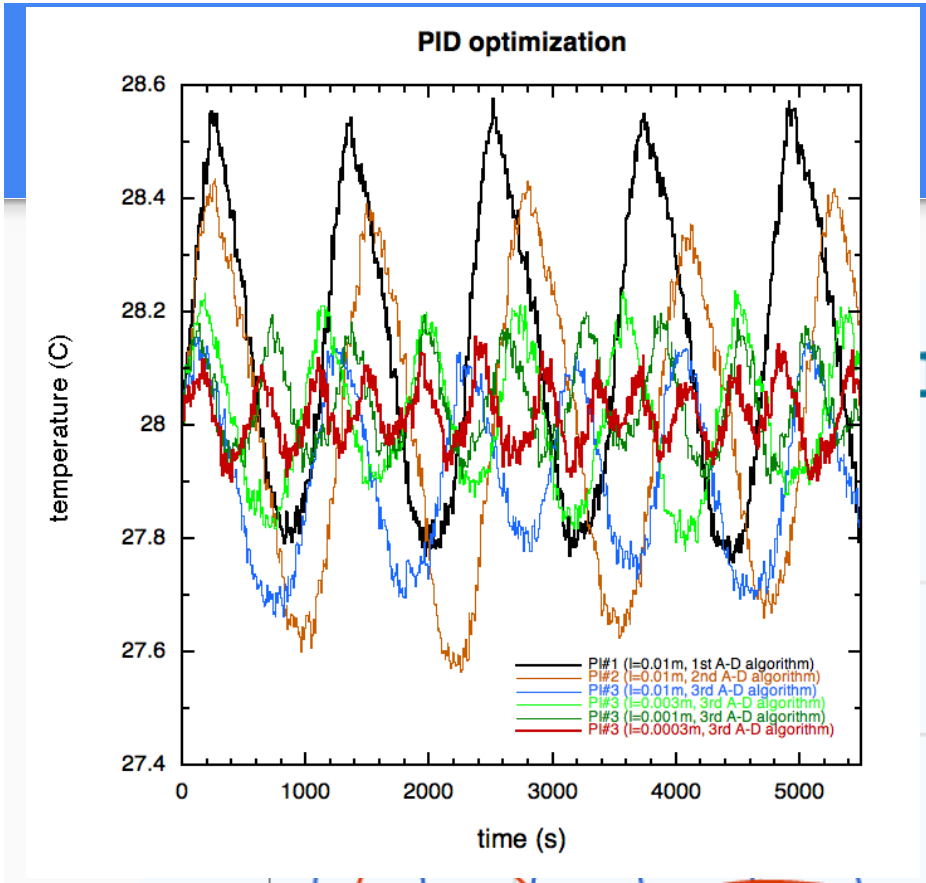




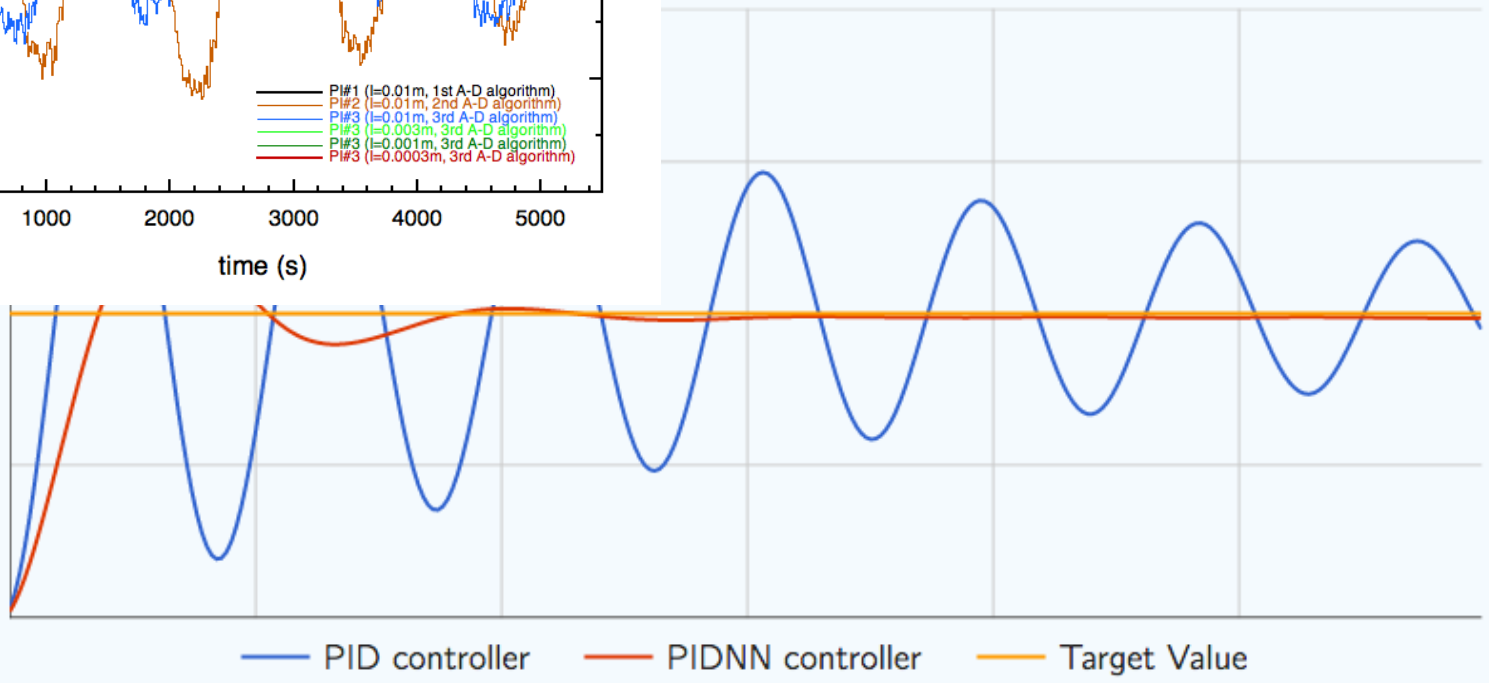
Another example: PID control



http://molefrog.com/pidnn-talk/



rformance



http://molefrog.com/pidnn-talk/

Another example: Background Rejection

Background rejection in NEXT using deep neural networks arXiv:1609.06202v1

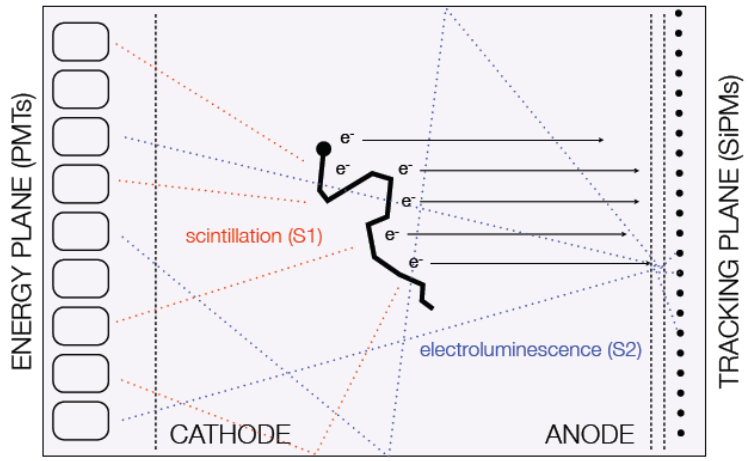
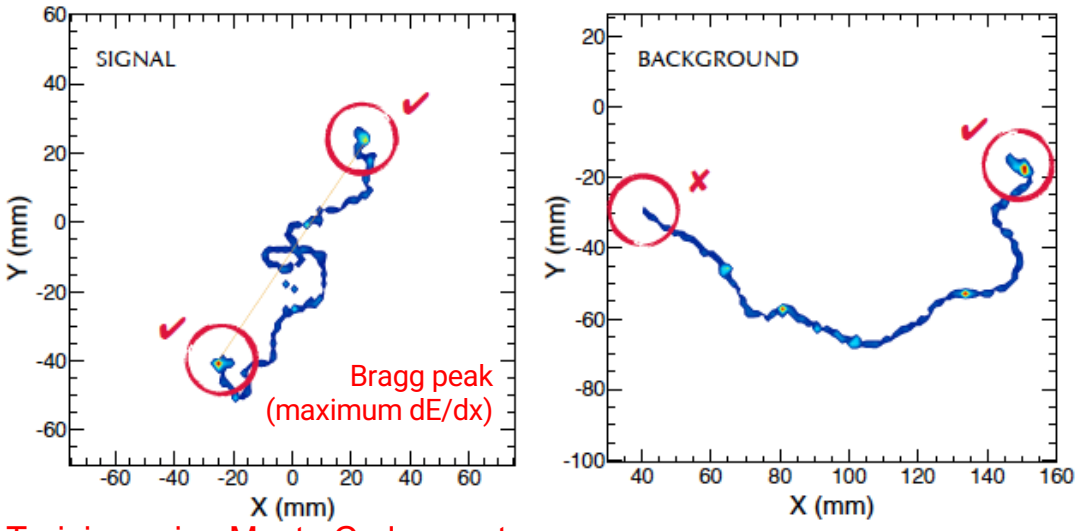
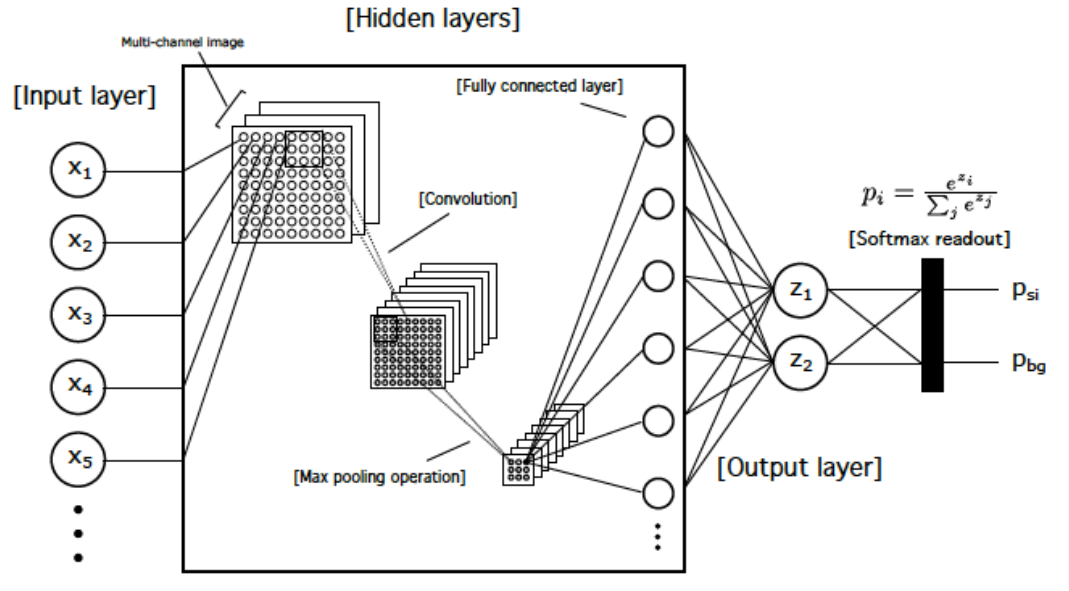
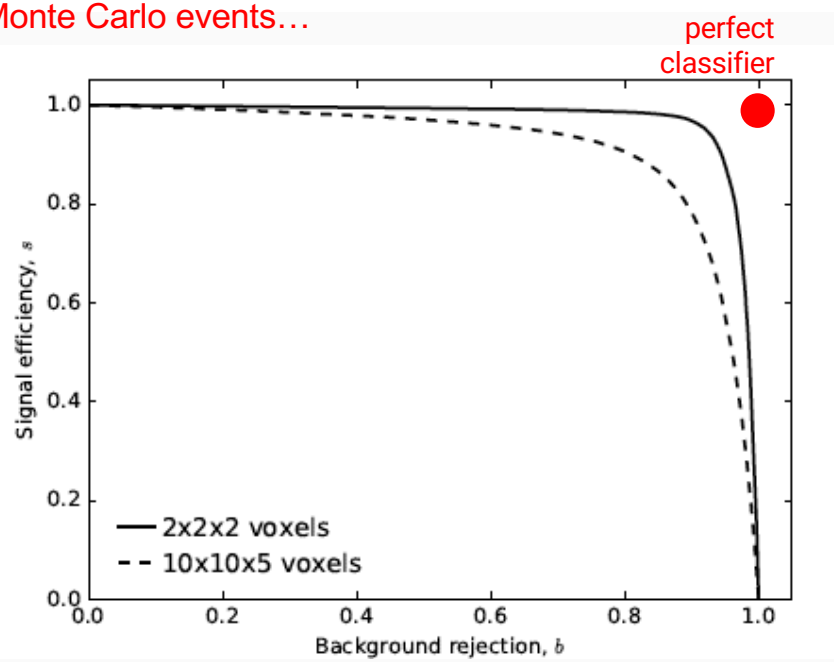


Figure 1. Principle of operation of an asymmetric HPXe TPC with EL readout. (Figure from [10].)



Training using Monte Carlo events…





Another example: Background Rejection

Background rejection in NEXT using deep neural networks arXiv:1609.06202v1

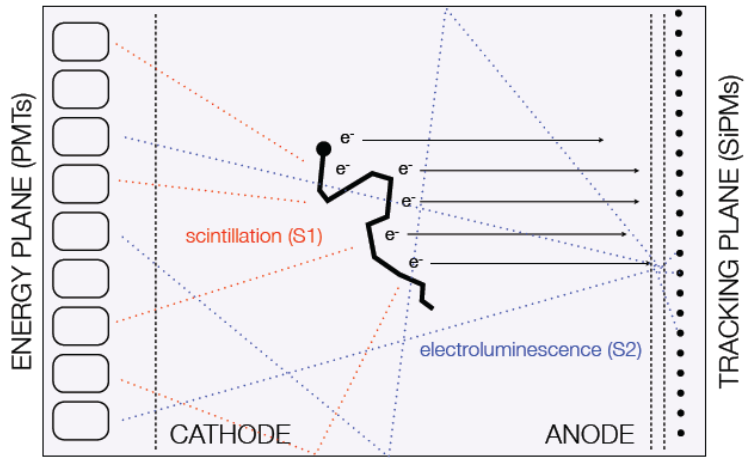
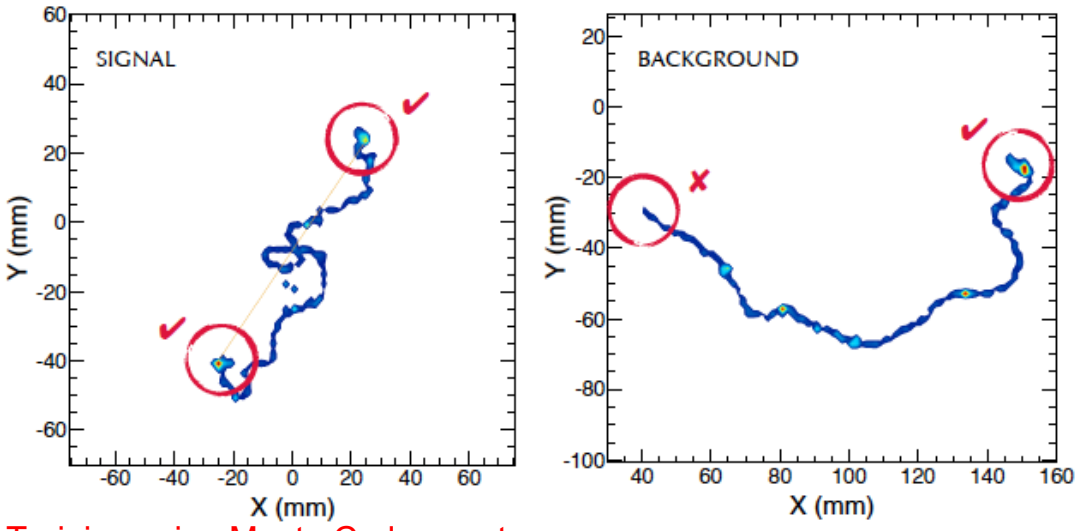
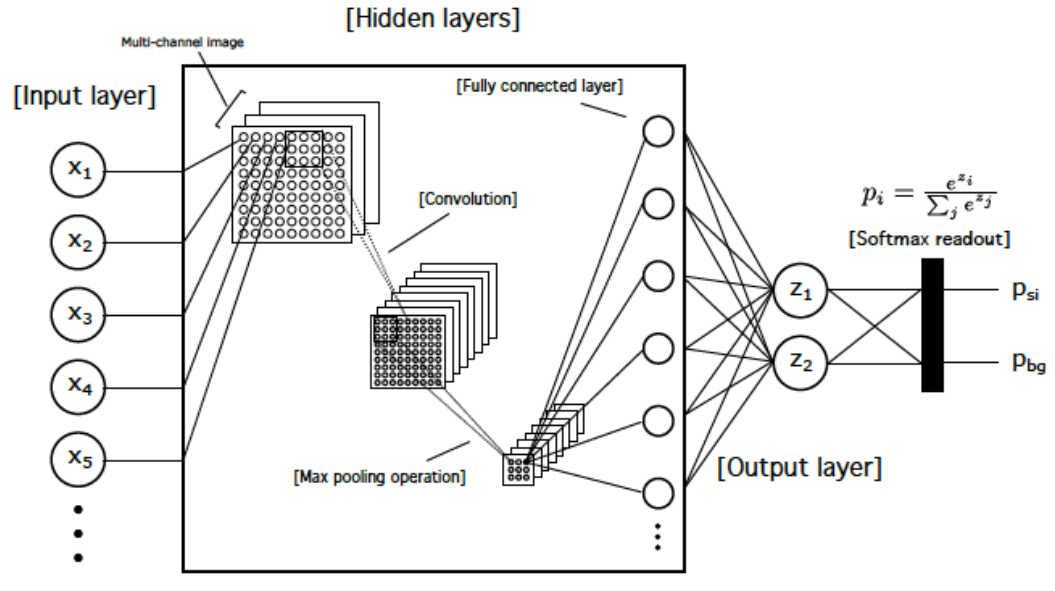


Figure 1. Principle of operation of an asymmetric HPXe TPC with EL readout. (Figure from [10].)



Training using Monte Carlo events…



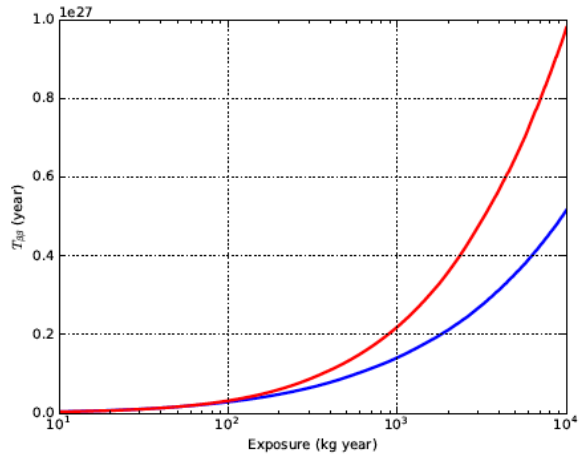
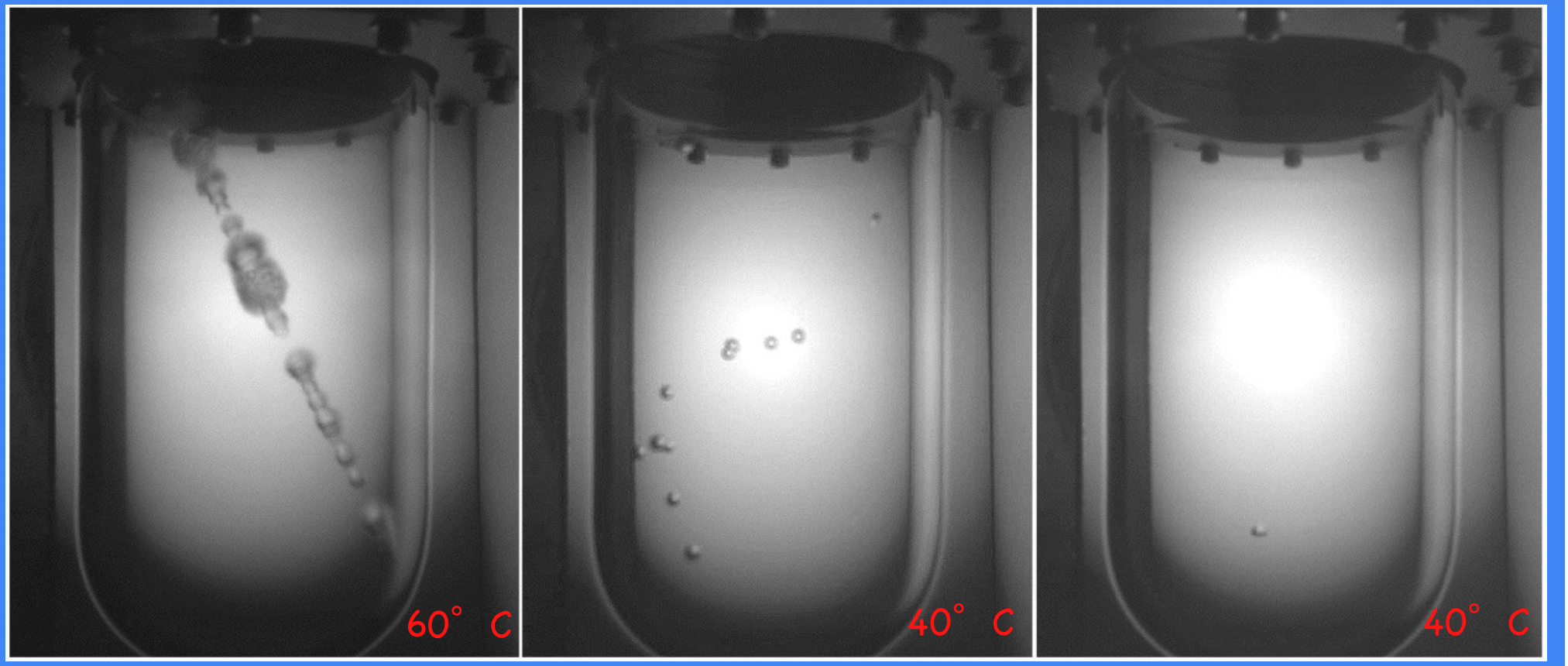


Figure 9. Sensitivity to the half-life $T_{1/2}^{0\nu\beta\beta}$ calculated using the Feldman-Cousins approach. The curves describe the NEXT-100 conventional analysis [7] (blue) and NEXT-100 with the improved DNN-based analysis with optimal figure of merit and low diffusion (red).

PICO: not your daddy's bubble chamber:

Conventional BC operation
(high superheat, MIP sensitive)

Low degree of superheat, sensitive to nuclear recoils only



muon

Neutron

WIMP ?

ultra-clean BC: Bolte *et al.*, NIM A577 (2007) 569
Science 319 (2008) 933

PICO approach to WIMP detection:

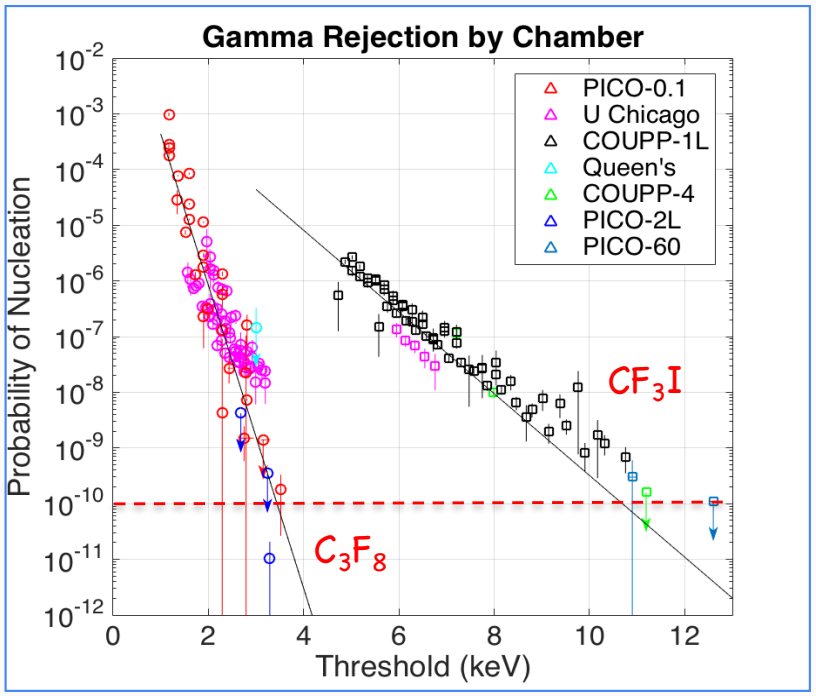
Seitz model of bubble nucleation (classical BC theory):

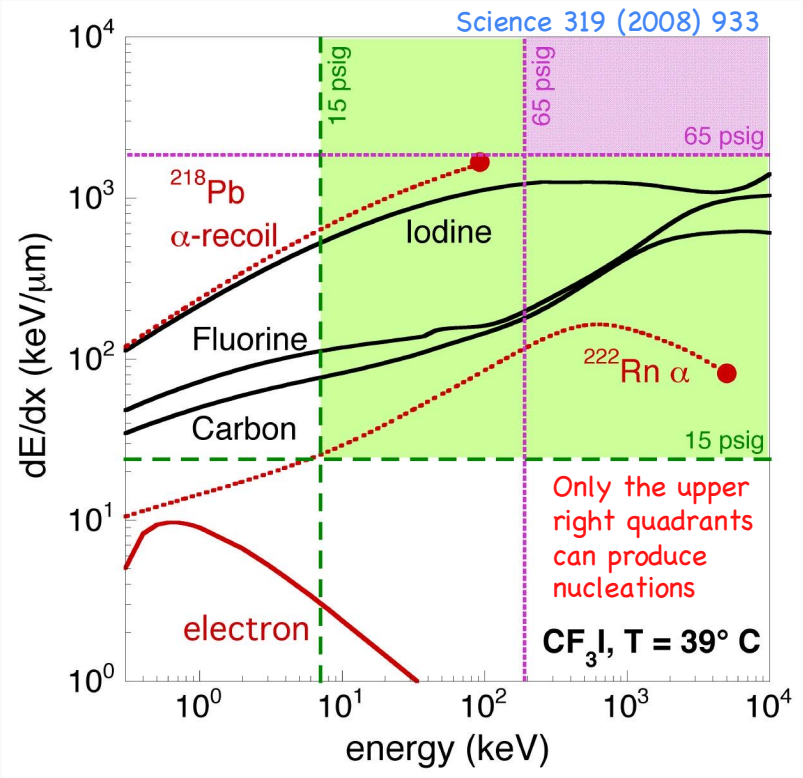
Threshold in deposited energy

$$E > E_c = 4\pi r_c^2 \left(\gamma - T\frac{\partial\gamma}{\partial T}\right) + \frac{4}{3}\pi r_c^3 \rho_v \frac{h_{fg}}{M} + \frac{4}{3}\pi r_c^3 P \quad , \quad r_c = 2\gamma/\Delta P$$

$$\mathrm{d}E/\mathrm{d}x > E_c/(ar_c)$$

Threshold also in stopping power, allows for efficient INTRINSIC MIP background rejection





- Best ER rejection of any WIMP detector.
- Scalability: built-in " amplification. Choice of three triggers: pressure, acoustic, motion.
- Excellent sensitivity to SD WIMP couplings (fluorine)
- High spatial granularity = additional n rejection mechanism
- Low cost, room temperature operation, safe refrigerants.

Listening to particles (yes, listening)

Glaser (1955)

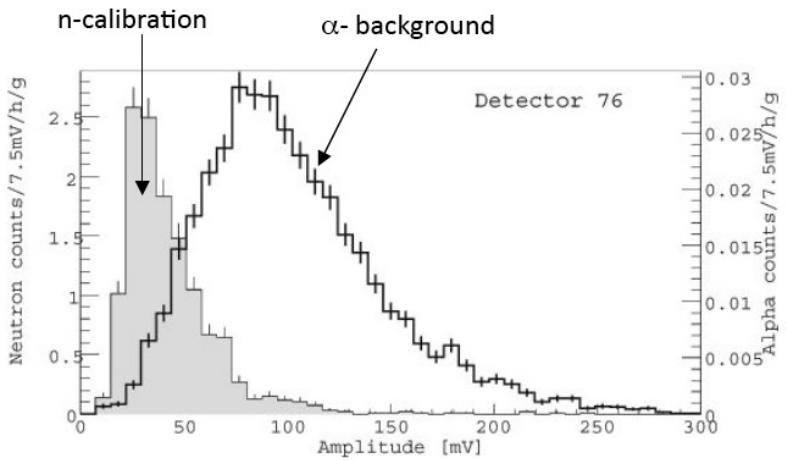
In order to see events more interesting than muons passing straight through the chamber, we took advantage of the violence of the eruption which produces an audible "plink" at each event. A General Electric variable-reluctance phonograph pickup was mounted with its stylus pressing against the wall of the chamber. Vibration signals occurring during the quiescent period after the expansion were allowed to trigger the lights and take pictures. In this way we saw tracks of particles passing through the chamber in various directions,

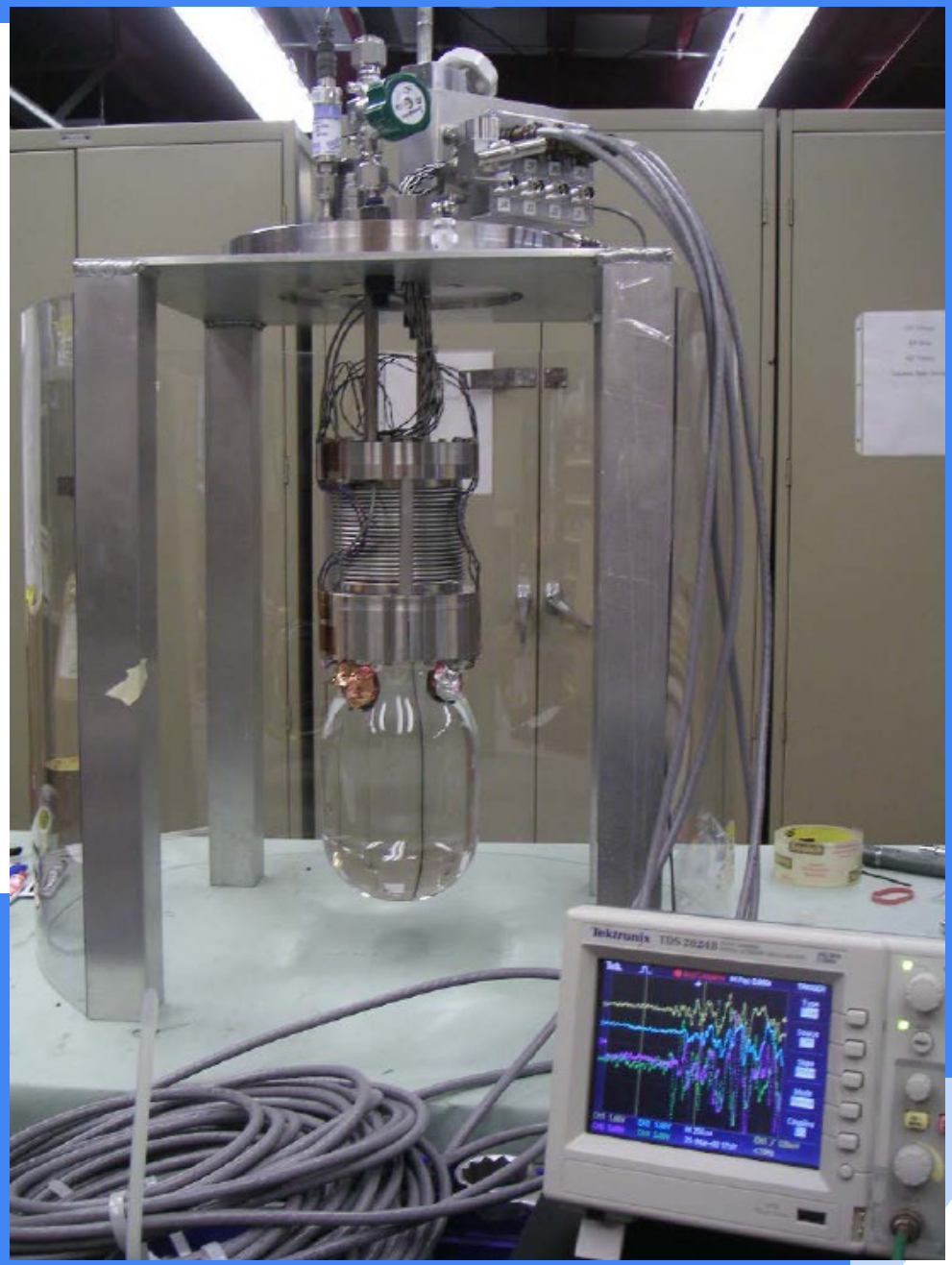
Martynyuk & Smirnova (1991)

The initial pressure in the volume V depends on the energy transmitted by the particle to that volume. Consequently, the characteristics of the acoustic pulse depend on the parameters of the particle responsible for formation of the bubble....

The parameters of these pulses must depend strongly on the characteristics of the particle.

PICASSO collab. (2009)





Listening to particles (yes, listening)

Glaser (1955)

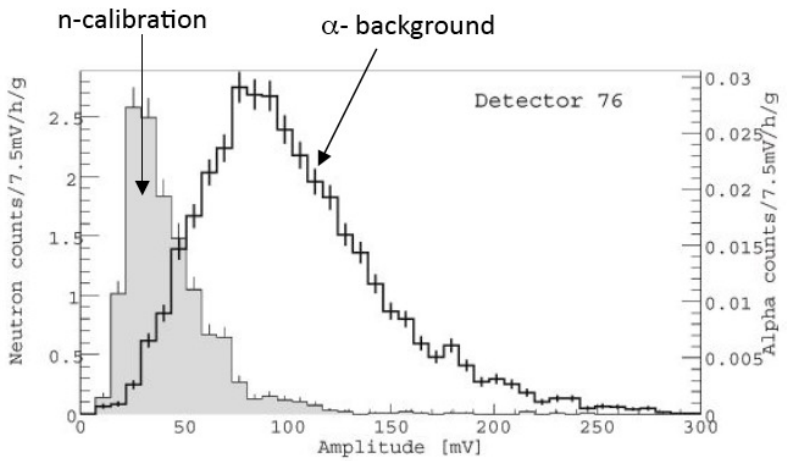
In order to see events more interesting than muons passing straight through the chamber, we took advantage of the violence of the eruption which produces an audible "plink" at each event. A General Electric variable-reluctance phonograph pickup was mounted with its stylus pressing against the wall of the chamber. Vibration signals occurring during the quiescent period after the expansion were allowed to trigger the lights and take pictures. In this way we saw tracks of particles passing through the chamber in various directions,

Martynyuk & Smirnova (1991)

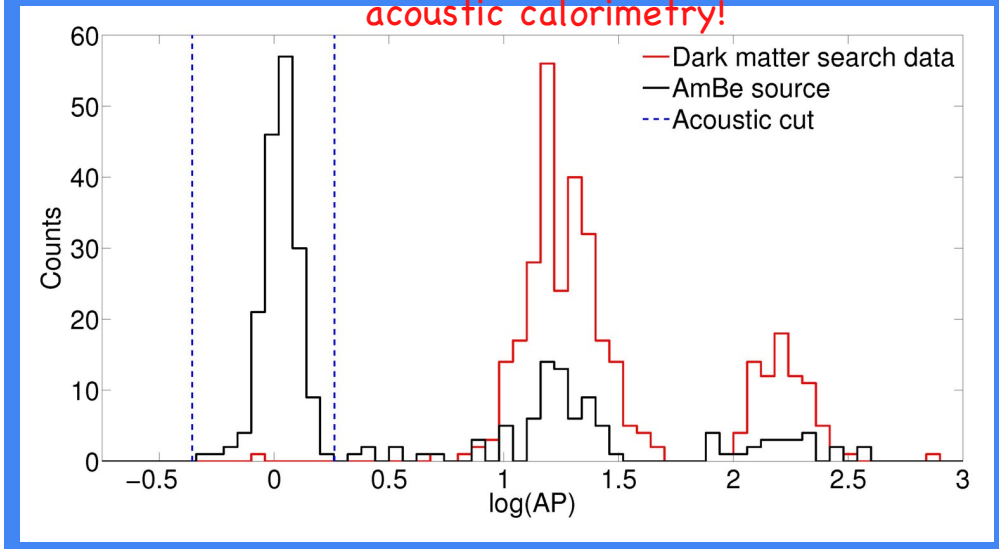
The initial pressure in the volume V depends on the energy transmitted by the particle to that volume. Consequently, the characteristics of the acoustic pulse depend on the parameters of the particle responsible for formation of the bubble....

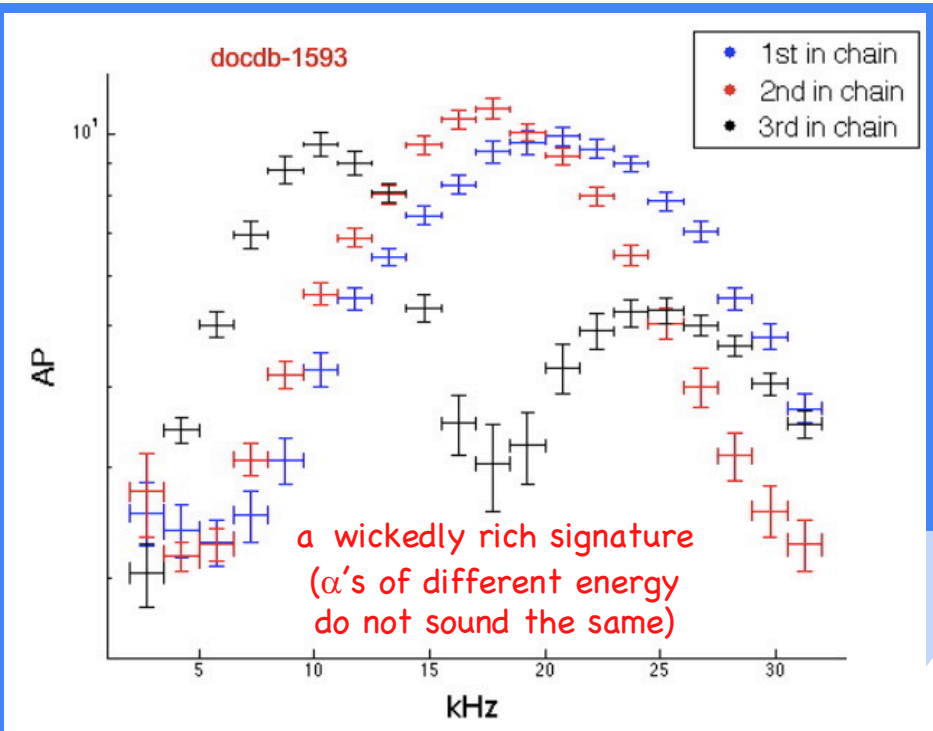
The parameters of these pulses must depend strongly on the characteristics of the particle.

PICASSO collab. (2009)



acoustic calorimetry!





Listening to particles (yes, listening)

IEEE Transactions on Nuclear Science, Vol.NS-25, No.1, February 1978
Pis'ma Zh. Eksp. Teor. Fiz. **25**, No. 5, 232–233 (5 March 1977)

Naive wave considerations[3,4] suggest that the sonic wave produced by the instantaneous expansion of this hot rod should have the following properties: (1) The half-wavelength $\lambda/2$ is on the order of the diameter d of the energy deposition. (2) The fundamental frequency f of the sound emitted is therefore c/λ, where c is the speed of sound and the period $\tau = 1/f$ ($c = 1.5$ mm/μsec in water). (3) For observation angles $\theta \lesssim \lambda/L$, the sonic wave from the cylindrical acoustic antenna is

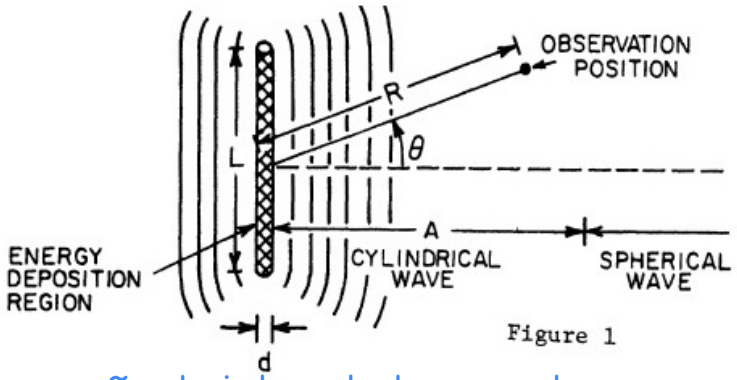
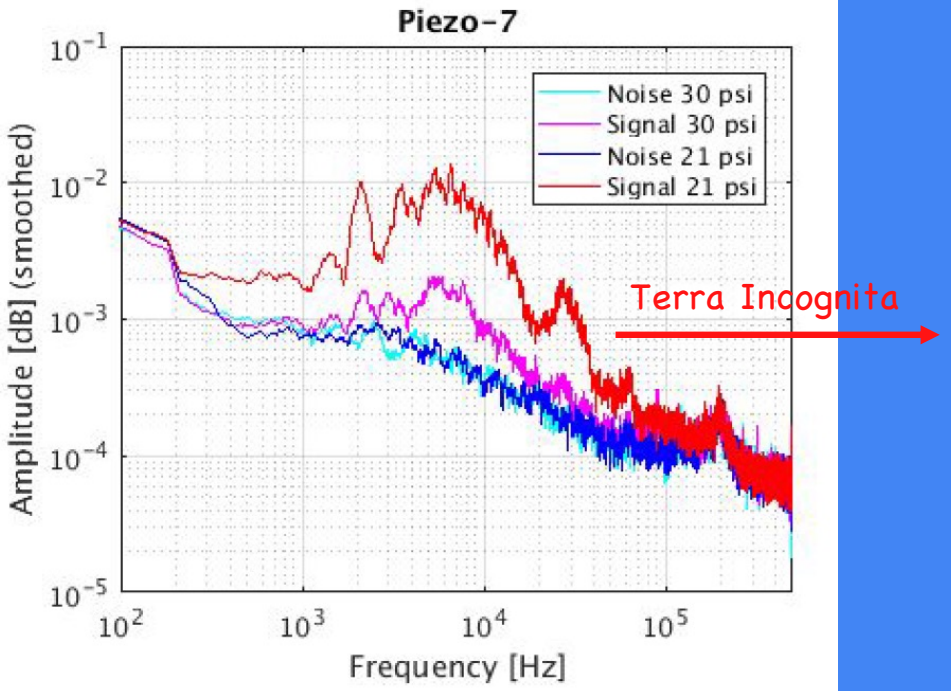
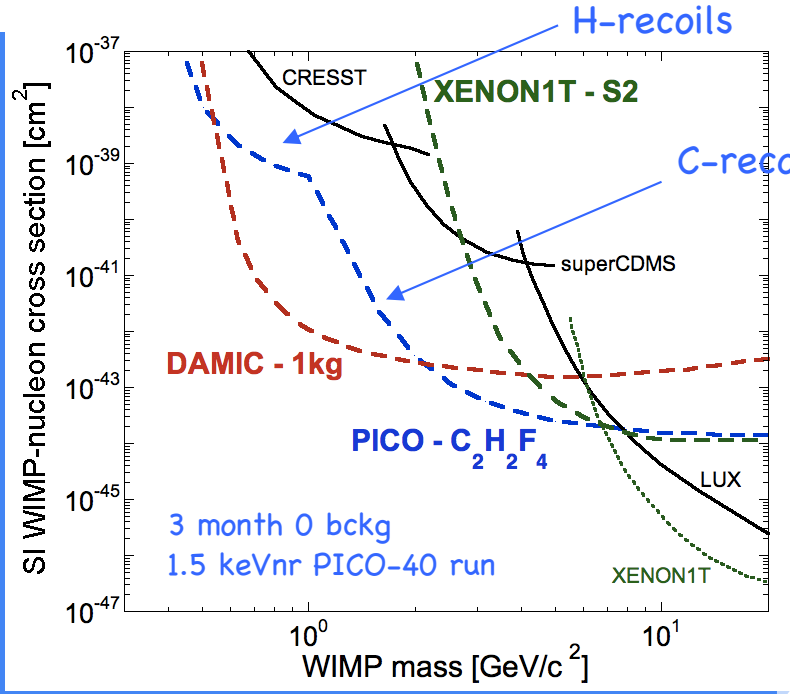


Figure 1

~scale-independent argument





Listening to particles (yes, listening)

IEEE Transactions on Nuclear Science, Vol.NS-25, No.1, February 1978
Pis'ma Zh. Eksp. Teor. Fiz. **25**, No. 5, 232–233 (5 March 1977)

Naive wave considerations[3,4] suggest that the sonic wave produced by the instantaneous expansion of this hot rod should have the following properties: (1) The half-wavelength $\lambda/2$ is on the order of the diameter d of the energy deposition. (2) The fundamental frequency f of the sound emitted is therefore c/λ, where c is the speed of sound and the period $\tau = 1/f$ ($c = 1.5$ mm/μsec in water). (3) For observation angles $\theta \lesssim \lambda/L$, the sonic wave from the cylindrical acoustic antenna is

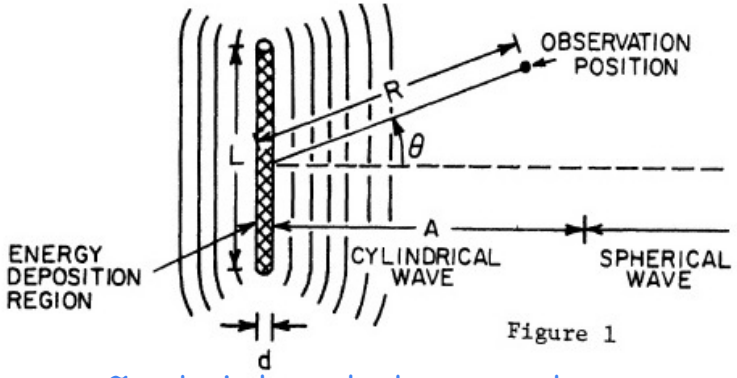
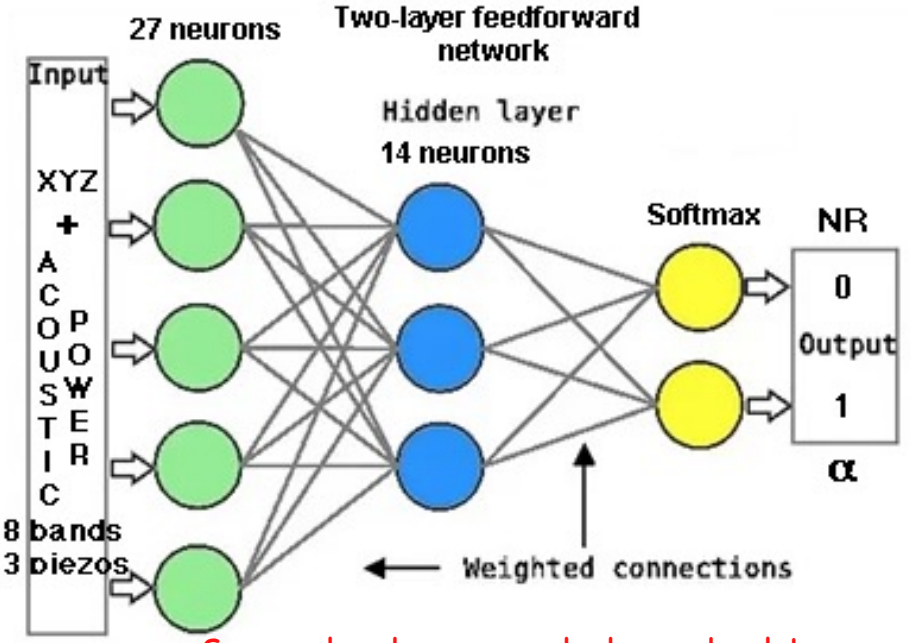
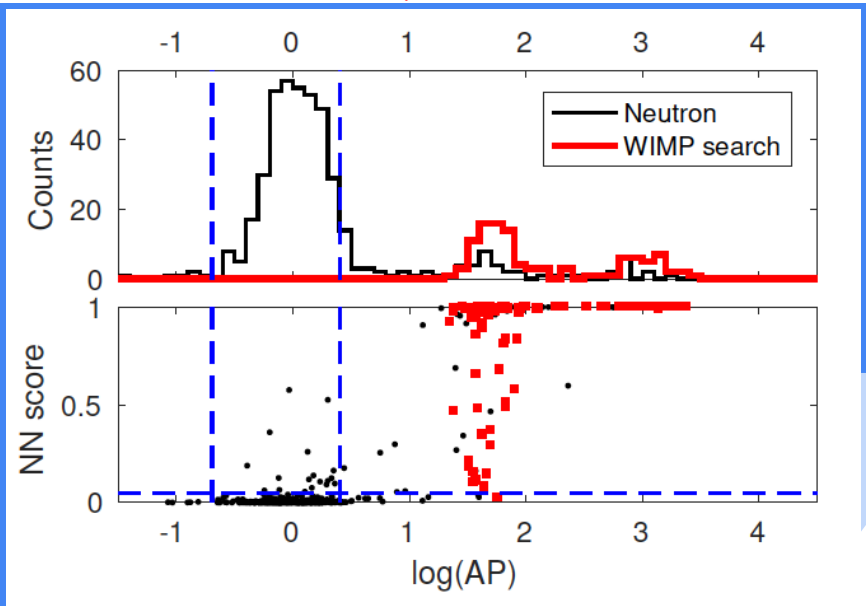


Figure 1

~scale-independent argument



Surrender to your robot overlords!



Listening to particles (yes, listening)

IEEE Transactions on Nuclear Science, Vol.NS-25, No.1, February 1978
Pis'ma Zh. Eksp. Teor. Fiz. **25**, No. 5, 232–233 (5 March 1977)

Naive wave considerations[3,4] suggest that the sonic wave produced by the instantaneous expansion of this hot rod should have the following properties: (1) The half-wavelength $\lambda/2$ is on the order of the diameter d of the energy deposition. (2) The fundamental frequency f of the sound emitted is therefore c/λ, where c is the speed of sound and the period $\tau = 1/f$ ($c = 1.5$ mm/μsec in water). (3) For observation angles $\theta \lesssim \lambda/L$, the sonic wave from the cylindrical acoustic antenna is

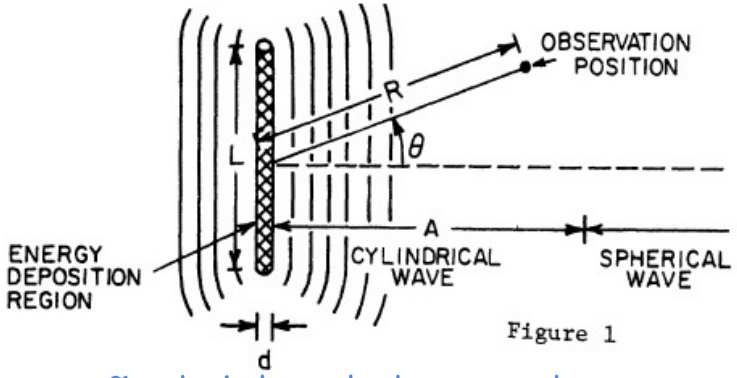
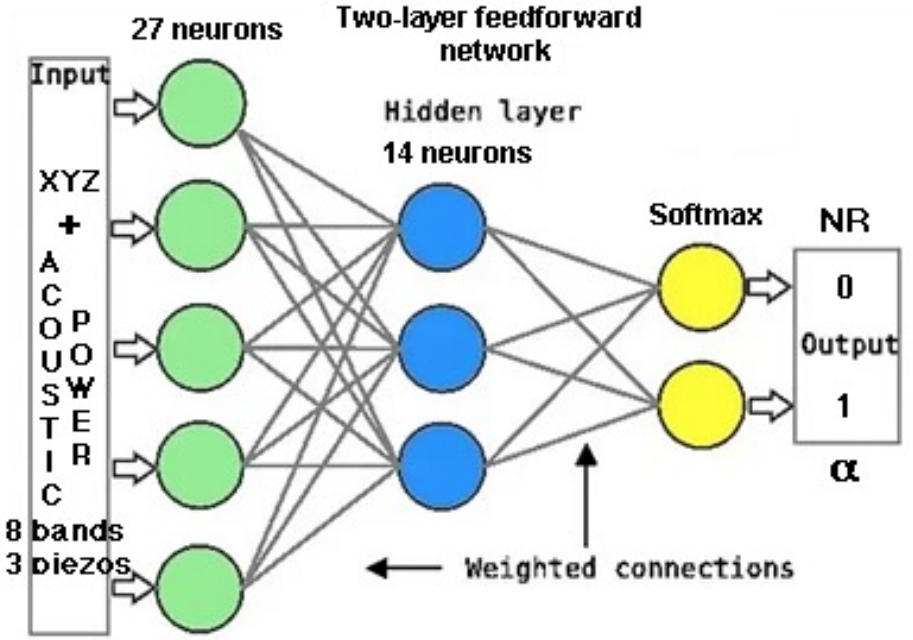
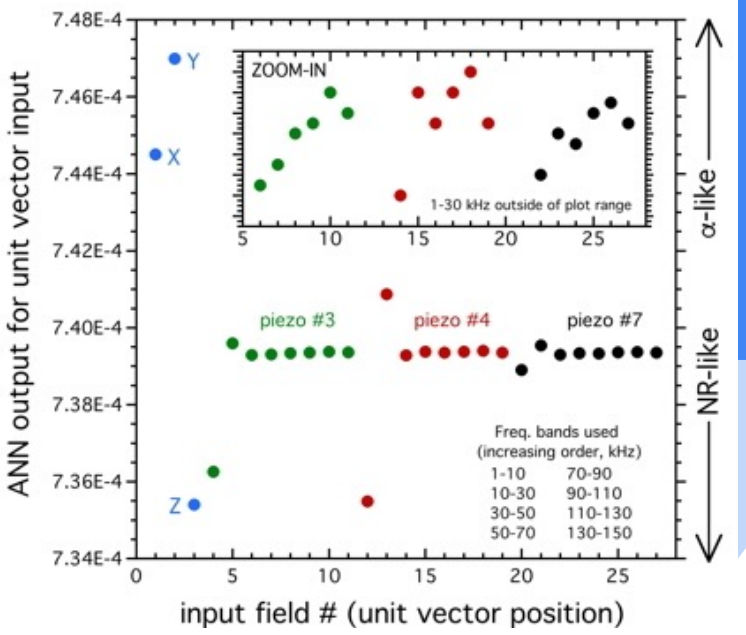


Figure 1

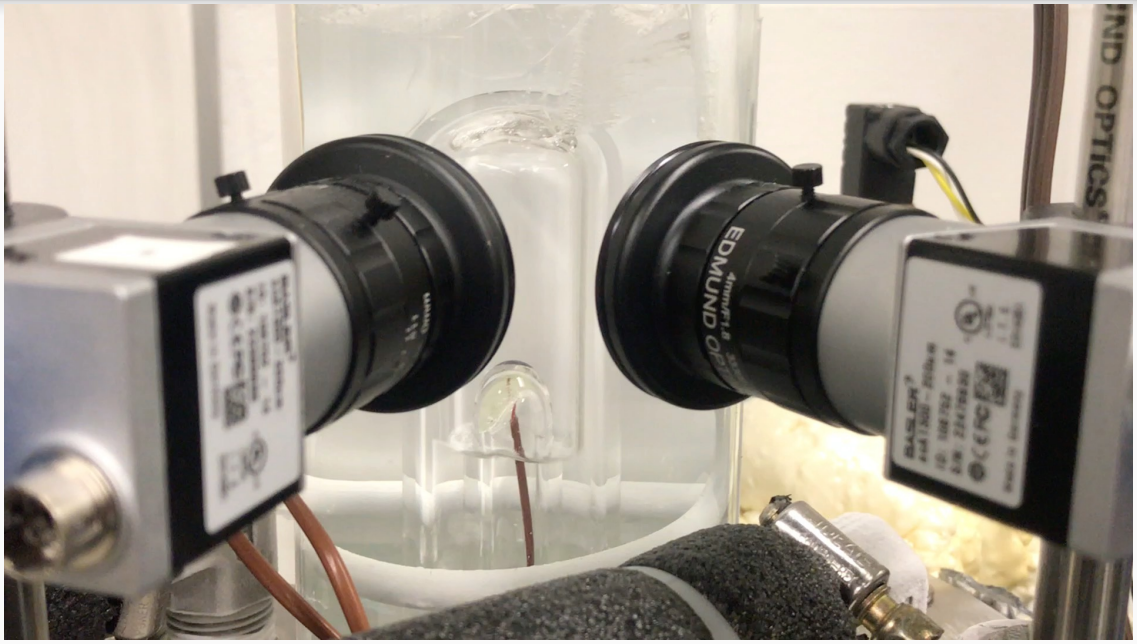
~scale-independent argument

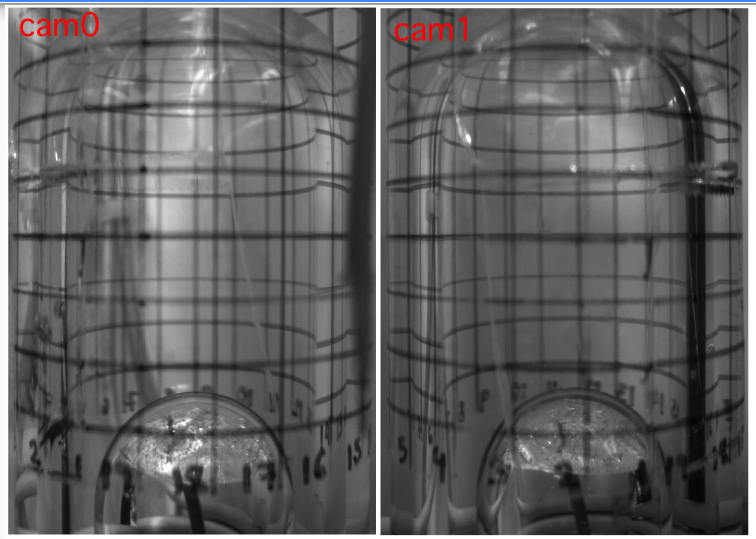


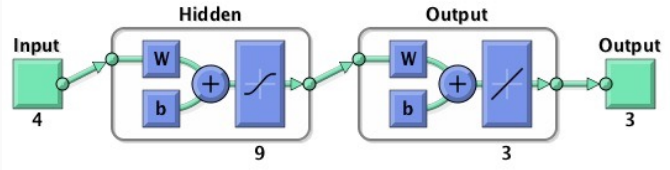
Peeering into the mind of an ANN:

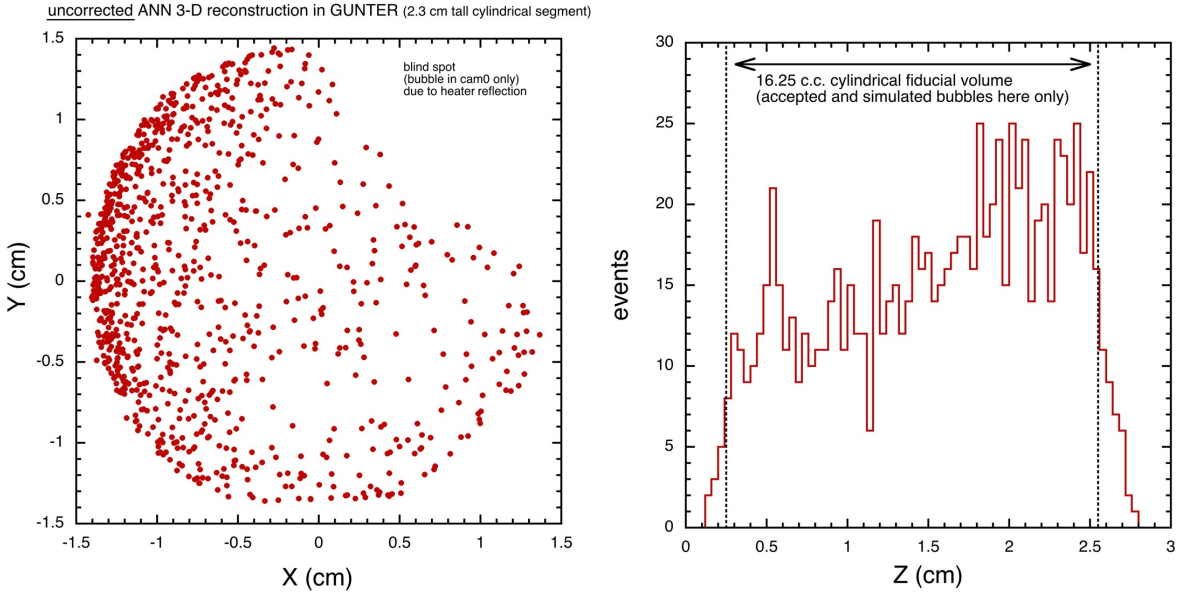


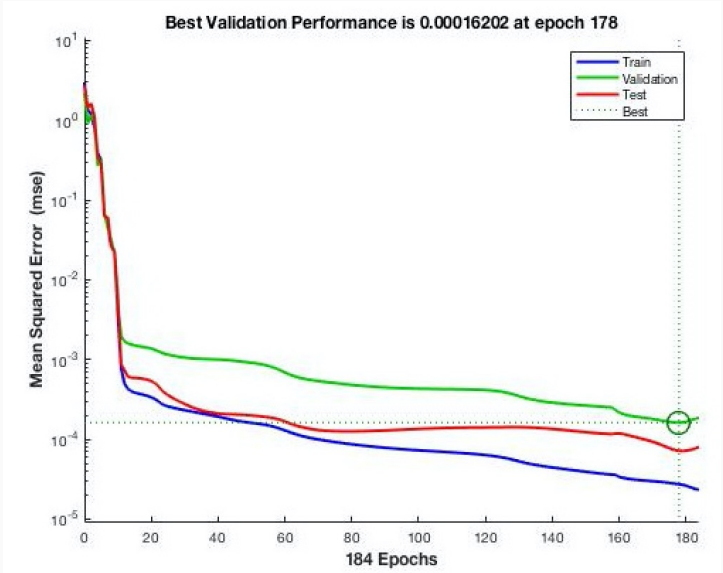
Another interesting application: stereoscopic vision (a NN in your brain does this)



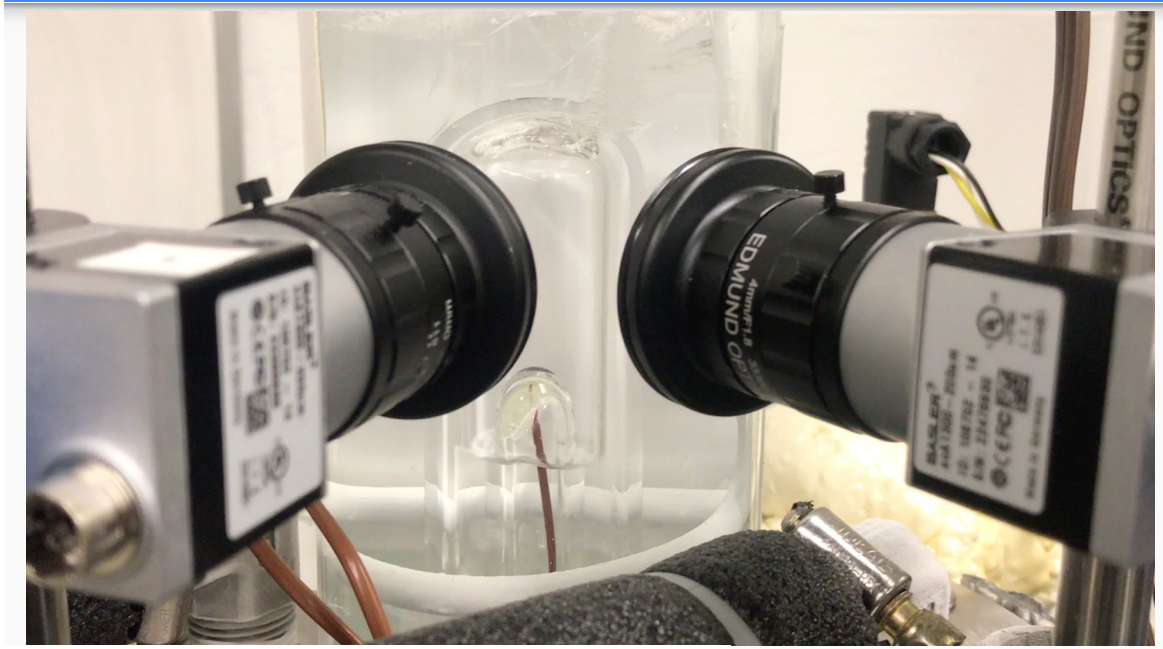


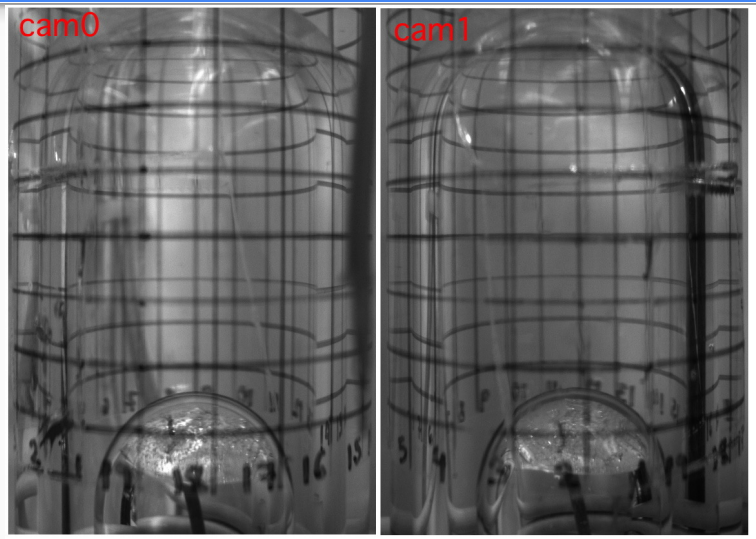


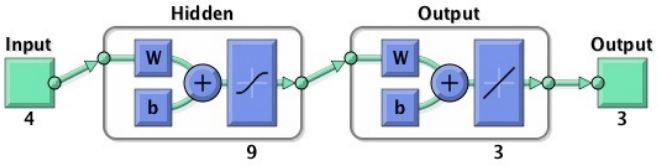




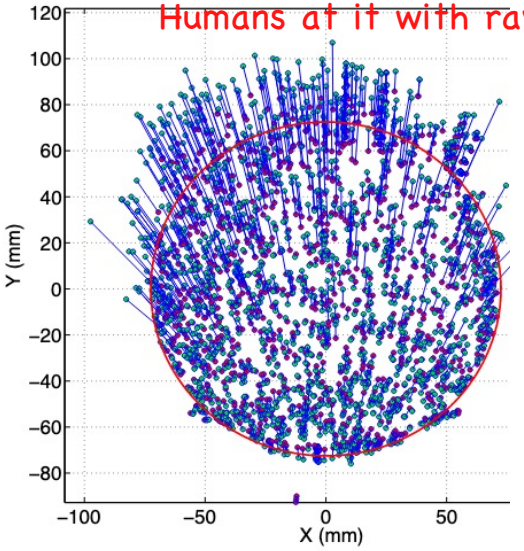
Another interesting application: stereoscopic vision (a NN in your brain does this)

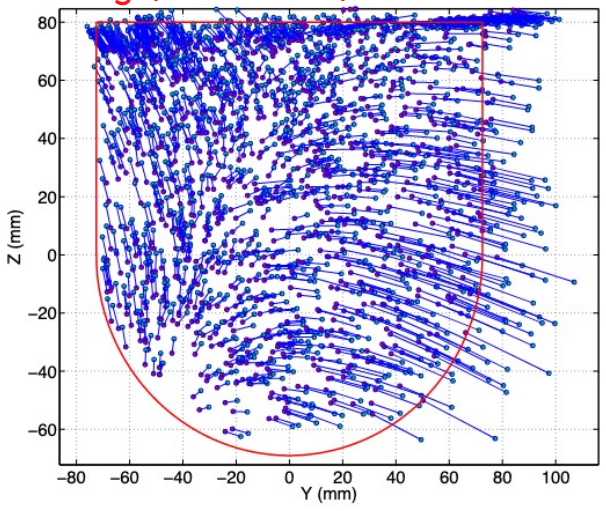






Humans at it with ray-tracing (the horror)





Z (cm)

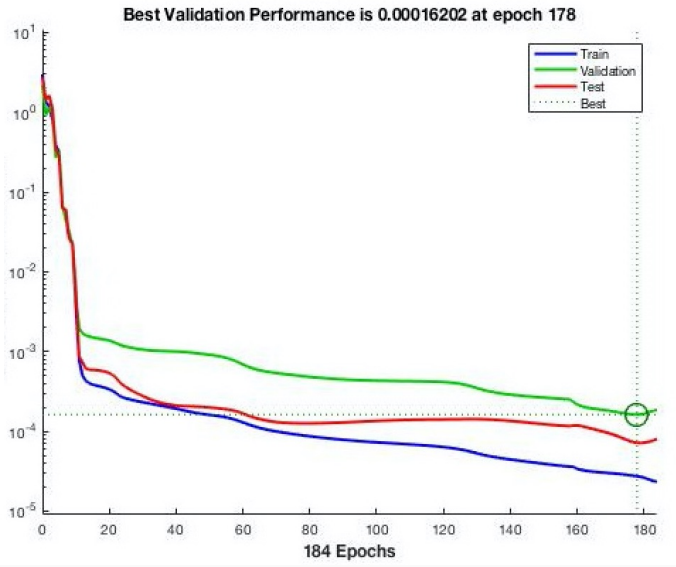
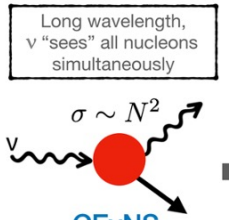


Figure 1: X-Y coordinates before and after tuning. The camera position is at the bottom of this figure. Bubbles have in general moved away from the right-hand camera (Cam0).

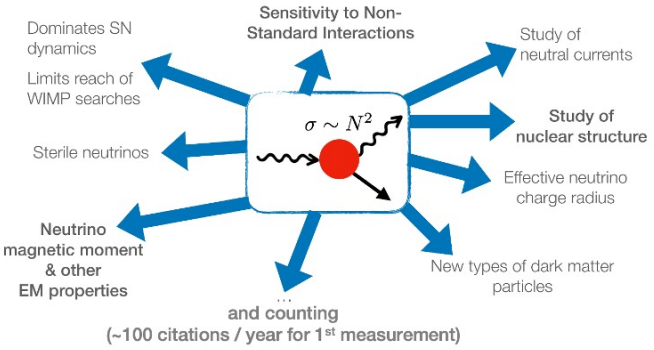
A dose of my own medicine (Cerenkov discrimination, MLP in Fourier space)

Long wavelength, ν "sees" all nucleons simultaneously



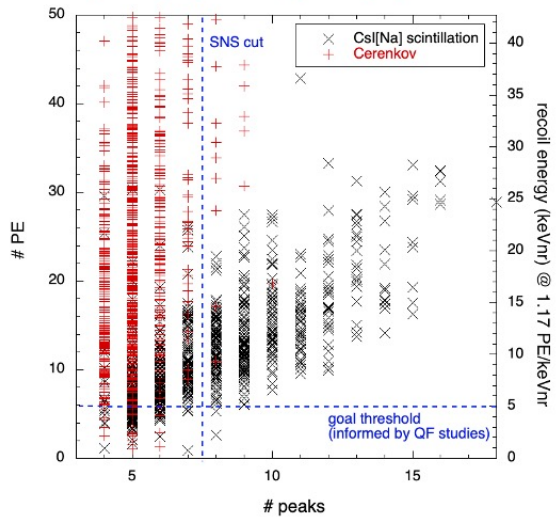
Cross section increases as N^2. (largest of all -> small detectors) CATCH: sub-keV to few keV recoils are only observable.

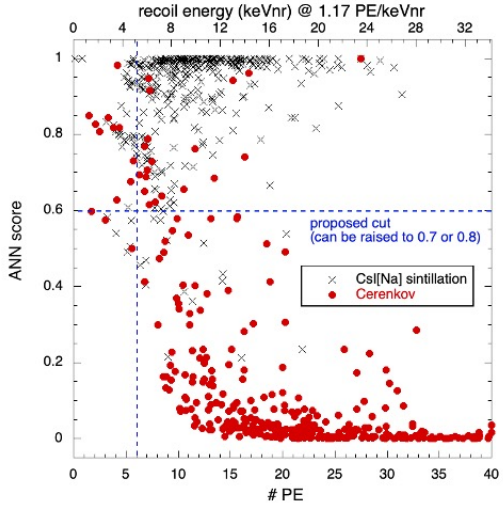
First measurement only 9 years ago (40 years after description)

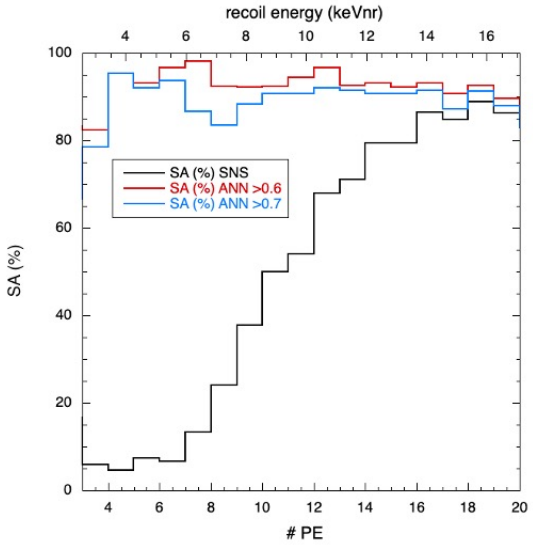


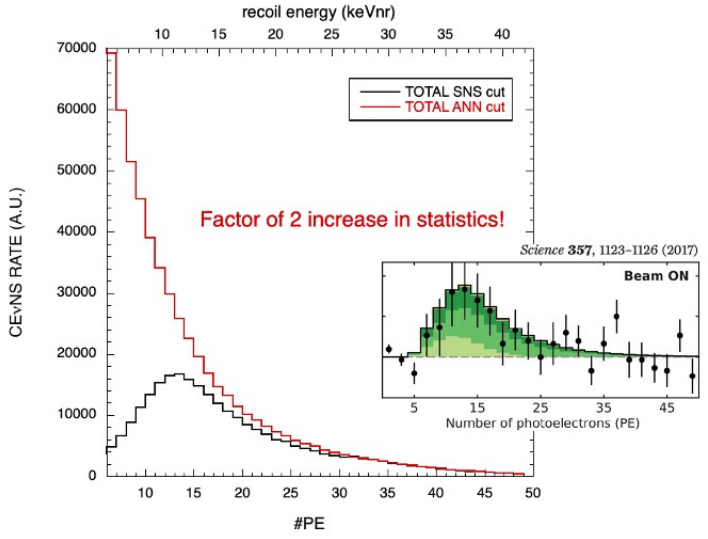
and counting
(~100 citations / year for 1st measurement)

Improving signal acceptance (SA) near threshold via ANNs:

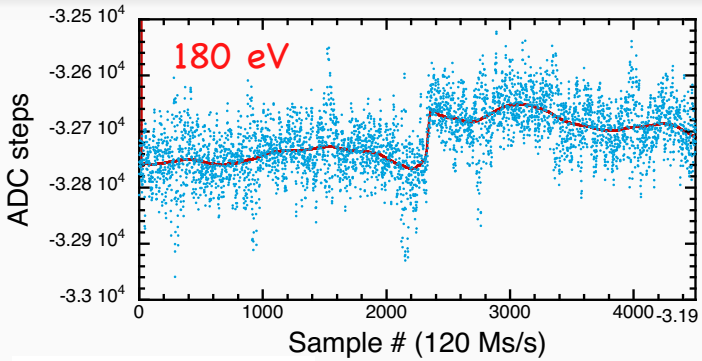




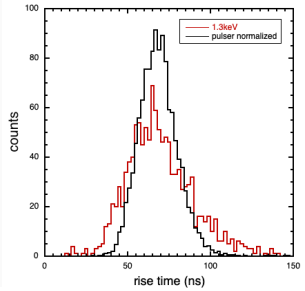




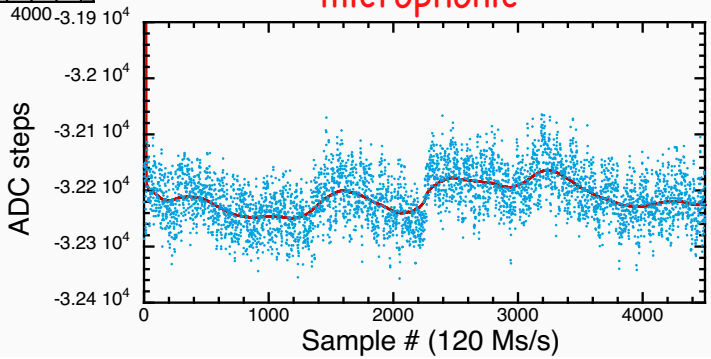
A dose of my own medicine (Cerenkov discrimination, MLP in Fourier space)

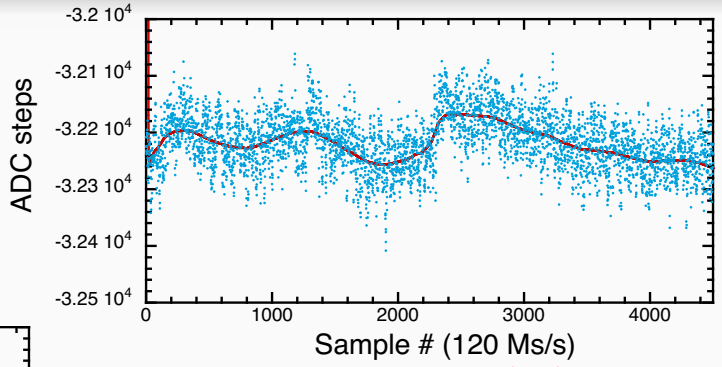


ν-like

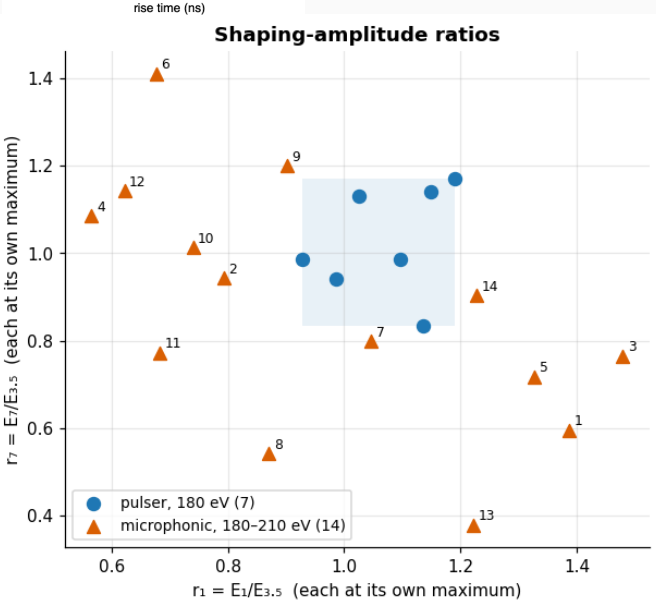


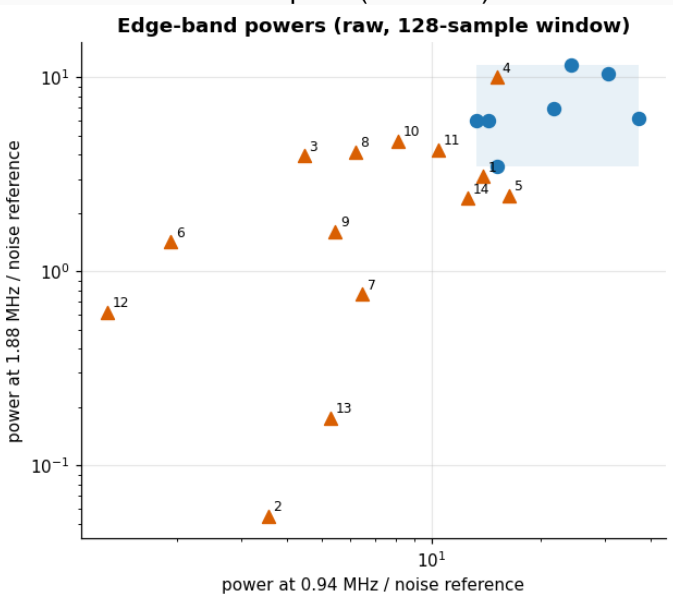
microphonic

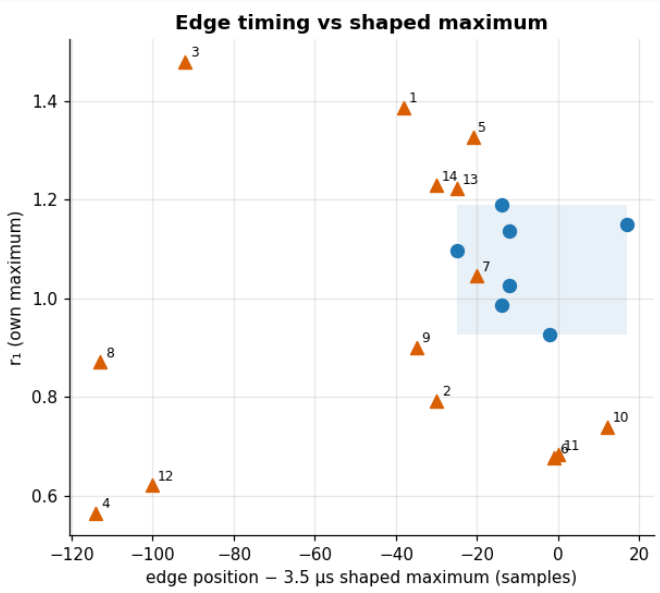




microphonic, but…

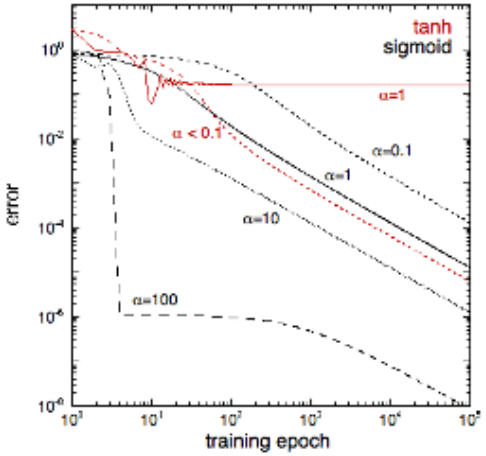






Shaded boxes: range spanned by the 7 pulser events. Ratios use a box-difference stand-in for your cusp; labels = microphonic trace number.

Off you go:



PHYS250 HOMEWORK #3 (due Nov. 15)

On your own (no help from Claude, other than for debugging):

* Build a single-layer perceptron as described in class, profiting from the matrix algebra and matrix manipulation commands of FORTRAN 90. Involve the commands MATMUL, SUM, TRANSPOSE. Train it in batch mode with the four examples mentioned in our slides, and test it with the other four remaining cases. Does it work? If you observe any limitations, explain their cause. Does it work for the example provided in the slides involving the XOR function?

* Compare the performance of the tanh and sigmoid transfer functions in your code (do not forget to change their derivative when doing this, also in the validation process). Is one faster than the other? (use the command CALL CPU_TIME( ) to time your code at given intervals). Does one work better than the other for the test inputs? If so, why?

* Optimize the learning rate separately for these two transfer functions. Do this by plotting the error vs. training epoch, as in the example above. Describe your observations.

* Does the performance (error vs. epoch, or error in validation set) improve if you expand the training set? (e.g., by adding one more case to it). If you scramble it? (stay at four training cases, but use other different cases for training). Do not forget to update the dimensions of your matrices when adding or subtracting new cases to the training set.

* Once your code is built, test it for other hidden patterns of your own that link inputs and outputs: in doing so, develop a feeling for the limitations of the single neuron perceptron. Comment on your observations.

(see additional tasks in Canvas)

Reading Materials:

Good Primer, covering most concepts discussed in class:

- Chapter 2 in “Soft Computing” by Tettamanzi & Tomassini (available from Canvas)

Idem, Chapter 1 in:

- http://neuralnetworksanddeeplearning.com/

Good general reference:

- http://cognet.mit.edu/book/neural-smithing
- (full single pdf also posted on Canvas)

Much deeper (Deep Learning):

- http://ofir.io/How-to-Start-Learning-Deep-Learning/
- http://cs231n.stanford.edu/
- http://www.deeplearningbook.org/

If you want to start doing cool stuff fast (use our UChicago MATLAB license):

- https://www.mathworks.com/products/neural-network.html
- https://www.mathworks.com/help/nnet/getting-started-with-neural-network-toolbox.html